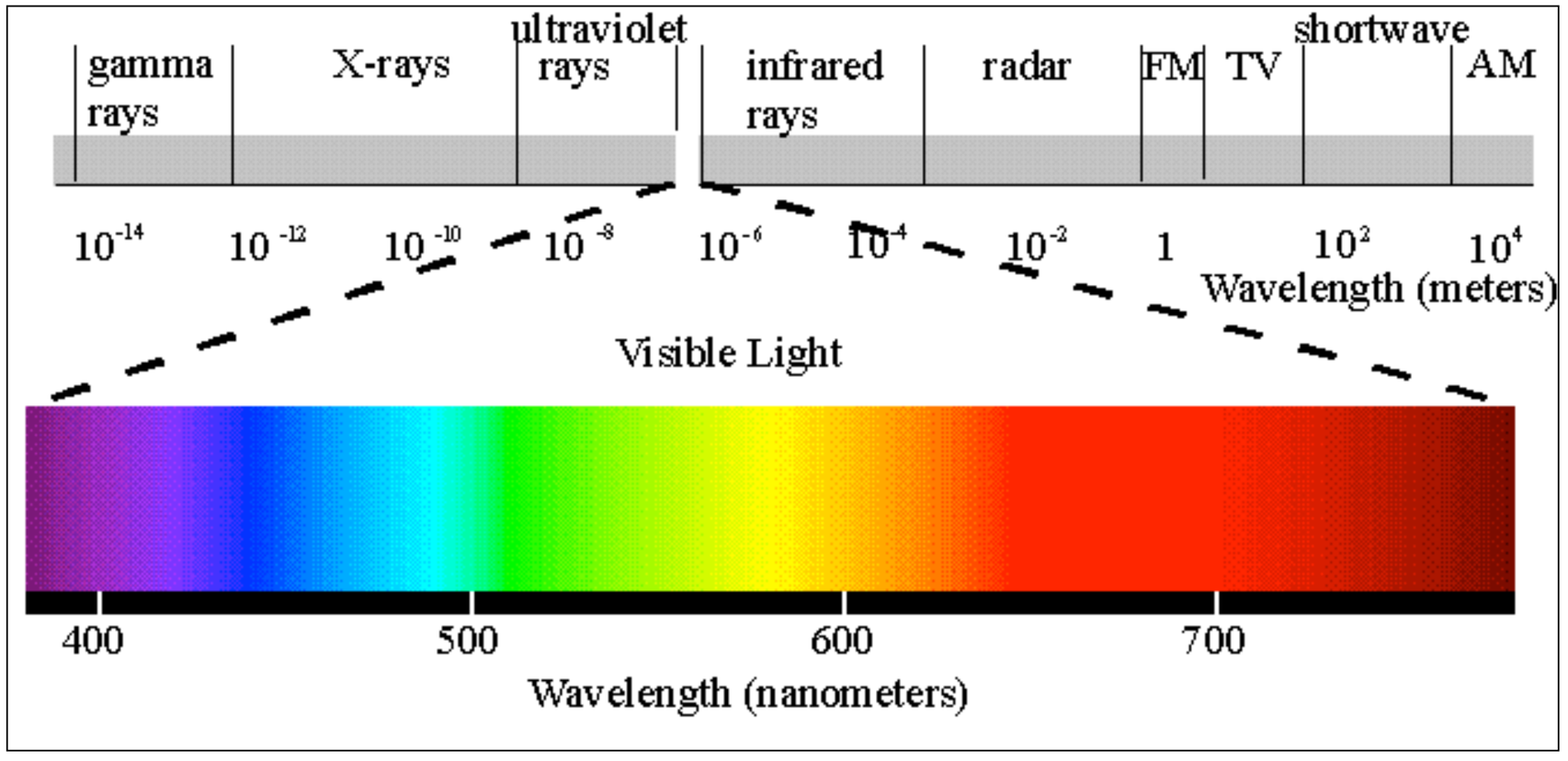
Lecture 15:

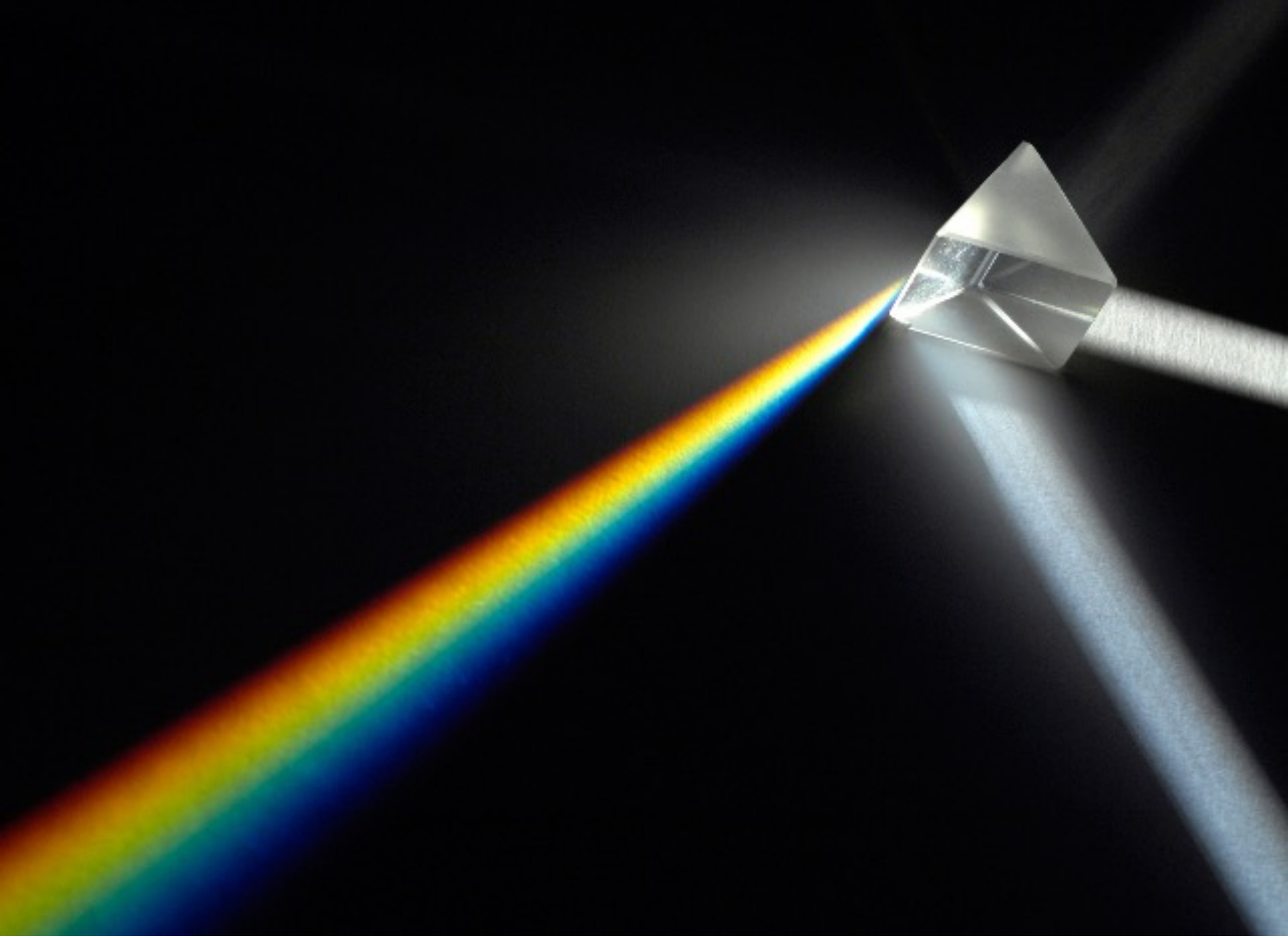
Color and Perception

Computer Graphics and Imaging
UC Berkeley CS184/284A

What is Light?

Radiation in a particular frequency range



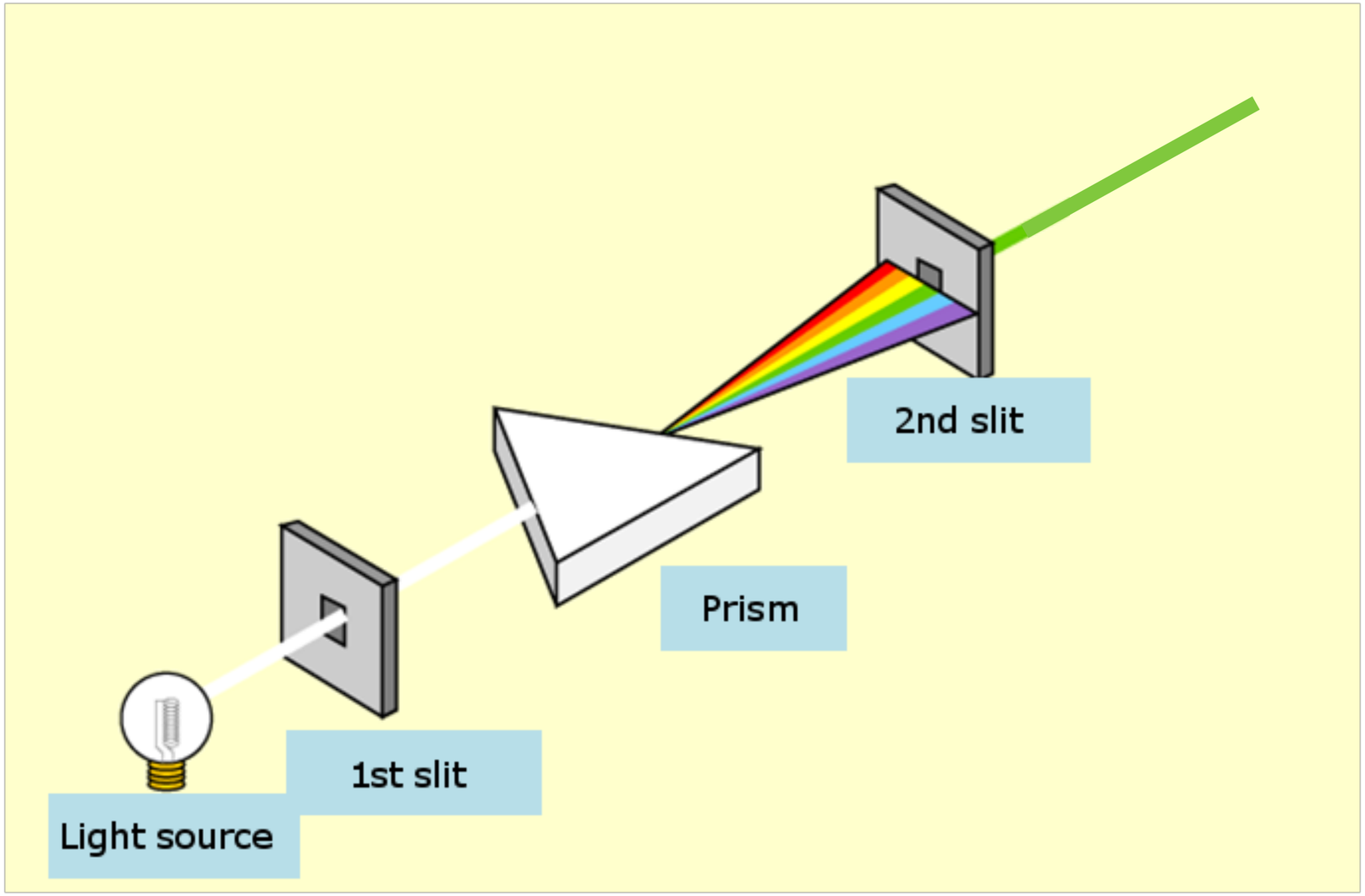


credit: Science Media Group.

?

credit: Science Media Group.

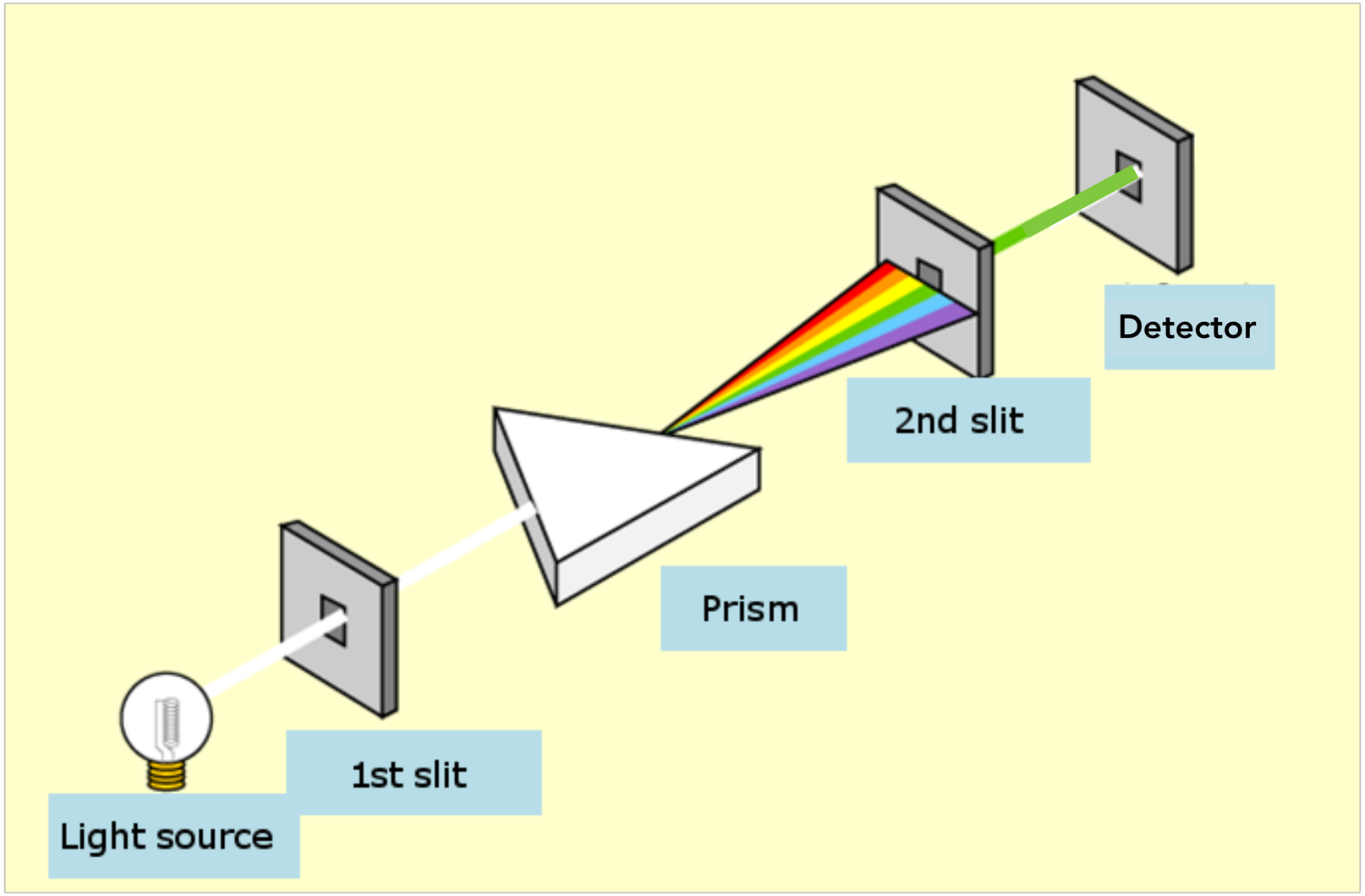
Monochromator



A monochromator delivers light of a single wavelength from a light source with broad spectrum. Control which wavelength by angle of prism.

Hegyi et al. "Introduction to Practical Biochemistry" http://elte.prompt.hu/sites/default/files/tananyagok/IntroductionToPracticalBiochemistry/ch04s07.html

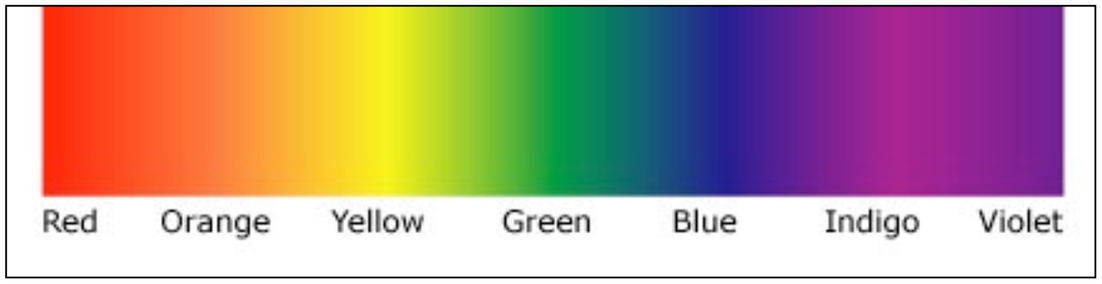
Spectrometer



For unknown light source, use a monochromator to isolate each wavelength of light for measurement

Hegyi et al. "Introduction to Practical Biochemistry" http://elte.prompt.hu/sites/default/files/tananyagok/IntroductionToPracticalBiochemistry/ch04s07.html

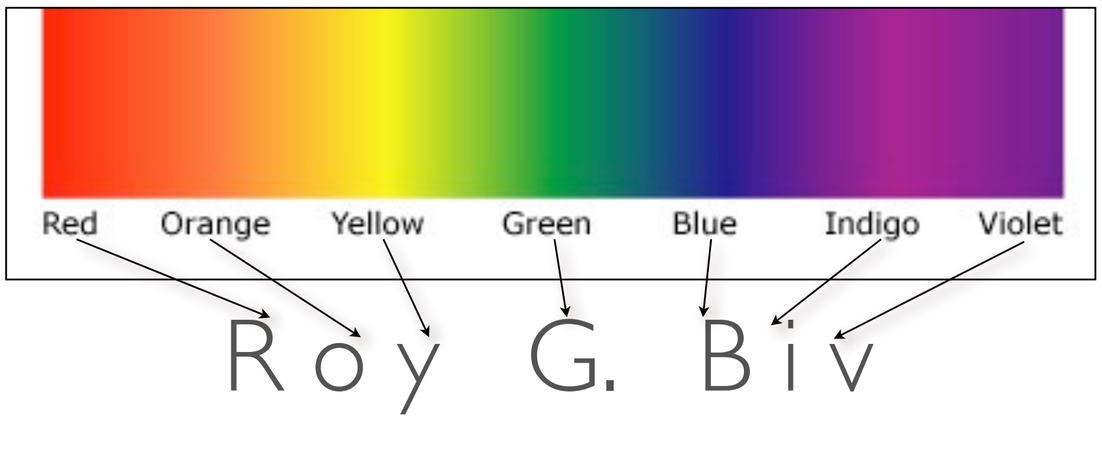
Spectral Colors



White LED

Reproductions only, can't show a real spectral color!

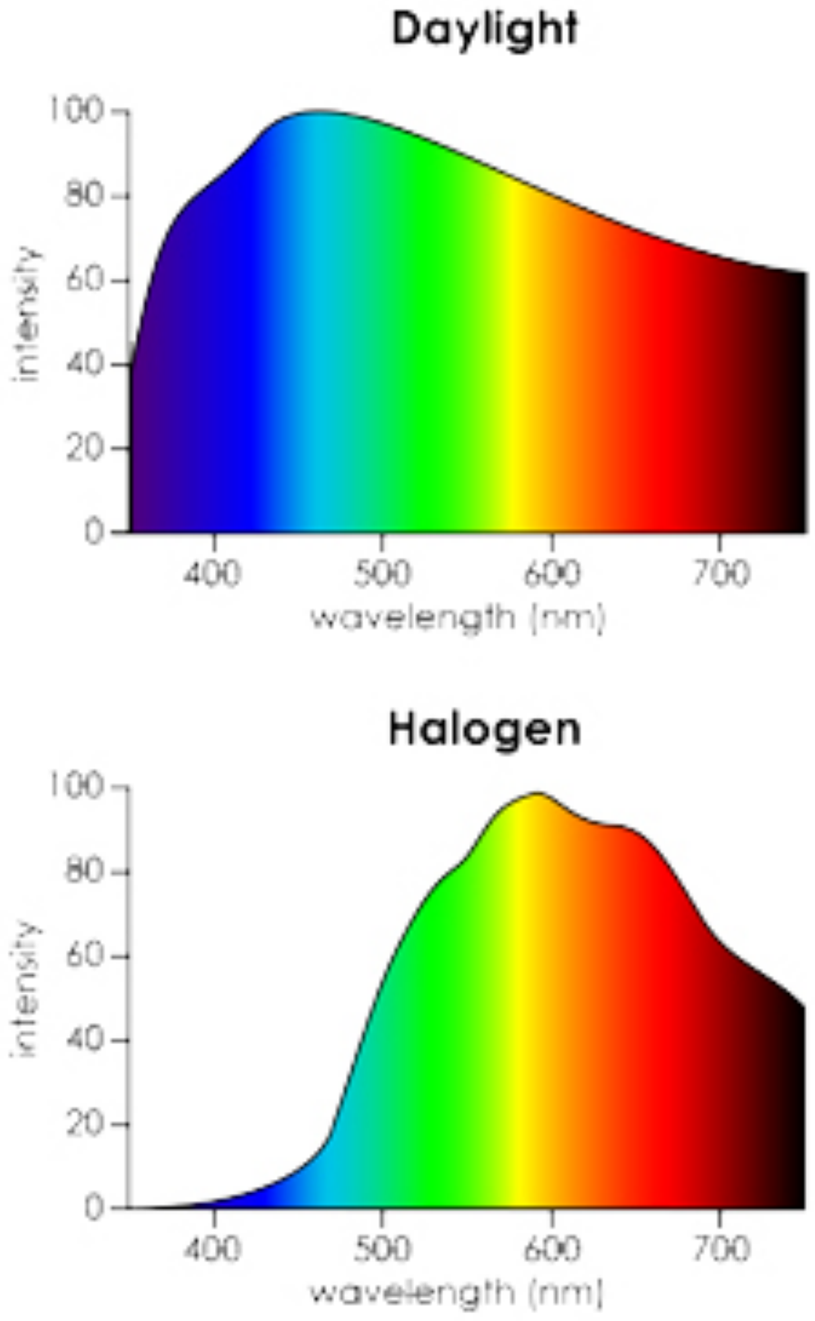
Spectral Colors

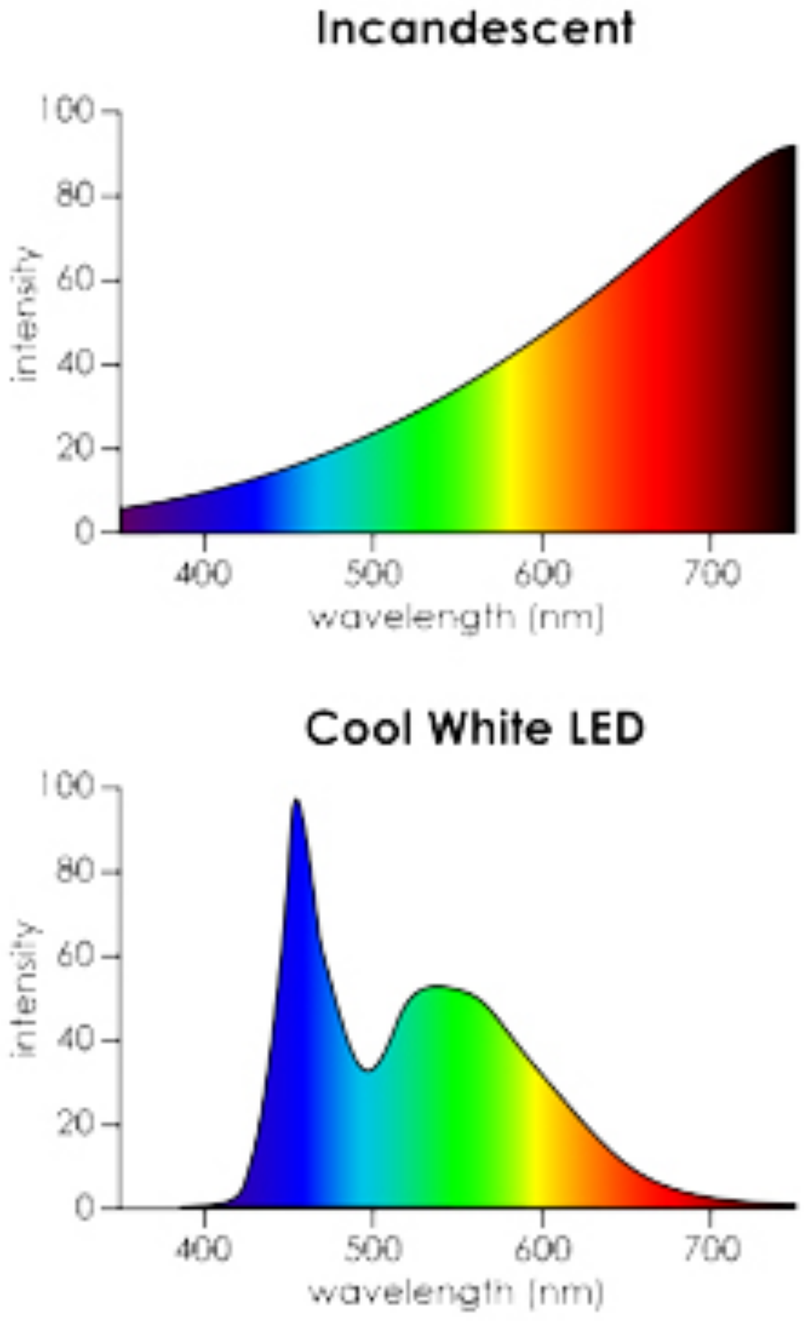


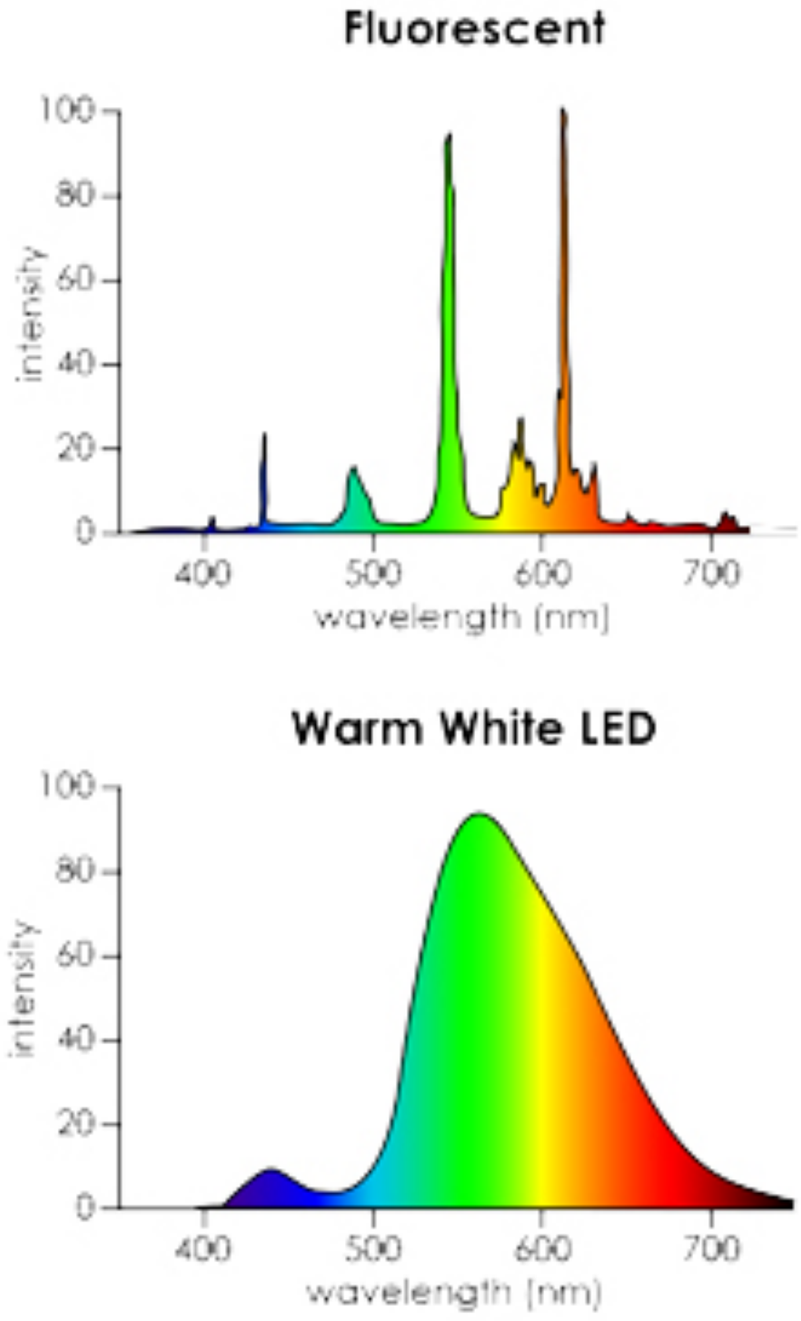
White LED

Reproductions only, can't show a real spectral color!

Other Colors





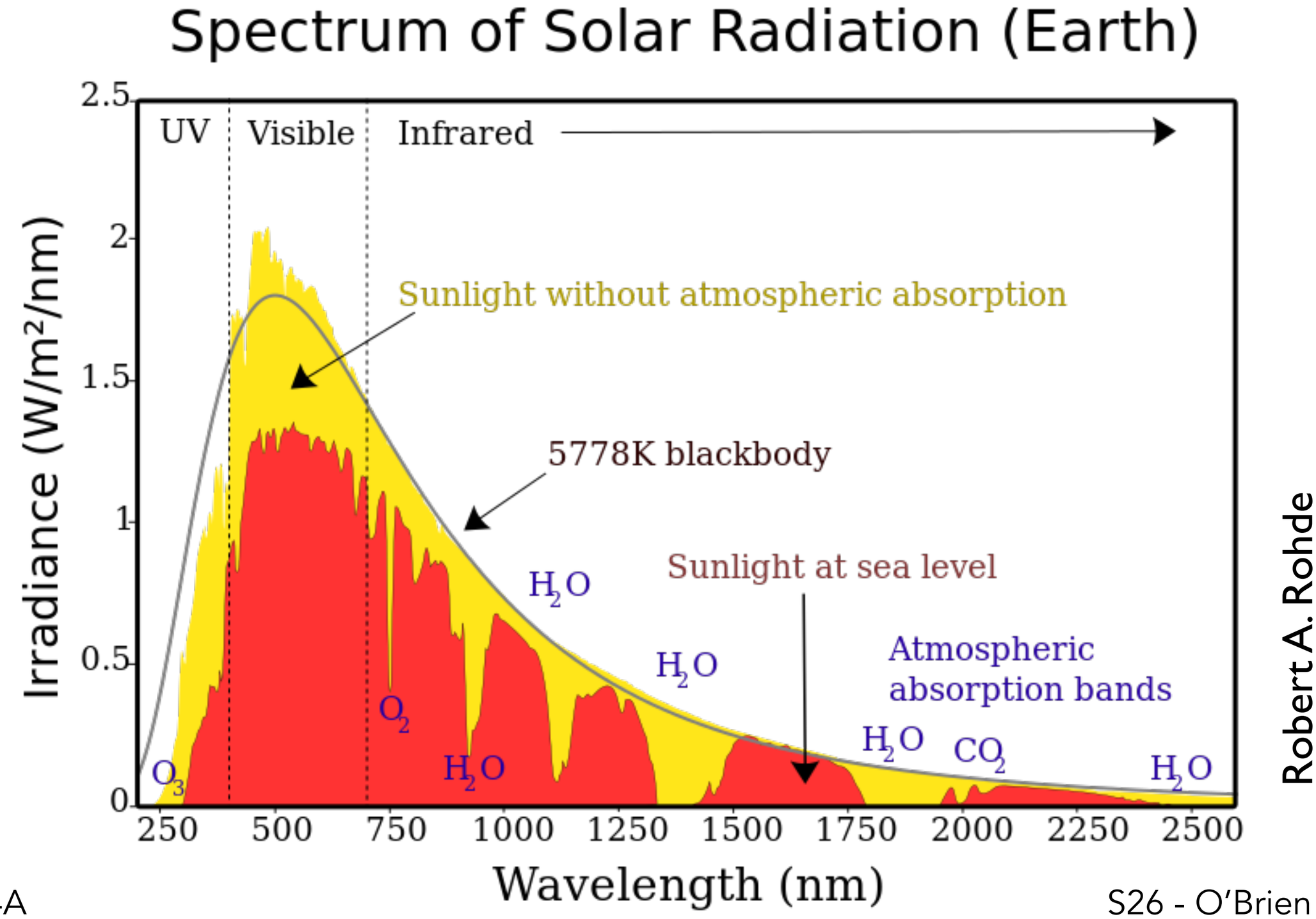


admesy
ADVANCED MEASUREMENT SYSTEMS

Other Colors

[Brian Wandell]

Other Colors



Other Colors

Most colors seen are a mix light of several frequencie

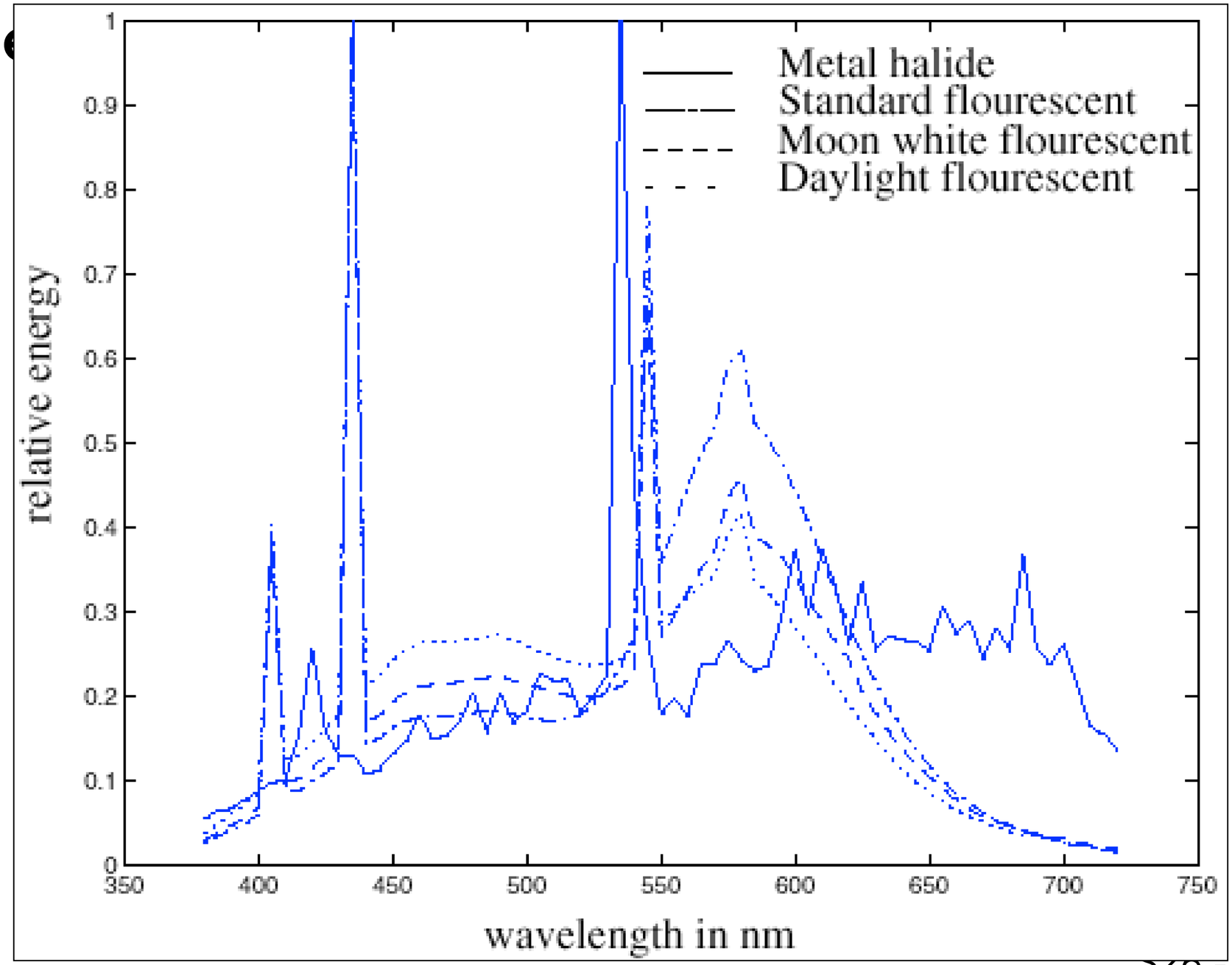


Image from David Forsyth

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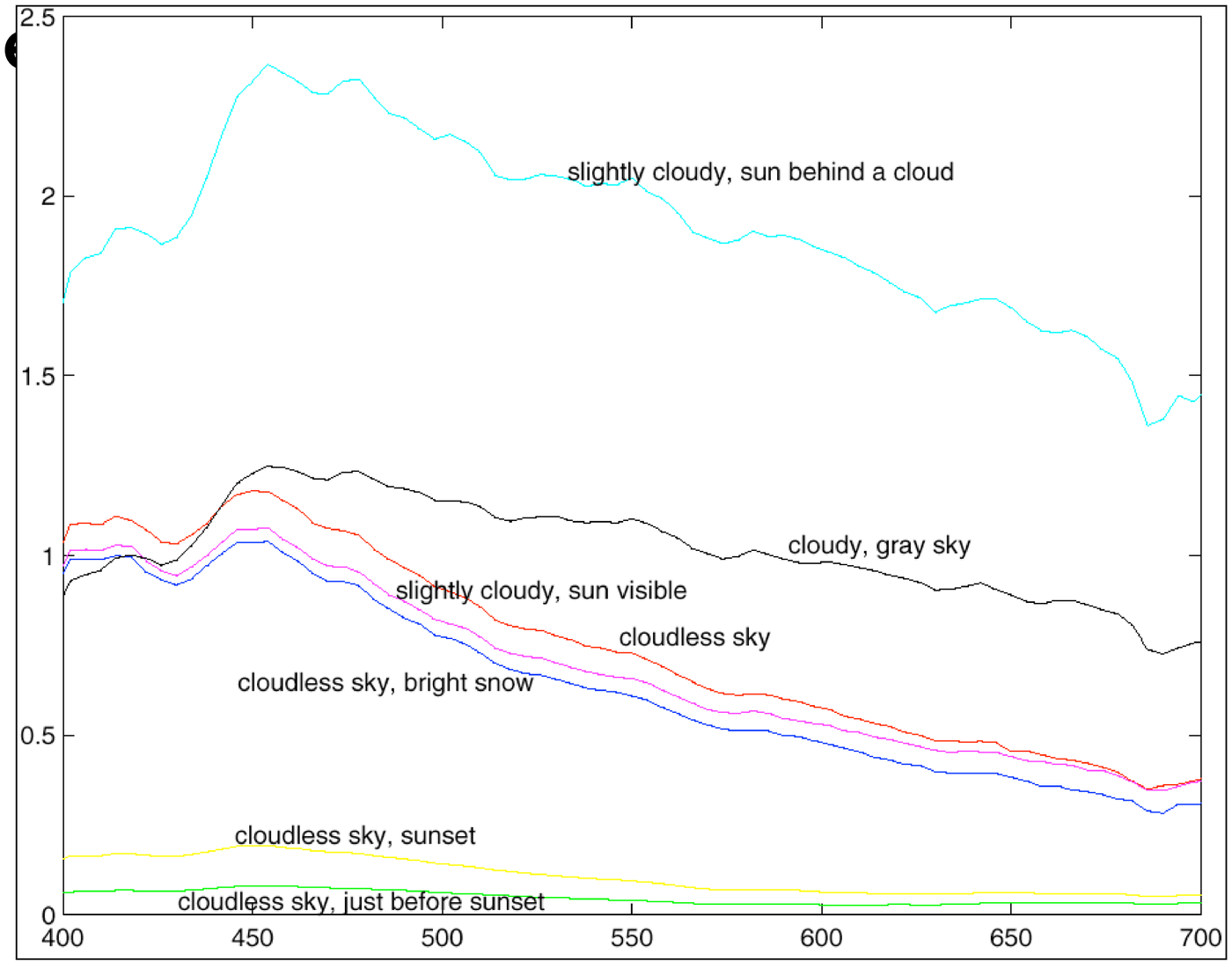

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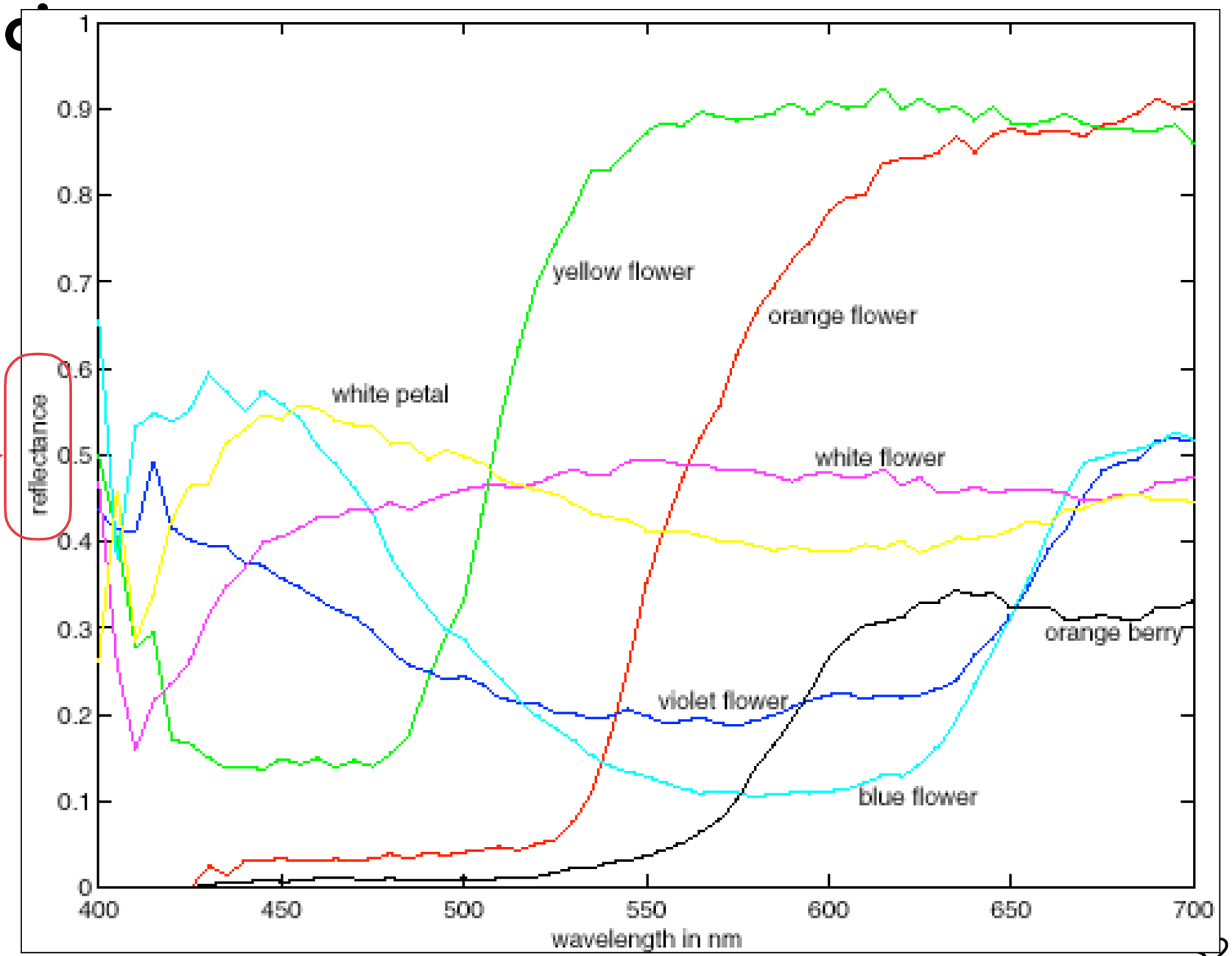
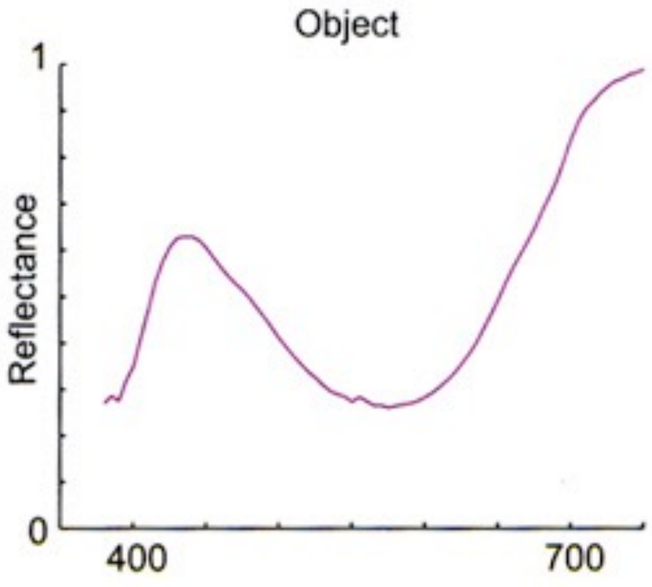
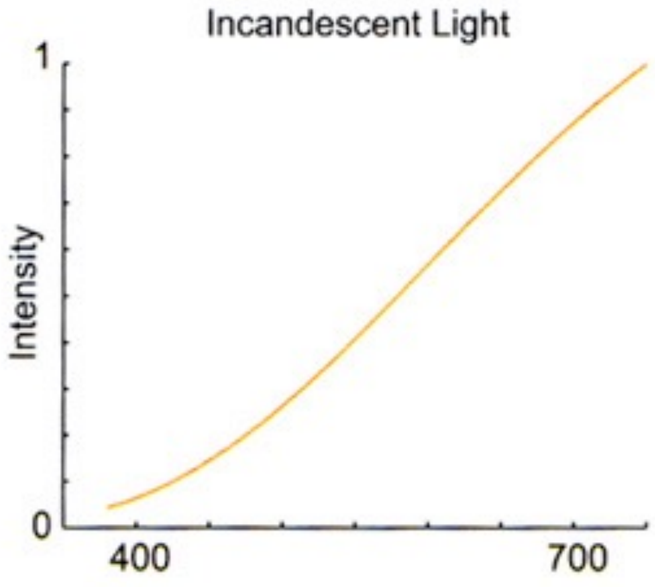


Image from David Forsyth

Reflected Light



X



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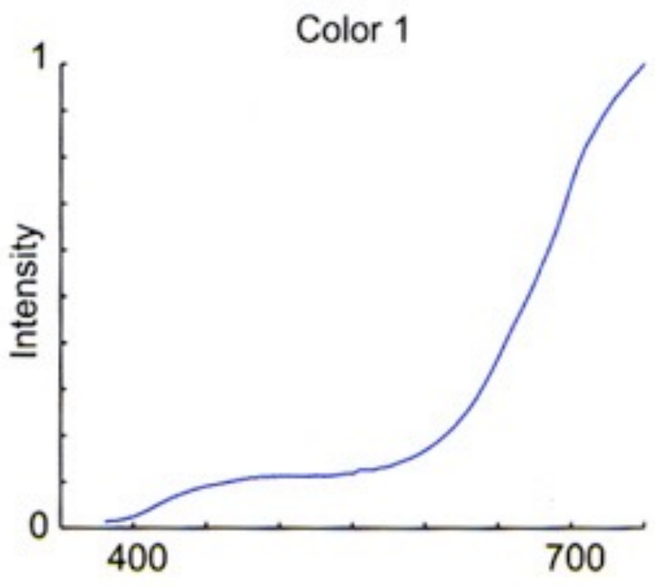
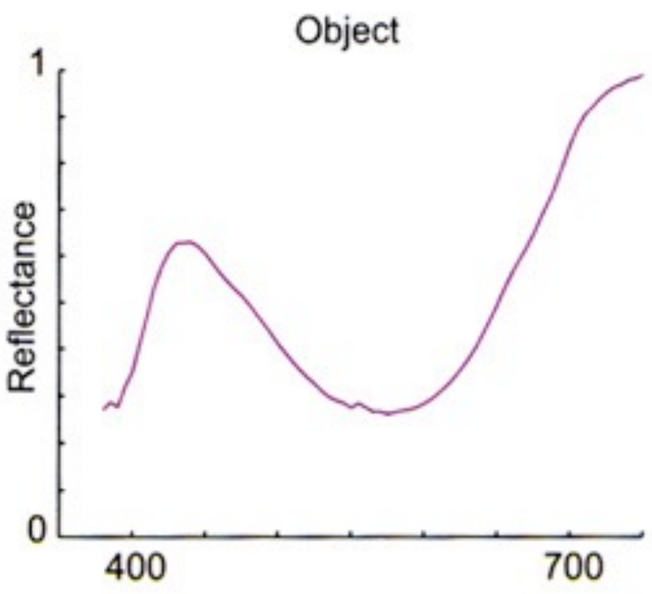
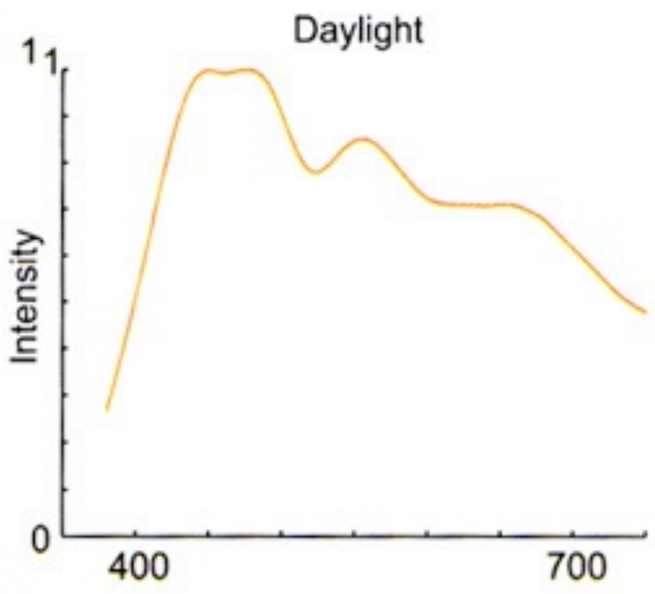


[Stone 2003]



X



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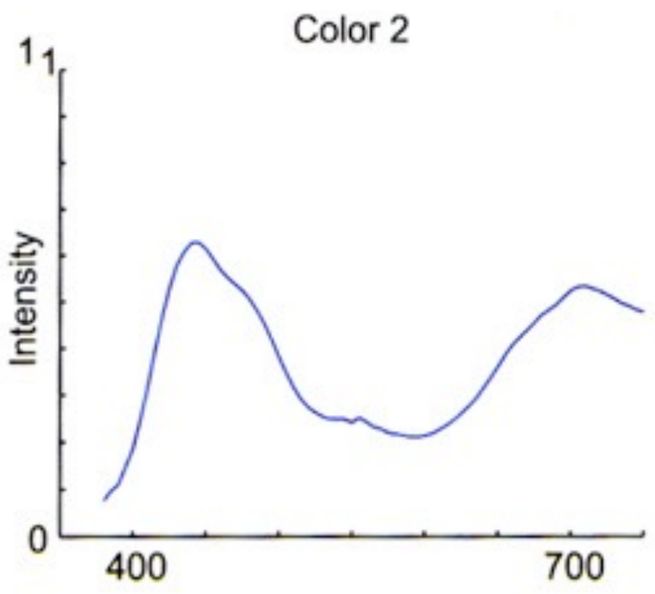
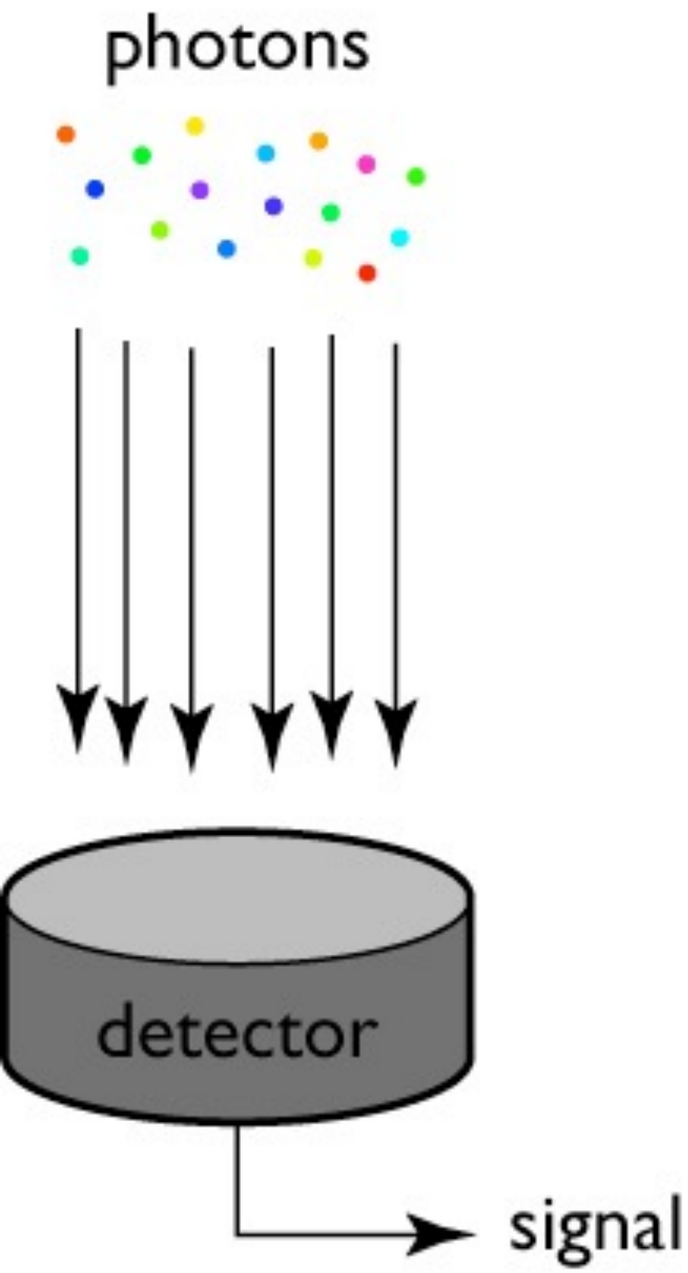
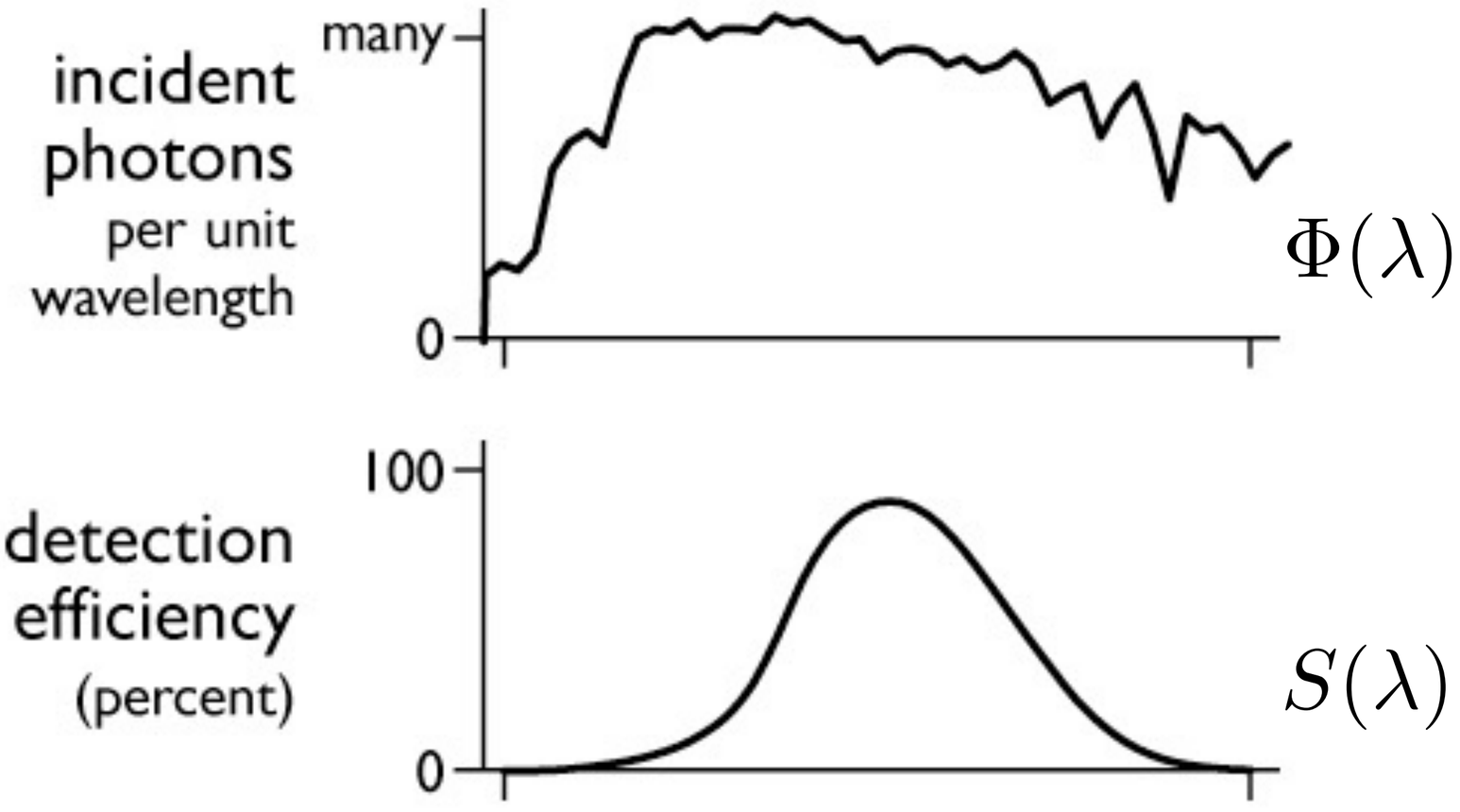


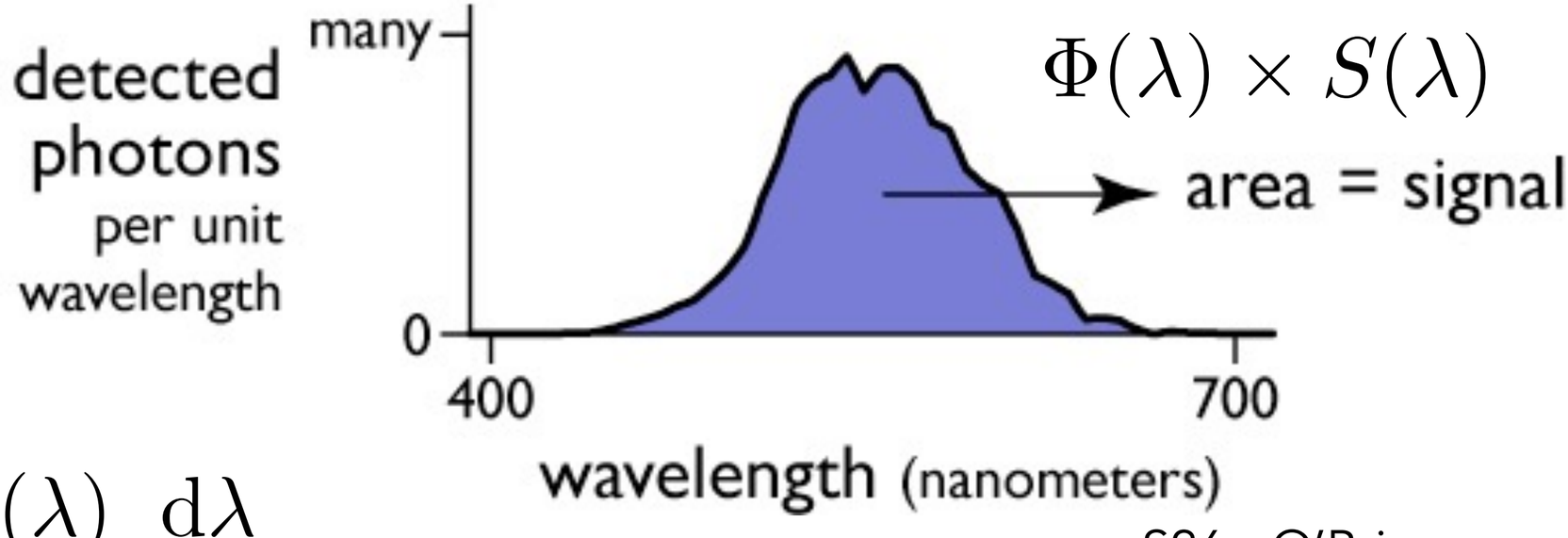
$L(\lambda)$ $R(\lambda)$ $\Phi(\lambda)$

$$\Phi(\lambda) = L(\lambda) \times R(\lambda)$$

Simple Model of a Detector







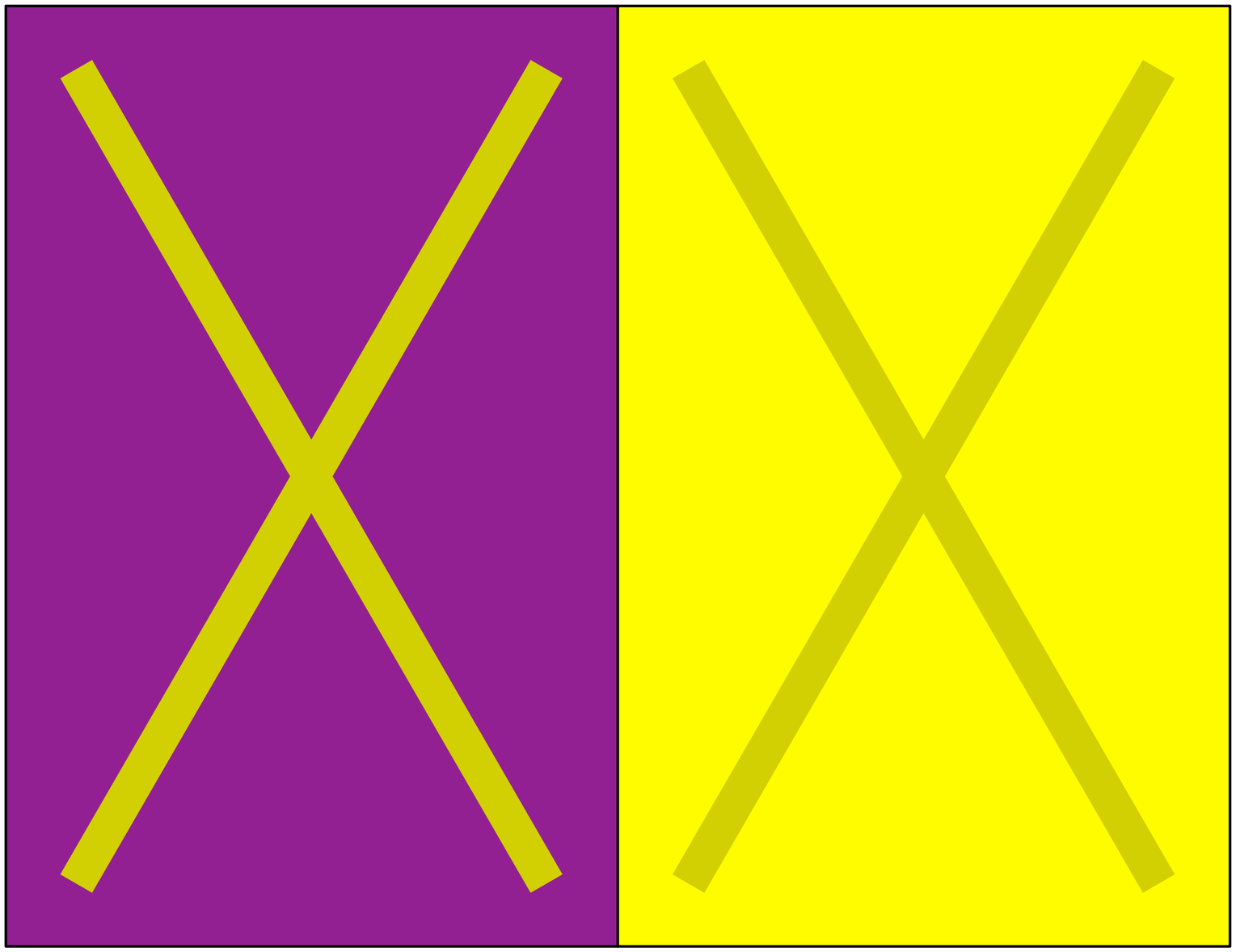
$$s = \int \Phi(\lambda) \times S(\lambda) \ \ \mathrm{d}\lambda$$

Perception -vs- Measurement

You do not “see” the spectrum of light

- Eyes make limited measurements
- Eyes physically adapt to circumstance
- You brain adapts in various ways also
- Weird psychological/psychophysical stuff also happens

Everything is Relative



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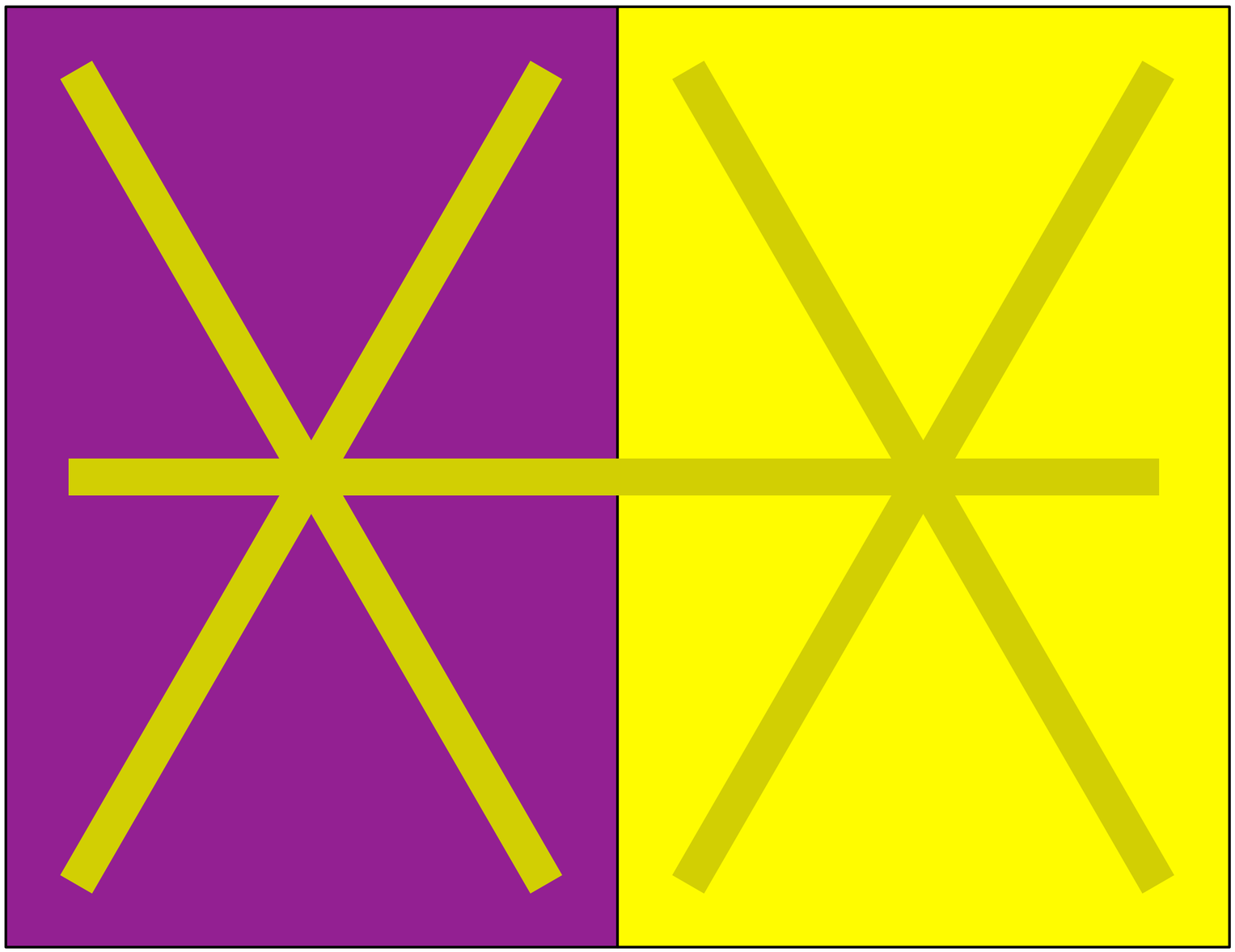
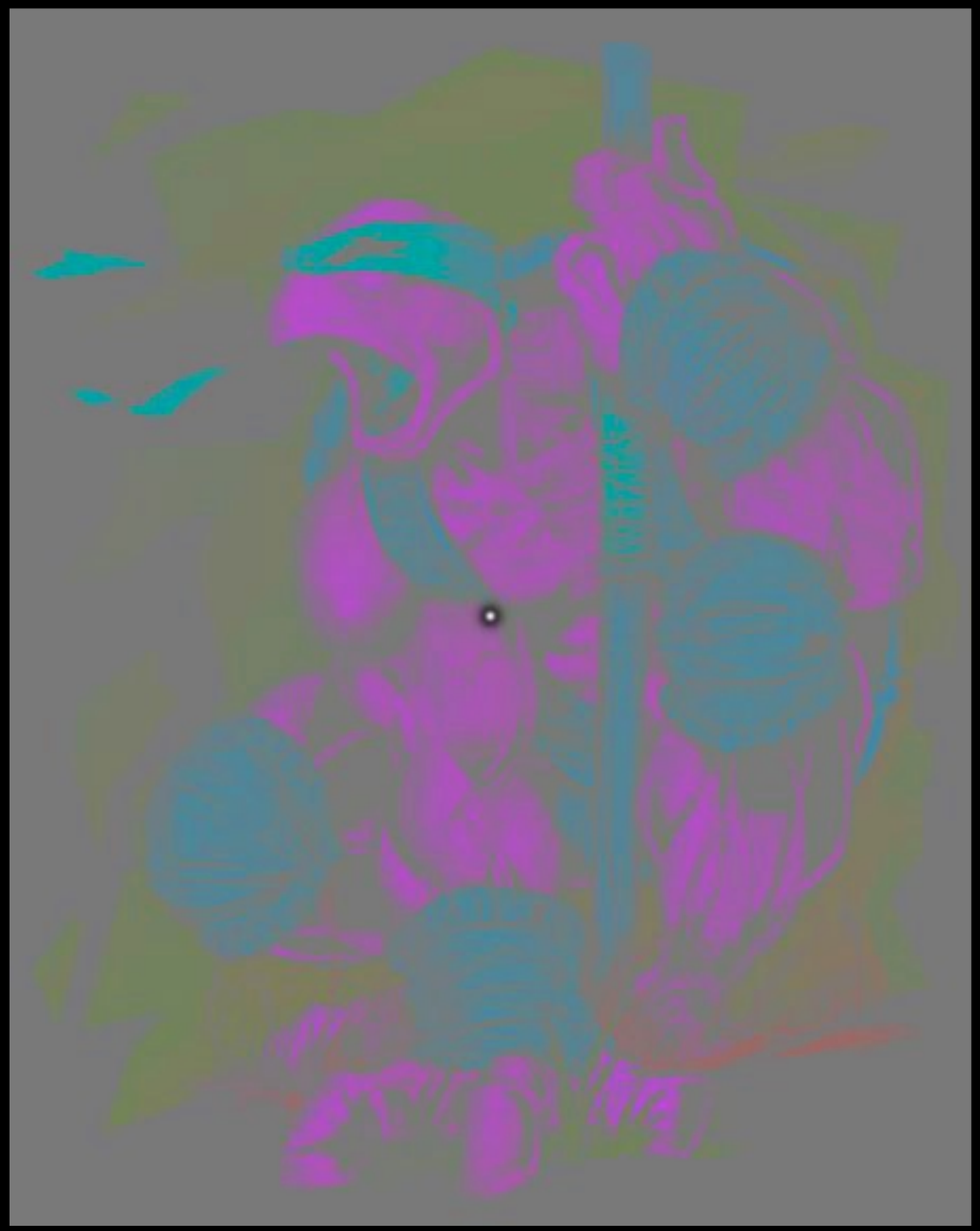
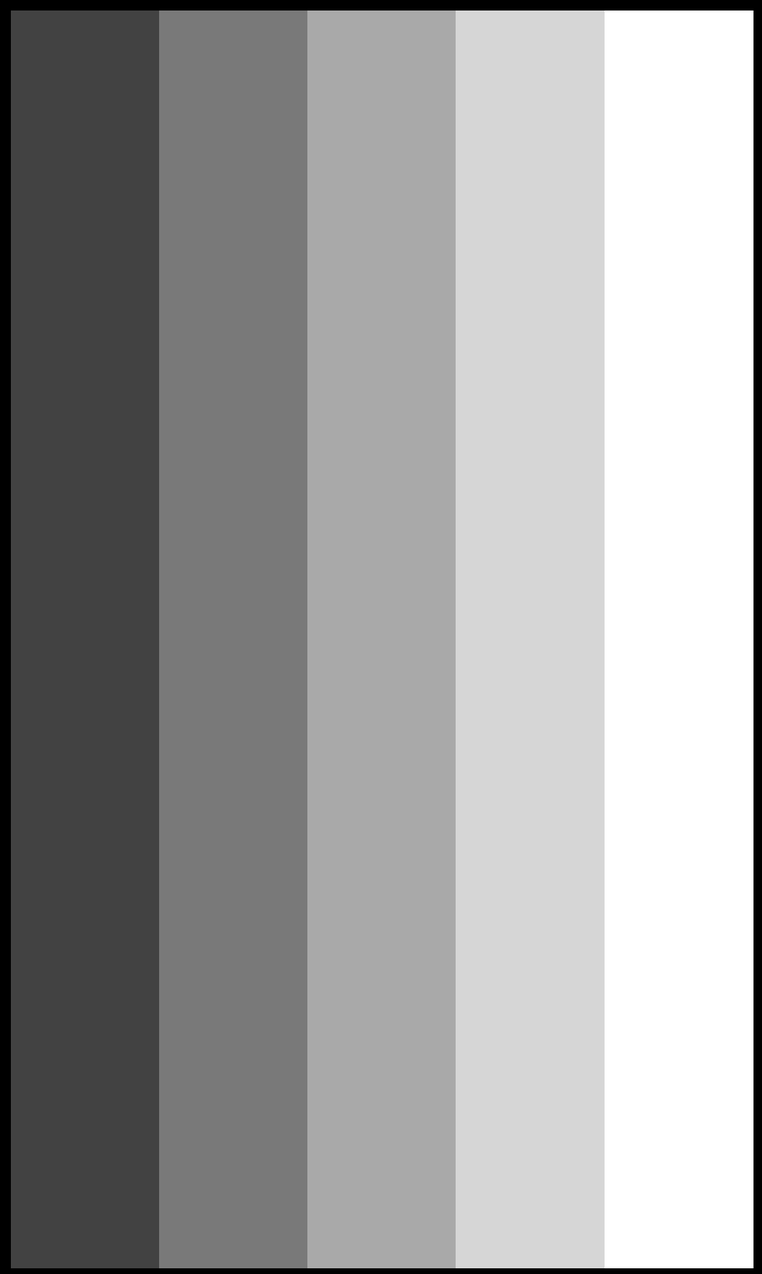


Adapt



Adapt

Mach Bands



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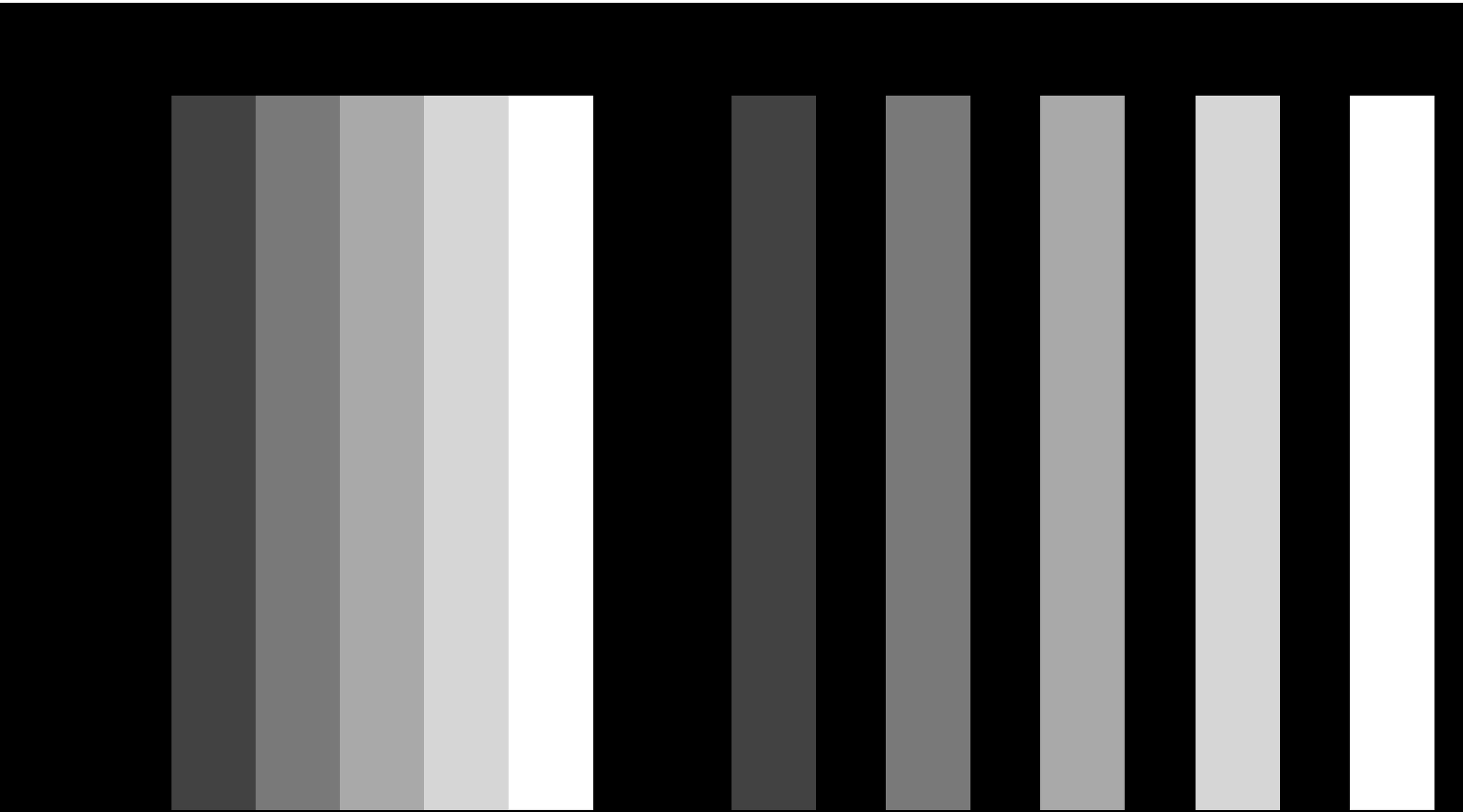
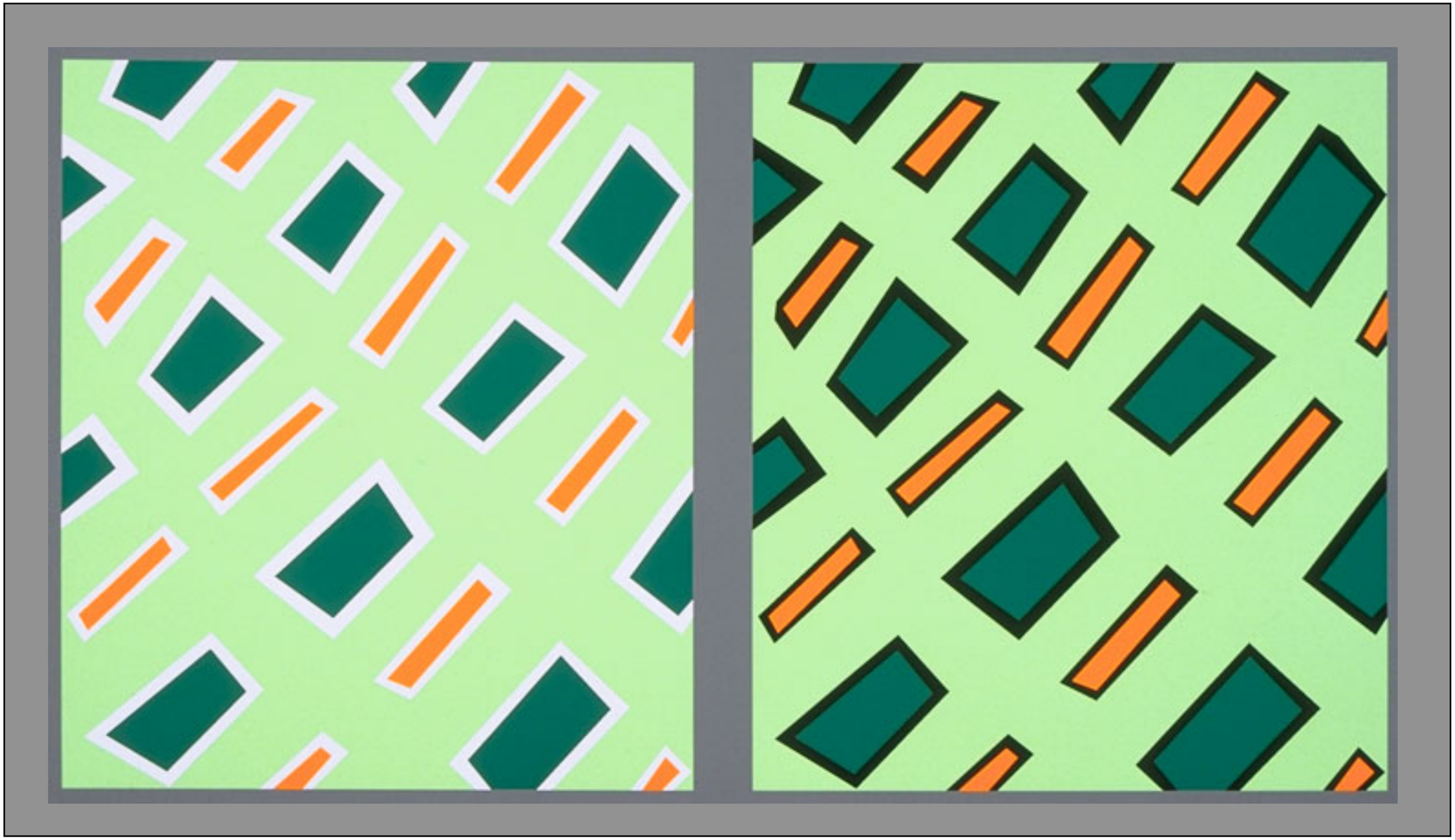


Everything's Still Relative

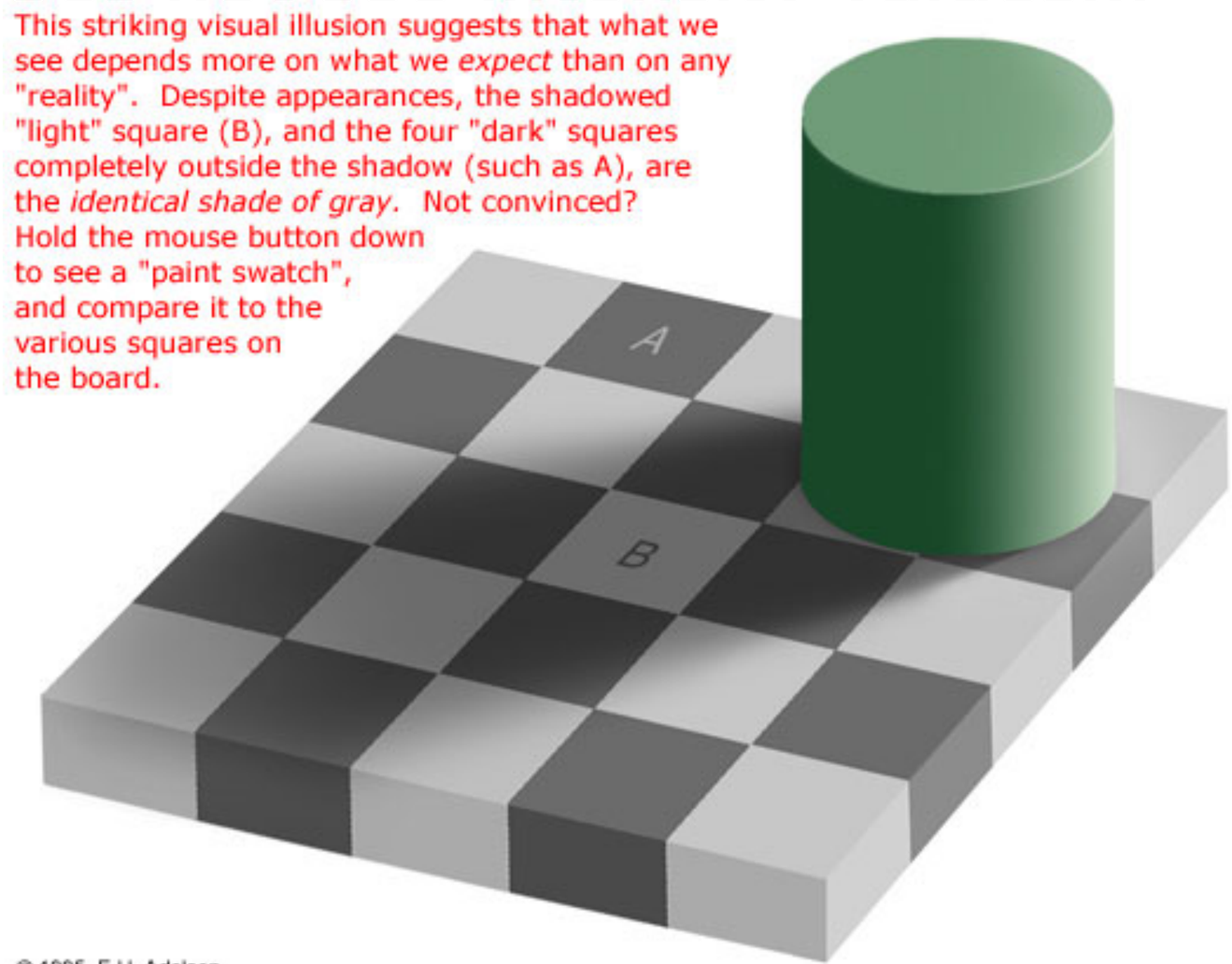
Everything's Still Relative

Bezold Effect



Perception

The eye does not see intensitv values...



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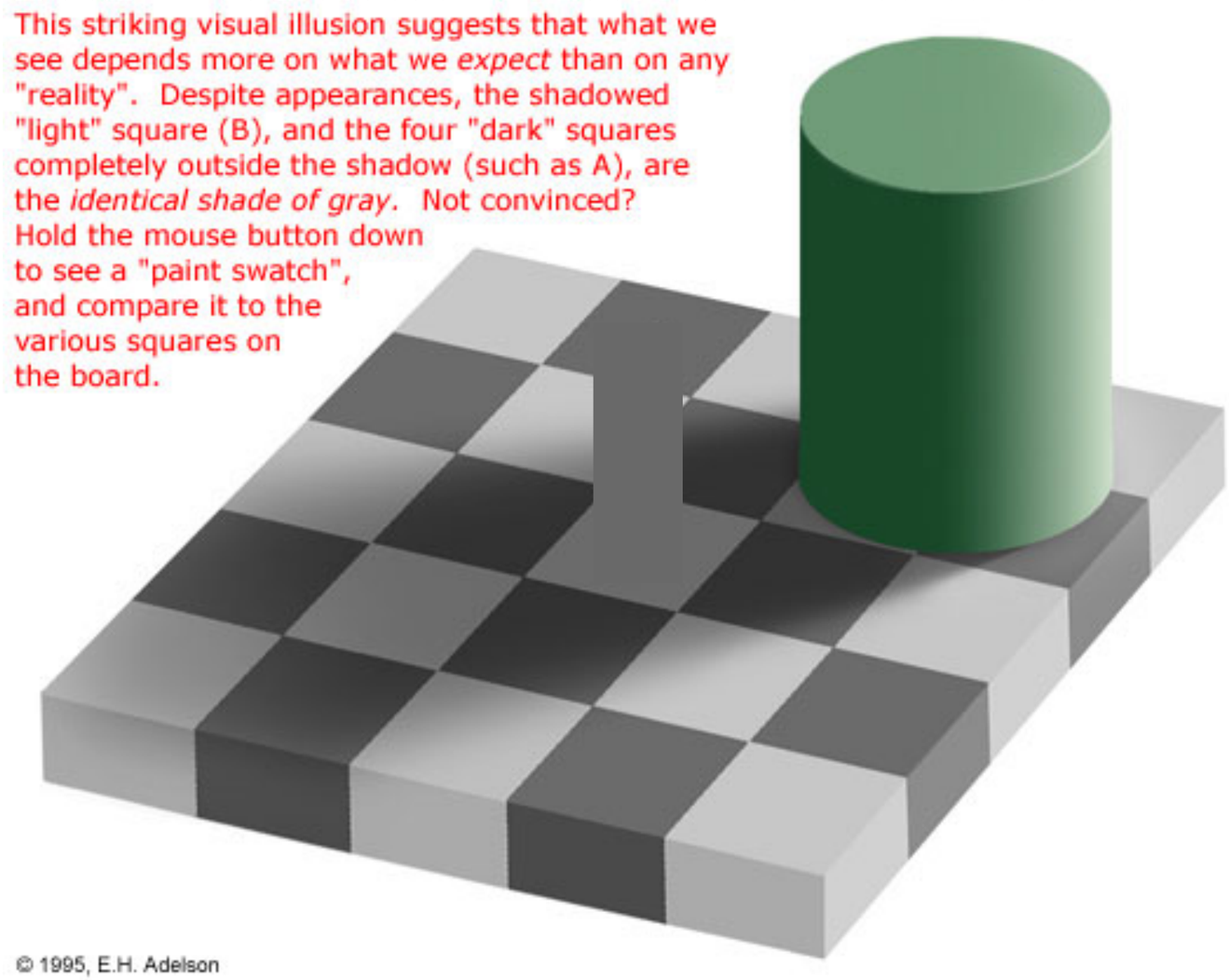

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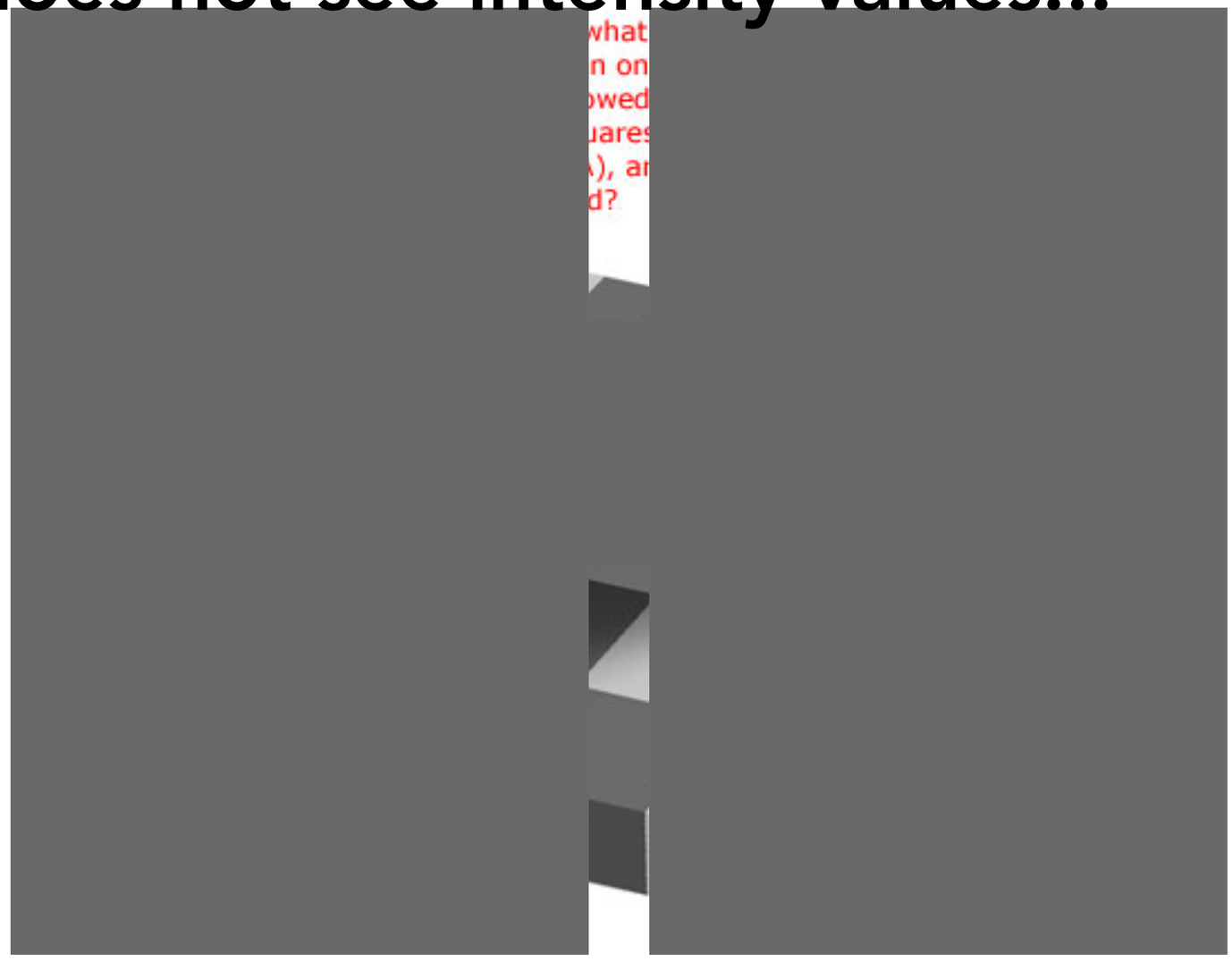

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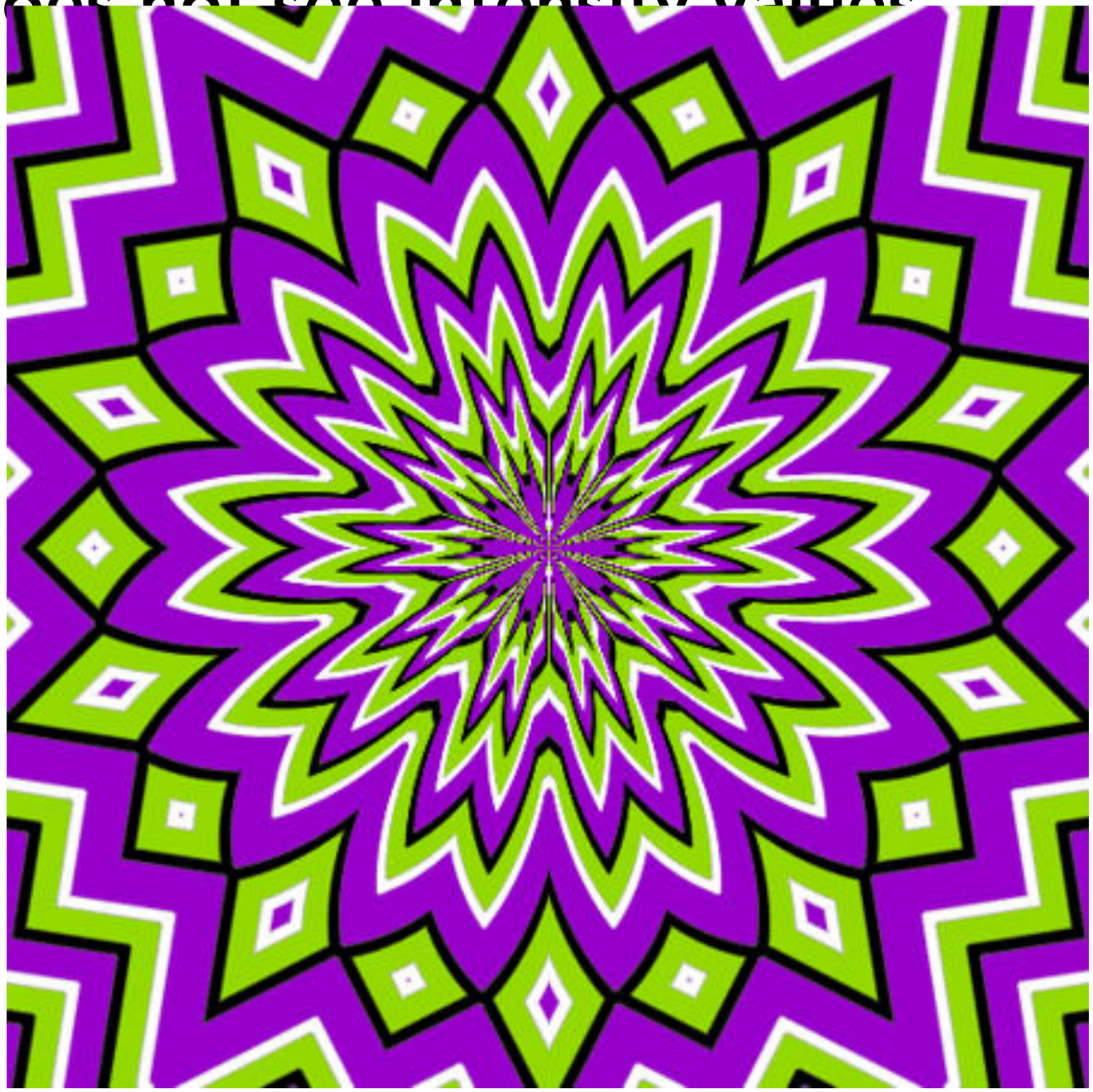
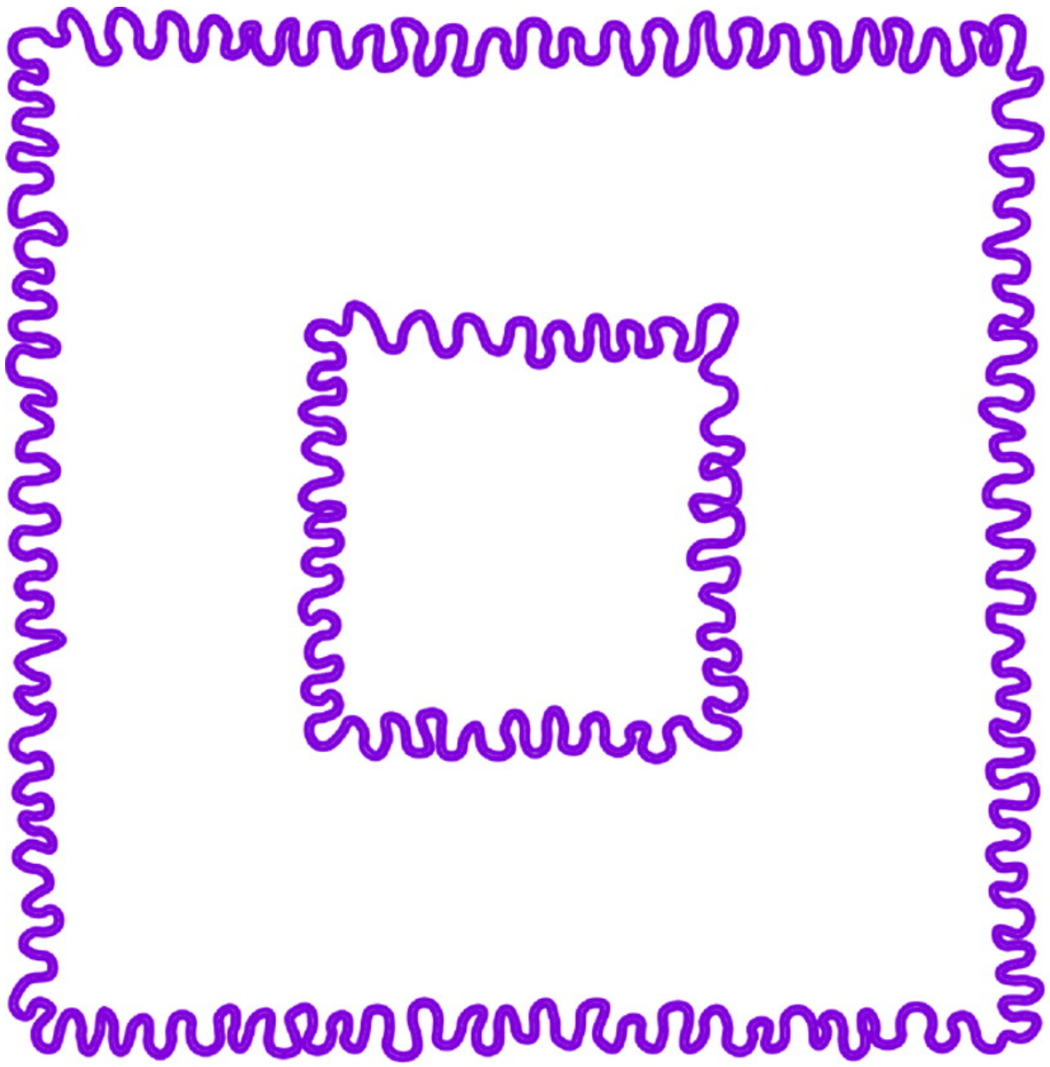
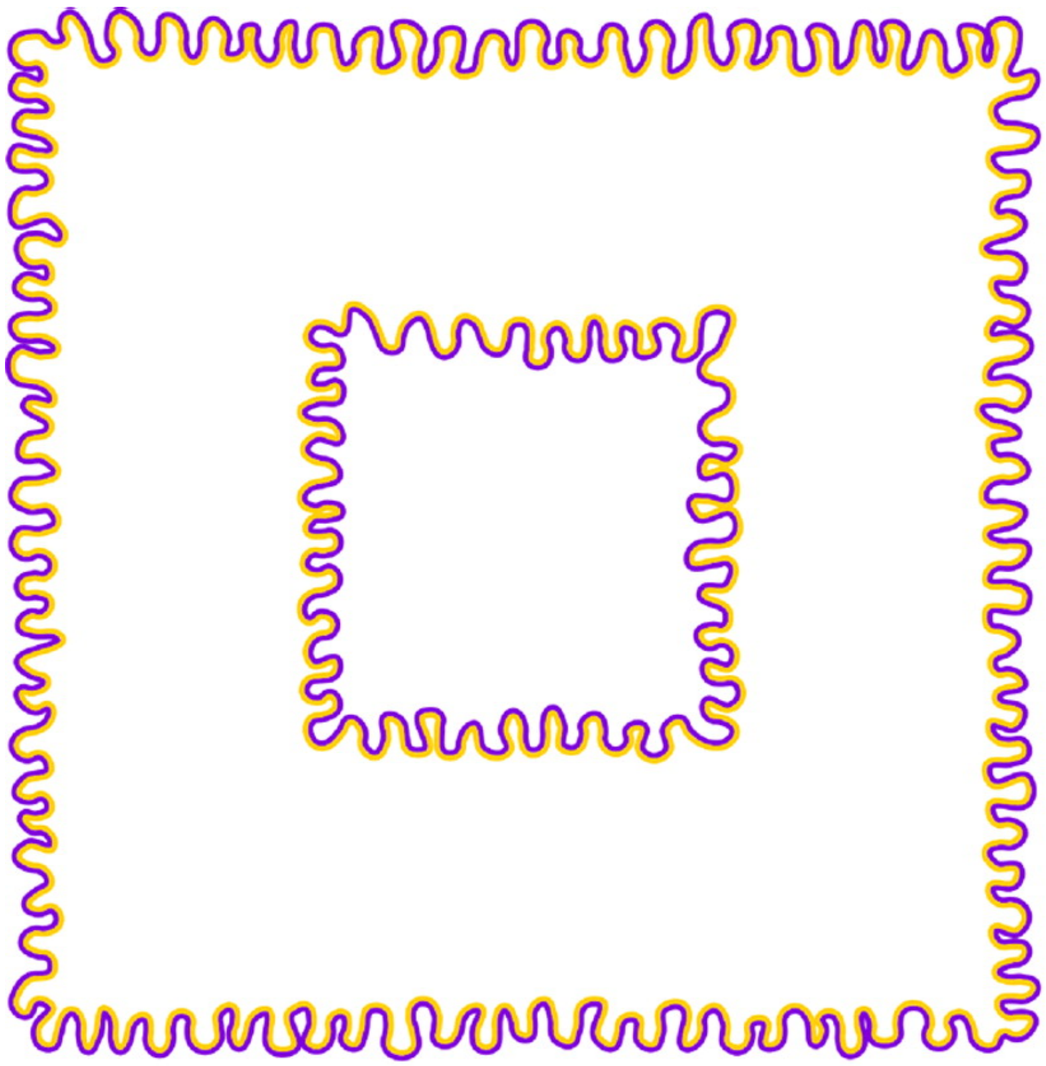


Watercolor Illusion



Pinna & Tanca, *JOV*, 2008

Watercolor Illusion



Pinna & Tanca, *JOV*, 2008

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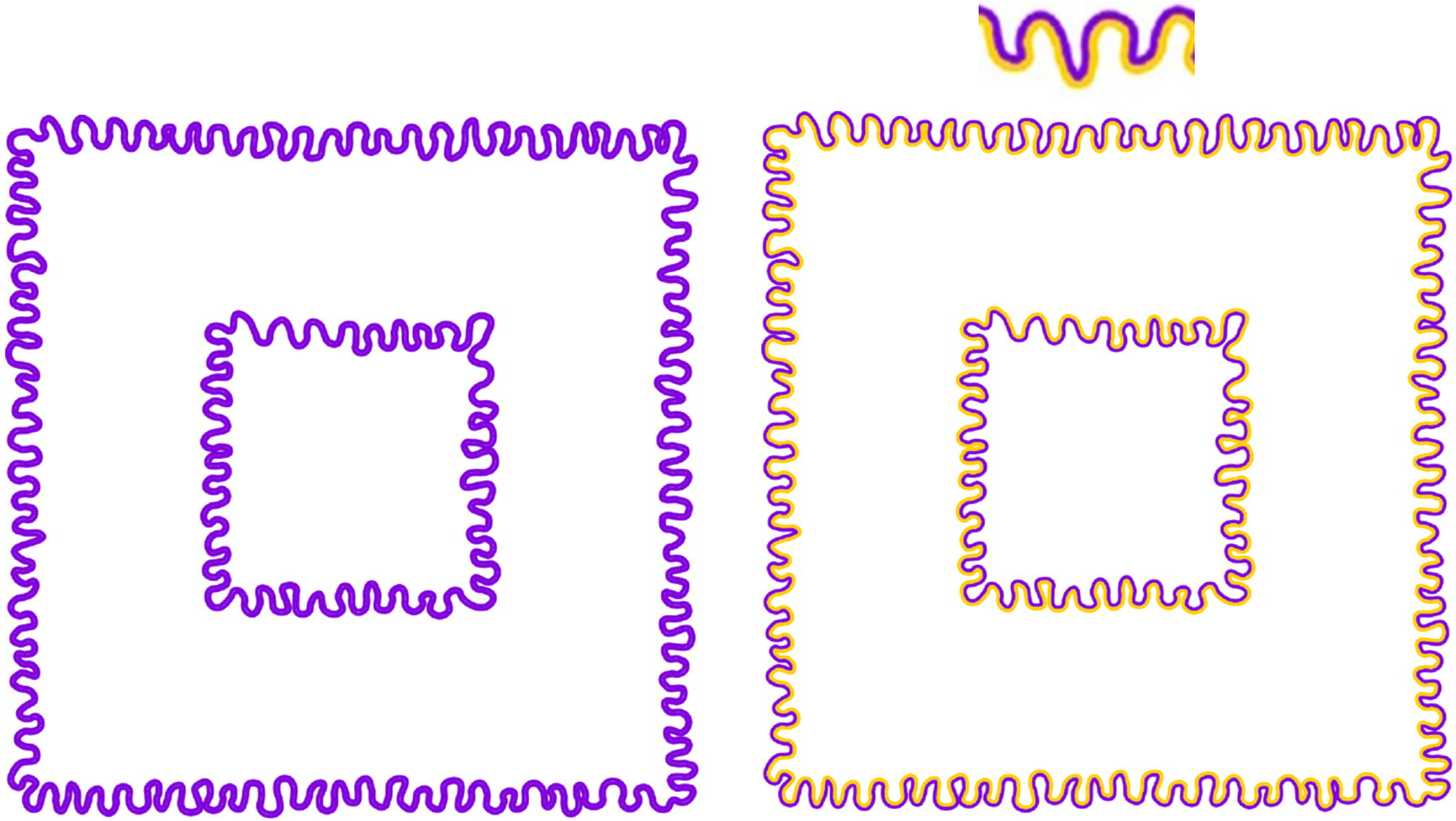


Pinna & Tanca, *JOV*, 2008

Eyes as Sensors

The human eye contains cells that sense light

- Rods
  - No color (sort of)
  - Spread over the retina
  - More sensitive
- Cones
  - Three types of cones
  - Each sensitive to different frequency distribution
  - Concentrated in fovea (center of the retina)
  - Less sensitive

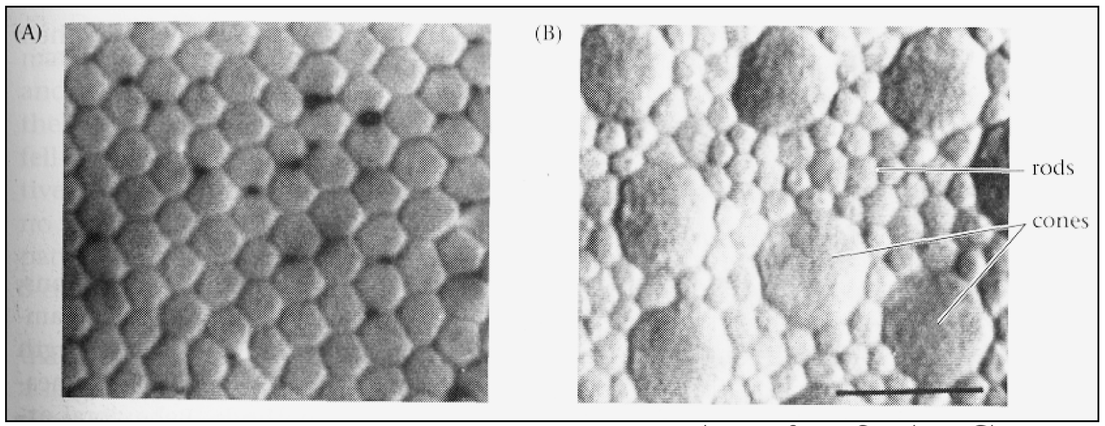
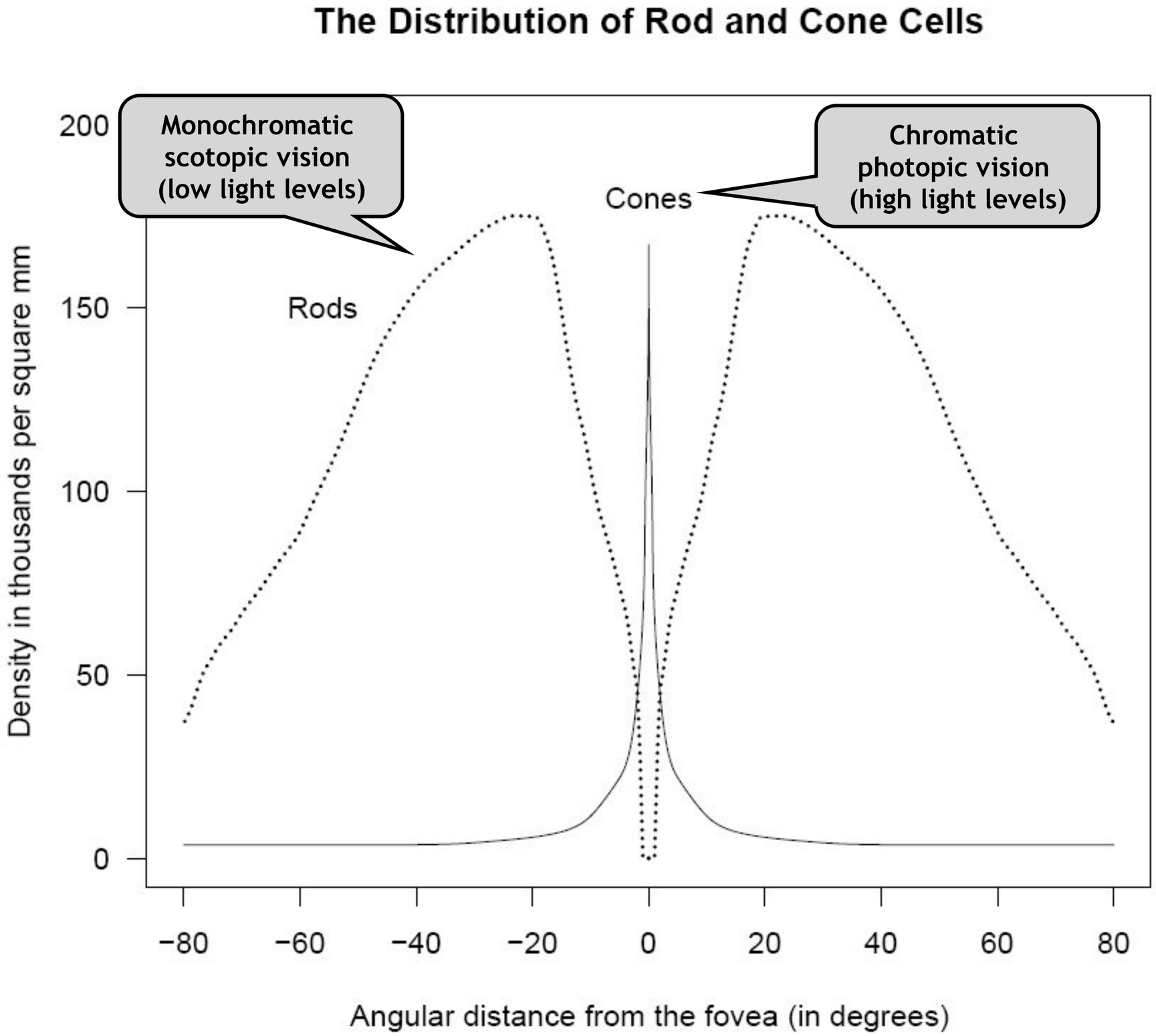
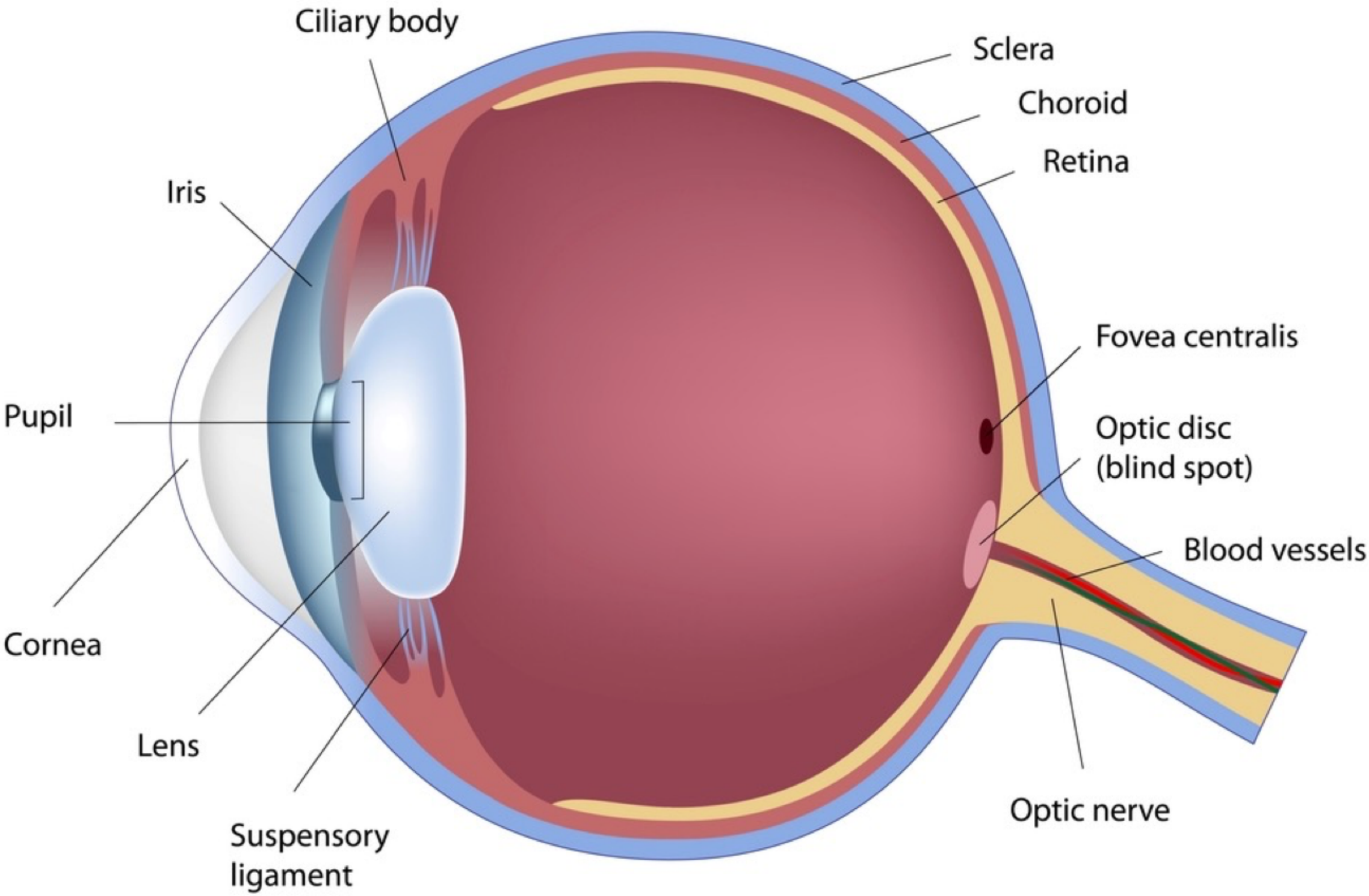


Image from Stephen Chenney

Eyes as Sensors



Eyes as Sensors



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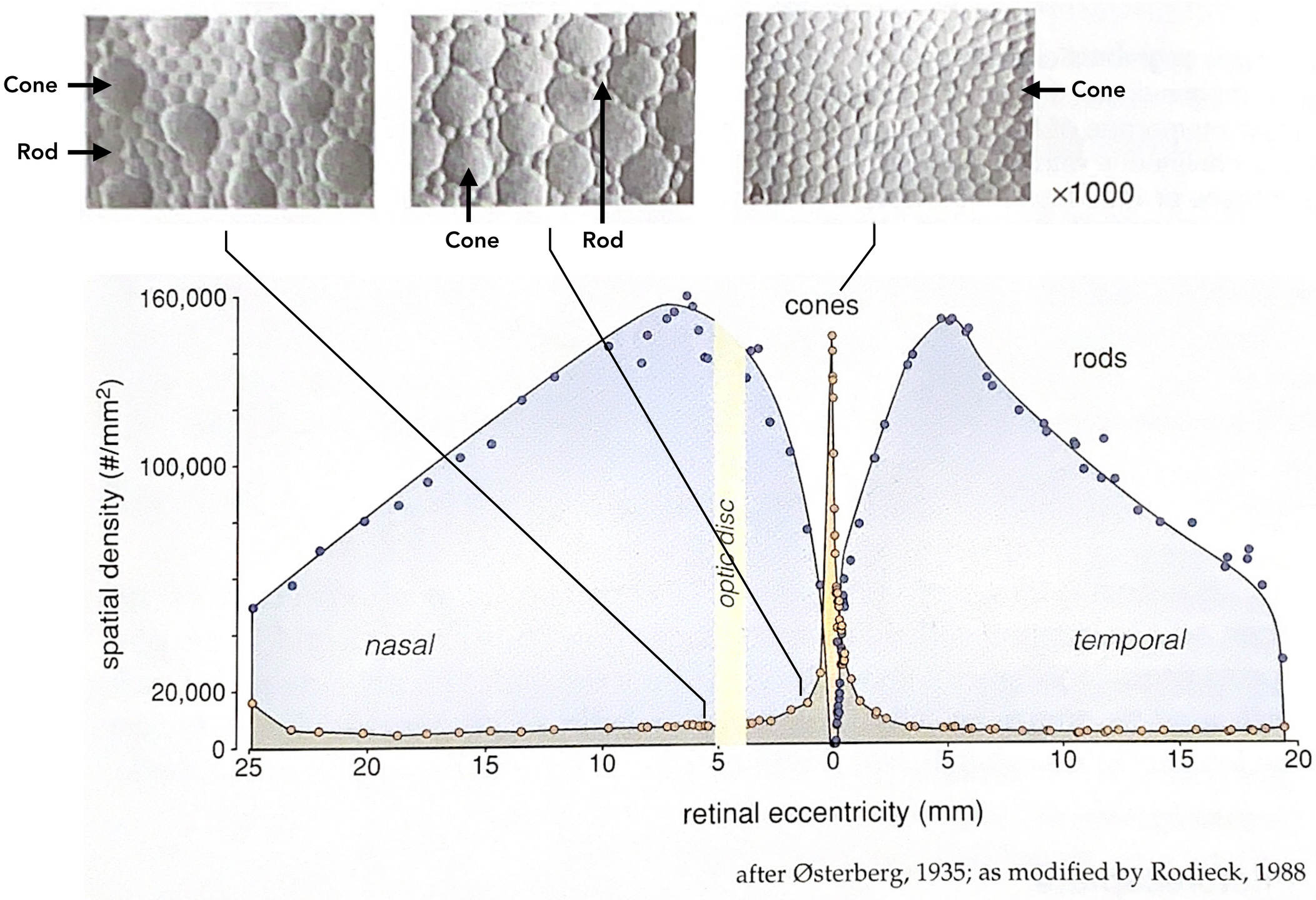
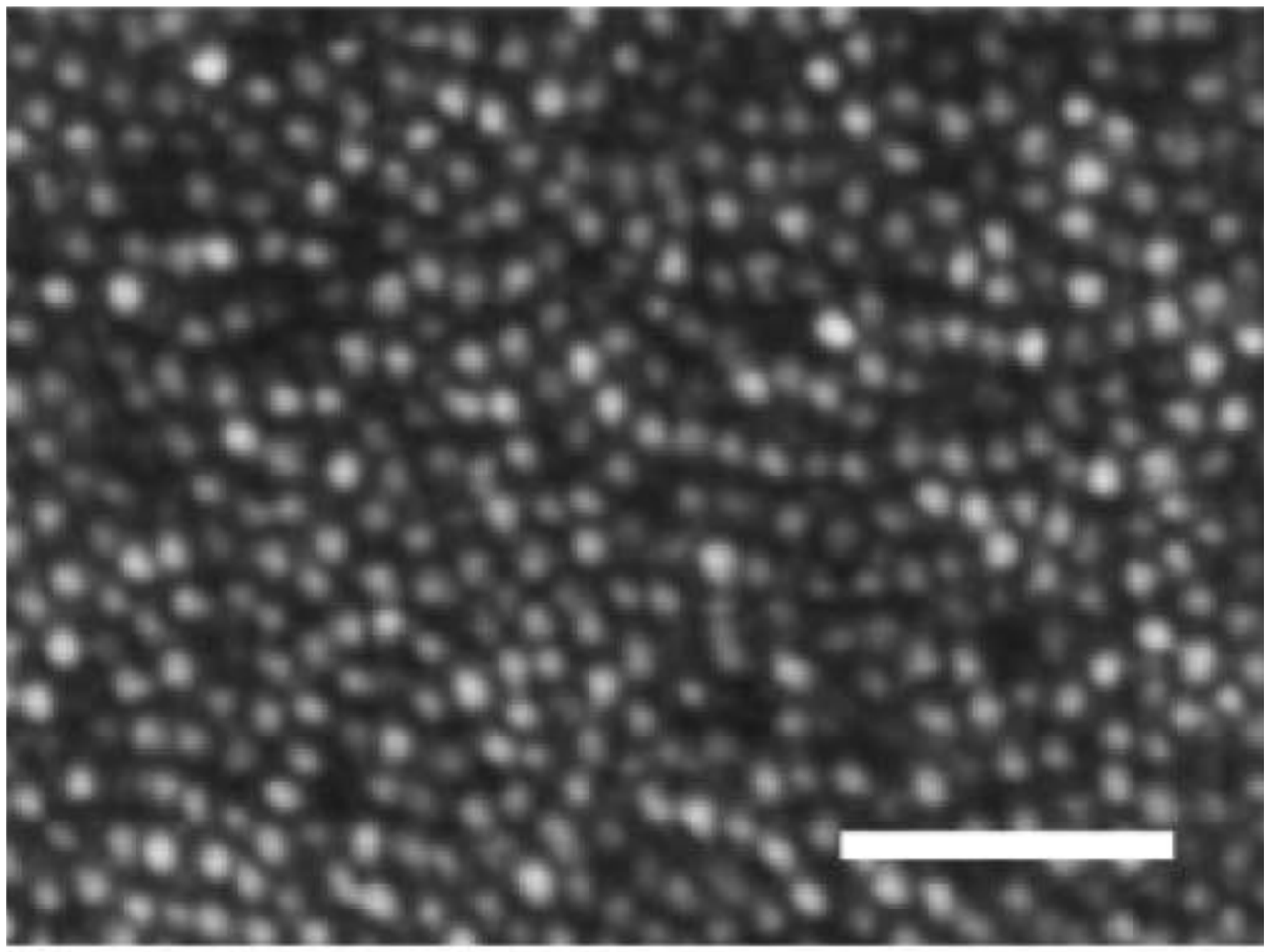
Photoreceptor Size and Distribution Vary Across Retina



after Østerberg, 1935; as modified by Rodieck, 1988

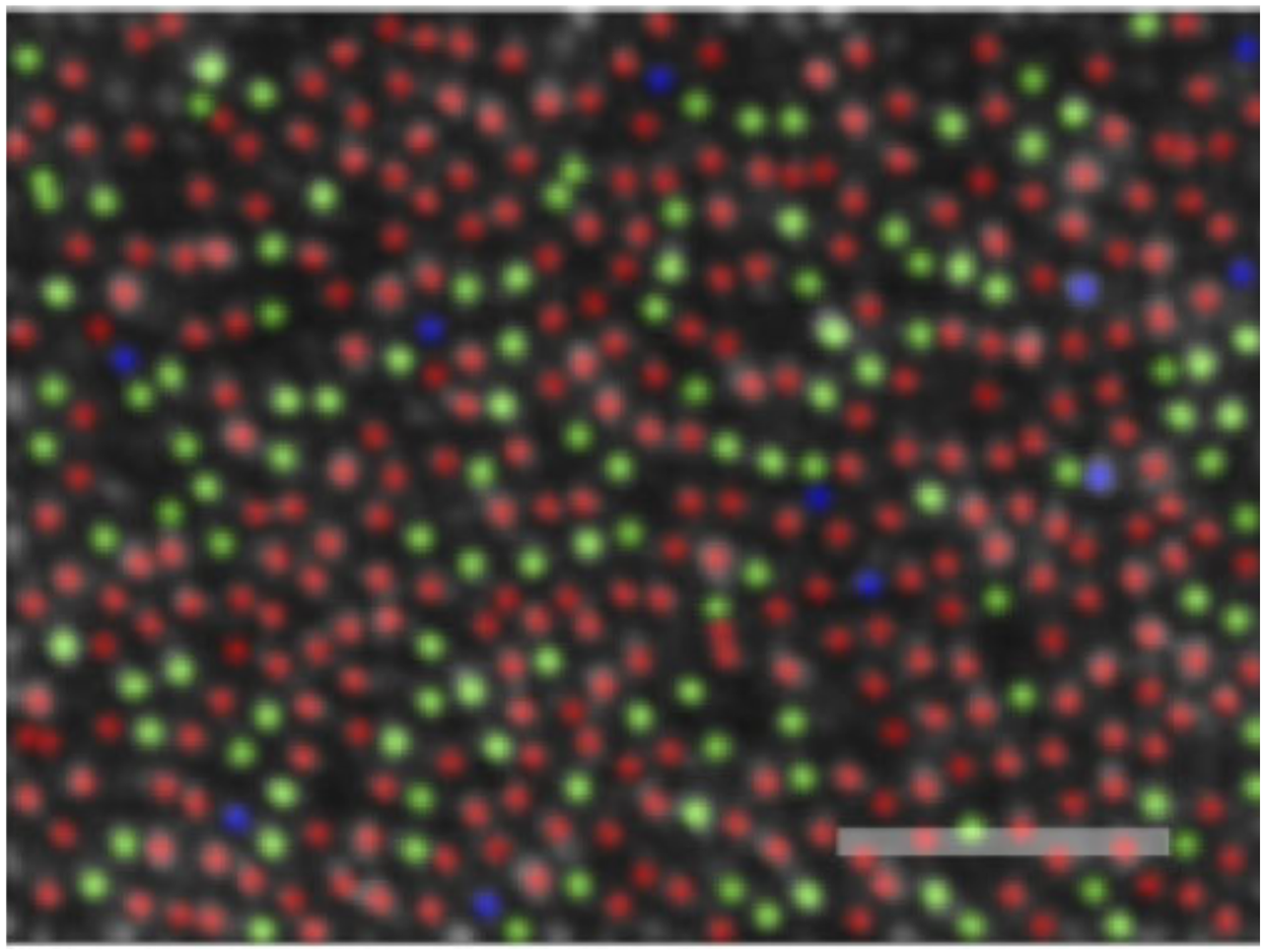
On the Retina, Three Types of Cone Cells



Sabesan Lab, UW. Pandiyan et al. 2020.

Credit: http://depts.washington.edu/sabaolab/

On the Retina, Three Types of Cone Cells



Sabesan Lab, UW. Pandiyan et al. 2020.

Credit: http://depts.washington.edu/sabaolab/

Cones

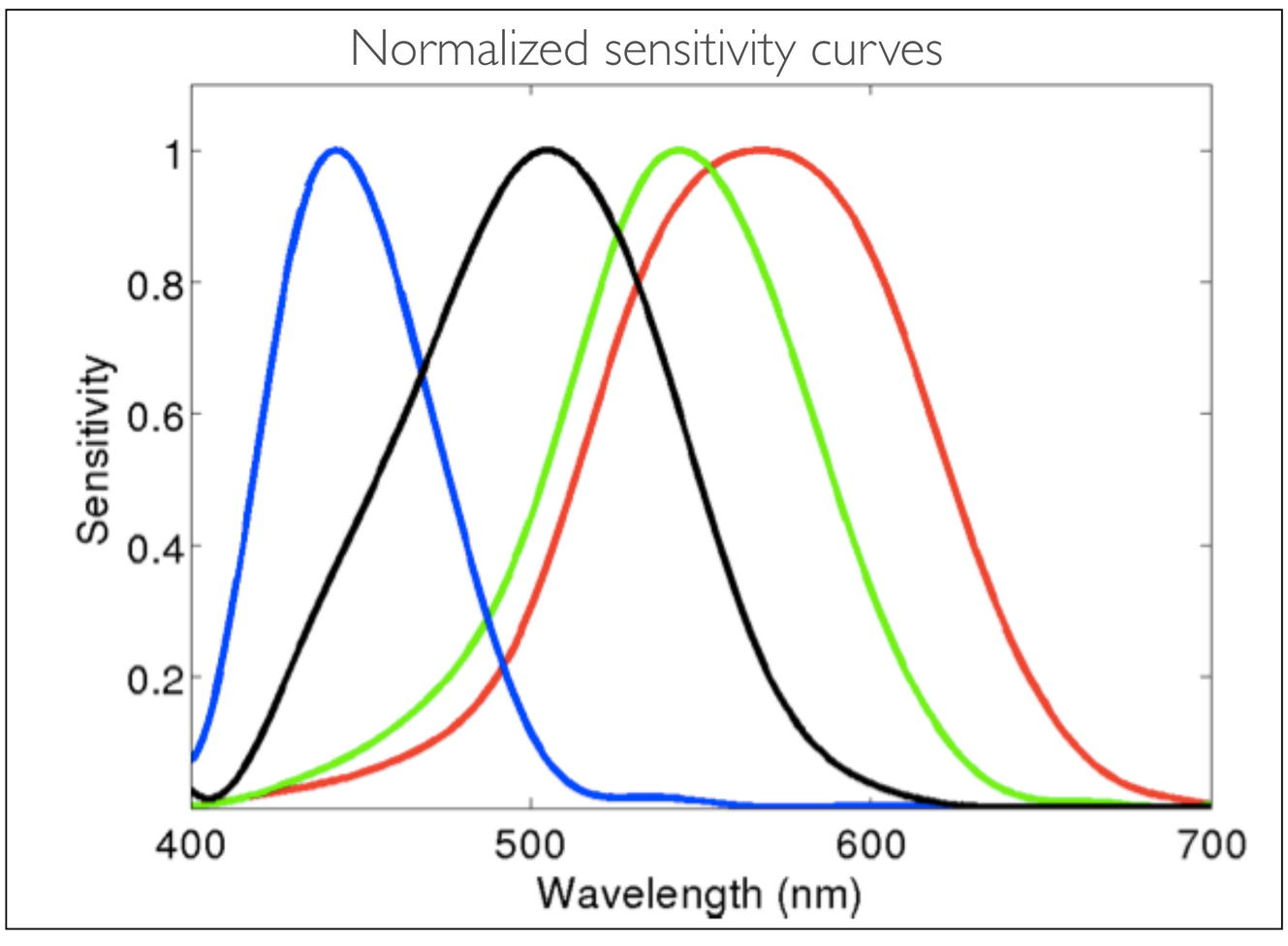
Each type of cone responds to different range of frequencies/wavelengths

- Long, medium, short

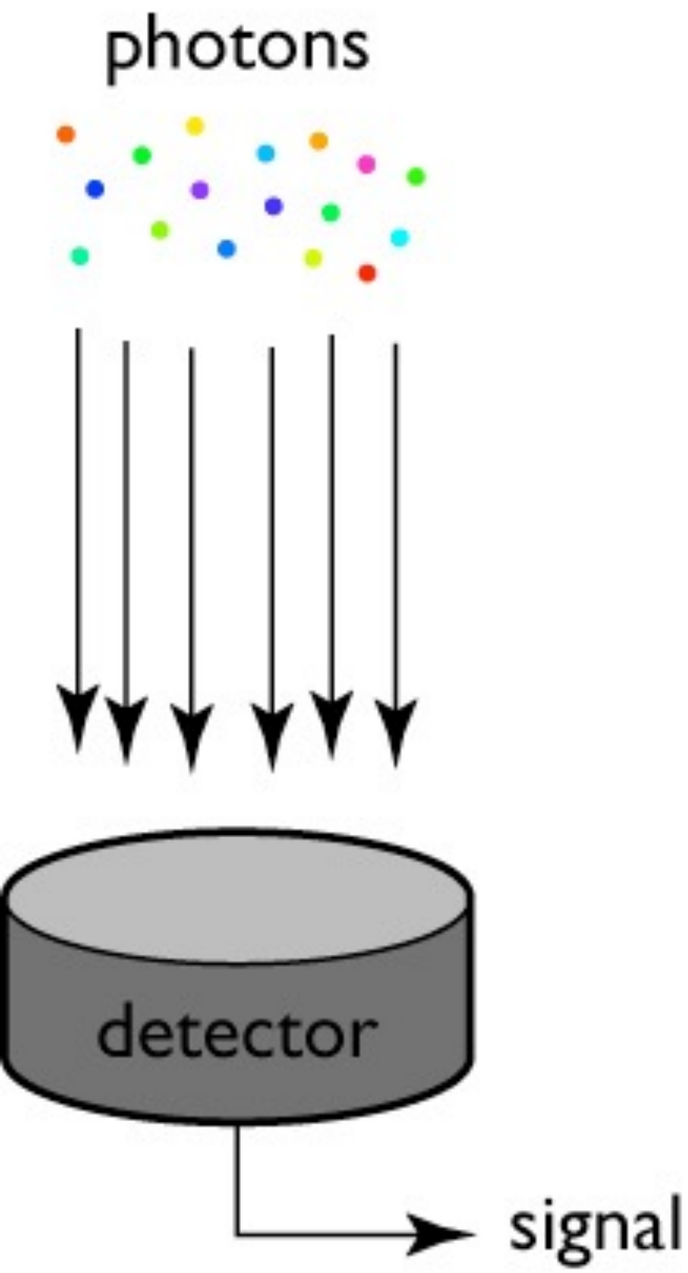
Also called by (incorrectly) color

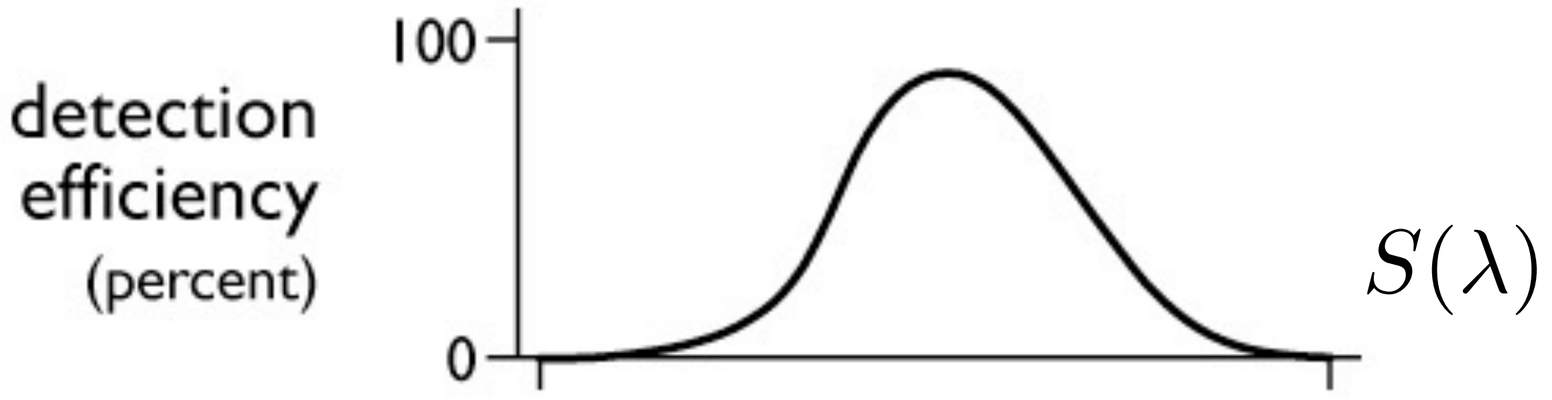
- Red, green, blue
- Misleading:

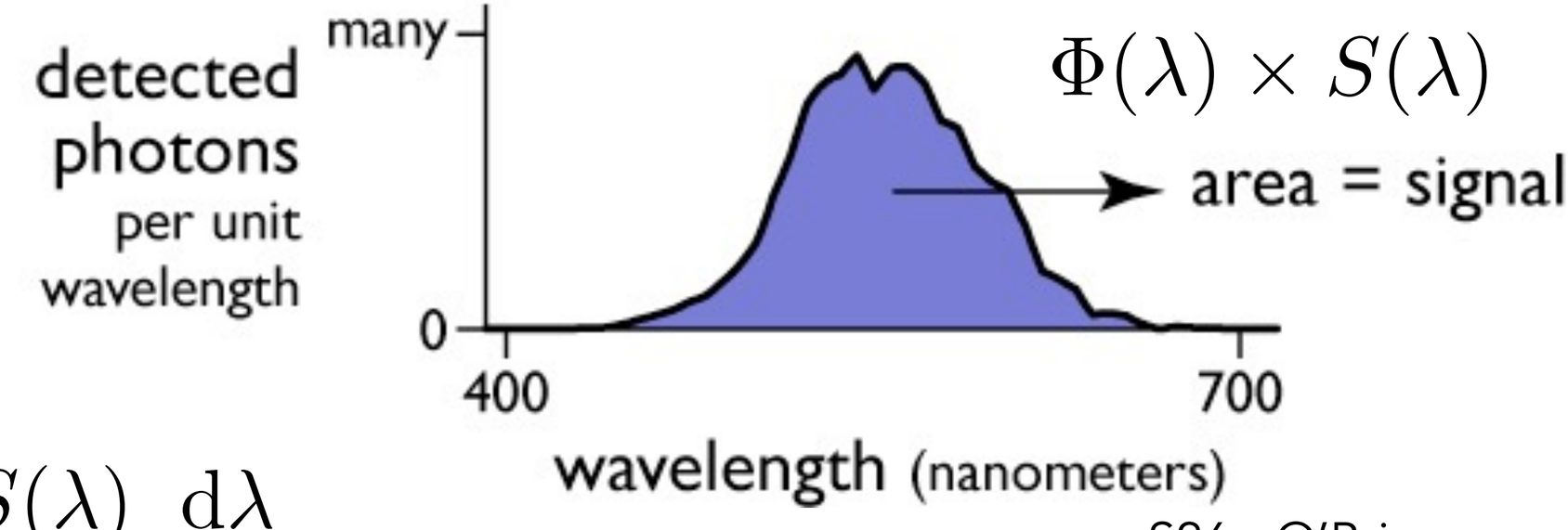
  “Red” does not mean your “red” cones are firing...



Recall







$$s = \int \Phi(\lambda) \times S(\lambda) \ \ \mathrm{d}\lambda$$

Cones

Response of a cone is given by a convolution integral :

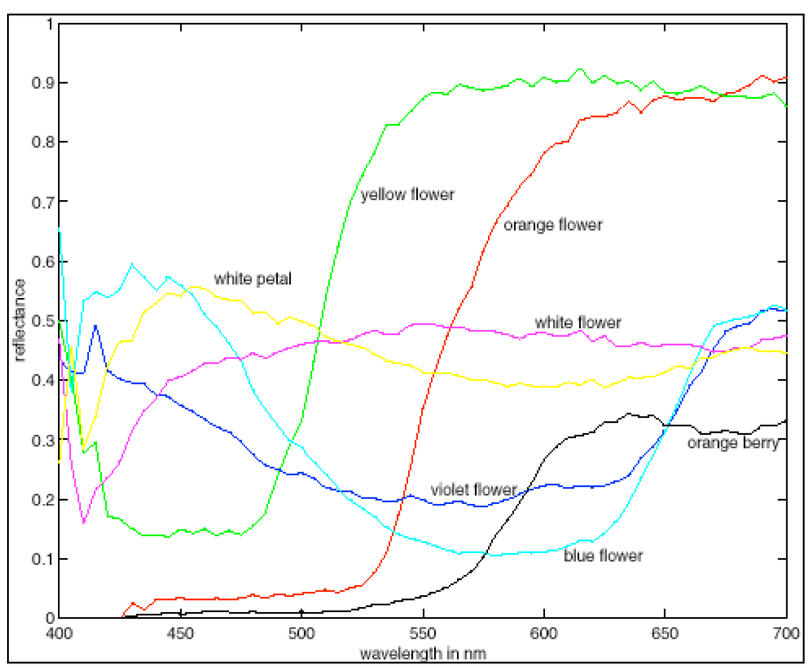
$$l = \int \Phi(\lambda) \times L(\lambda) \ \ \mathrm{d}\lambda$$

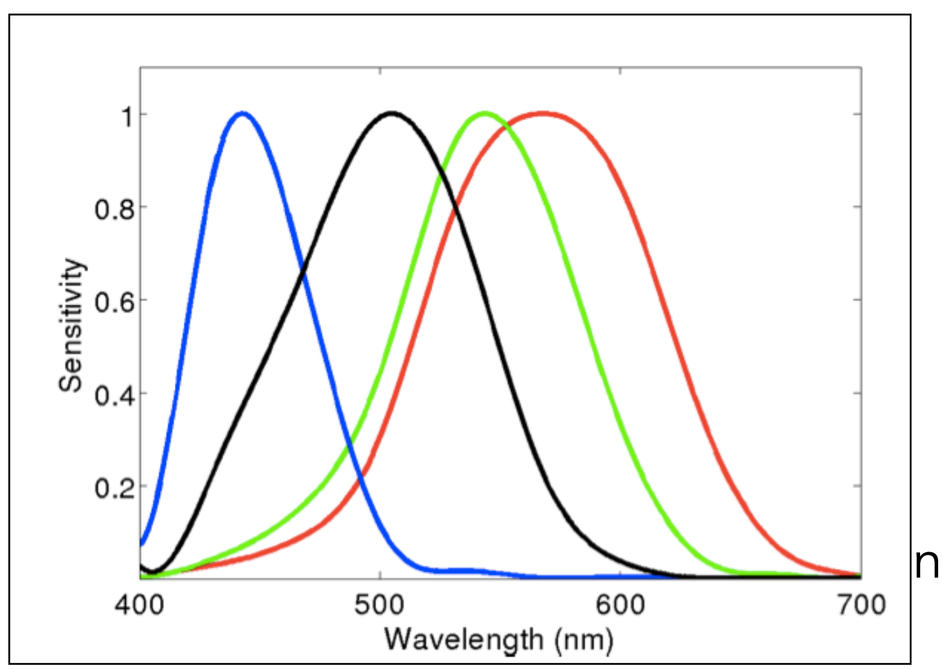
$$m = \int \Phi(\lambda) \times M(\lambda) \ \ \mathrm{d}\lambda$$

$$s = \int \Phi(\lambda) \times S(\lambda) \ \ \mathrm{d}\lambda$$

continuous version of a dot product

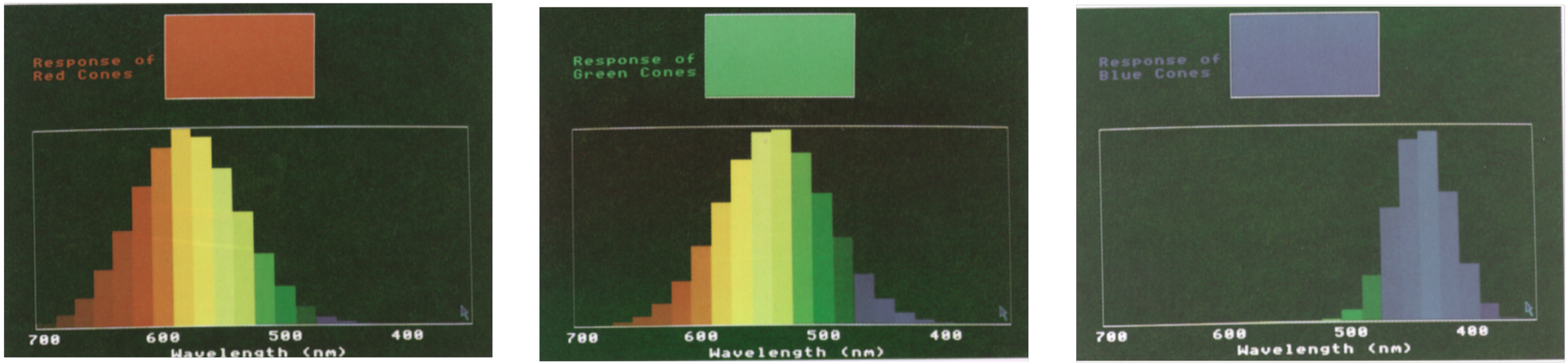
$$l = \sum \Phi_i \times L_i$$





Cones

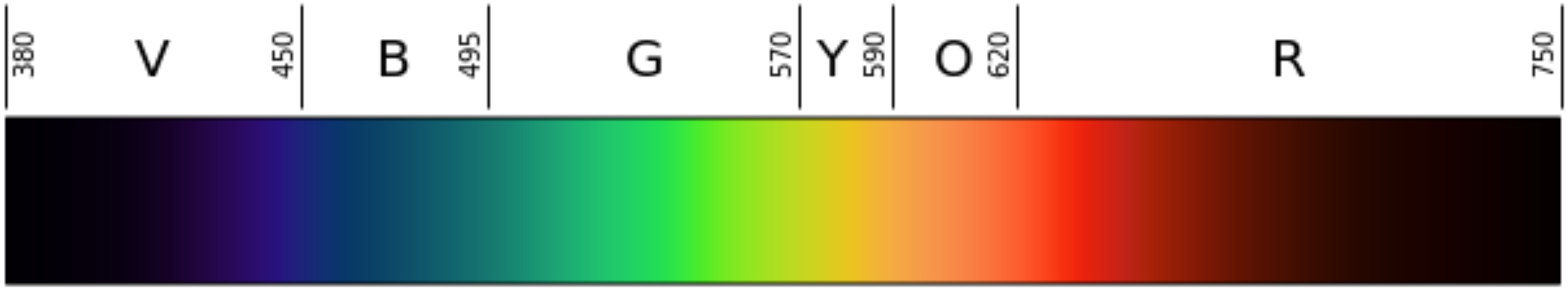
You can see that “red” and “green” respond to more more than just red and green...



Images from David Forsyth

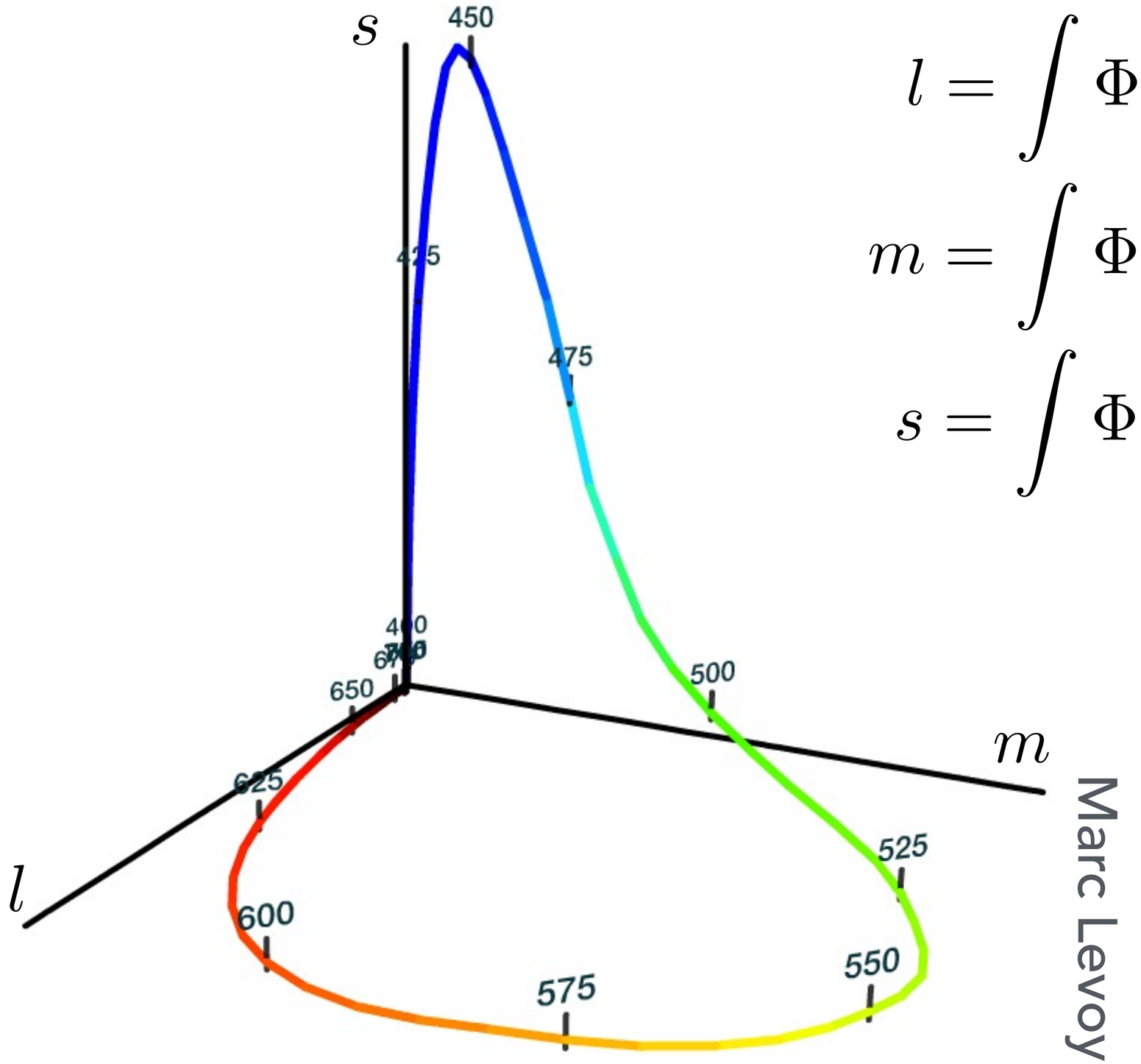
Cones

Our perception of color is not evenly spaced in wavelength.



Images from David Forsyth

Cones



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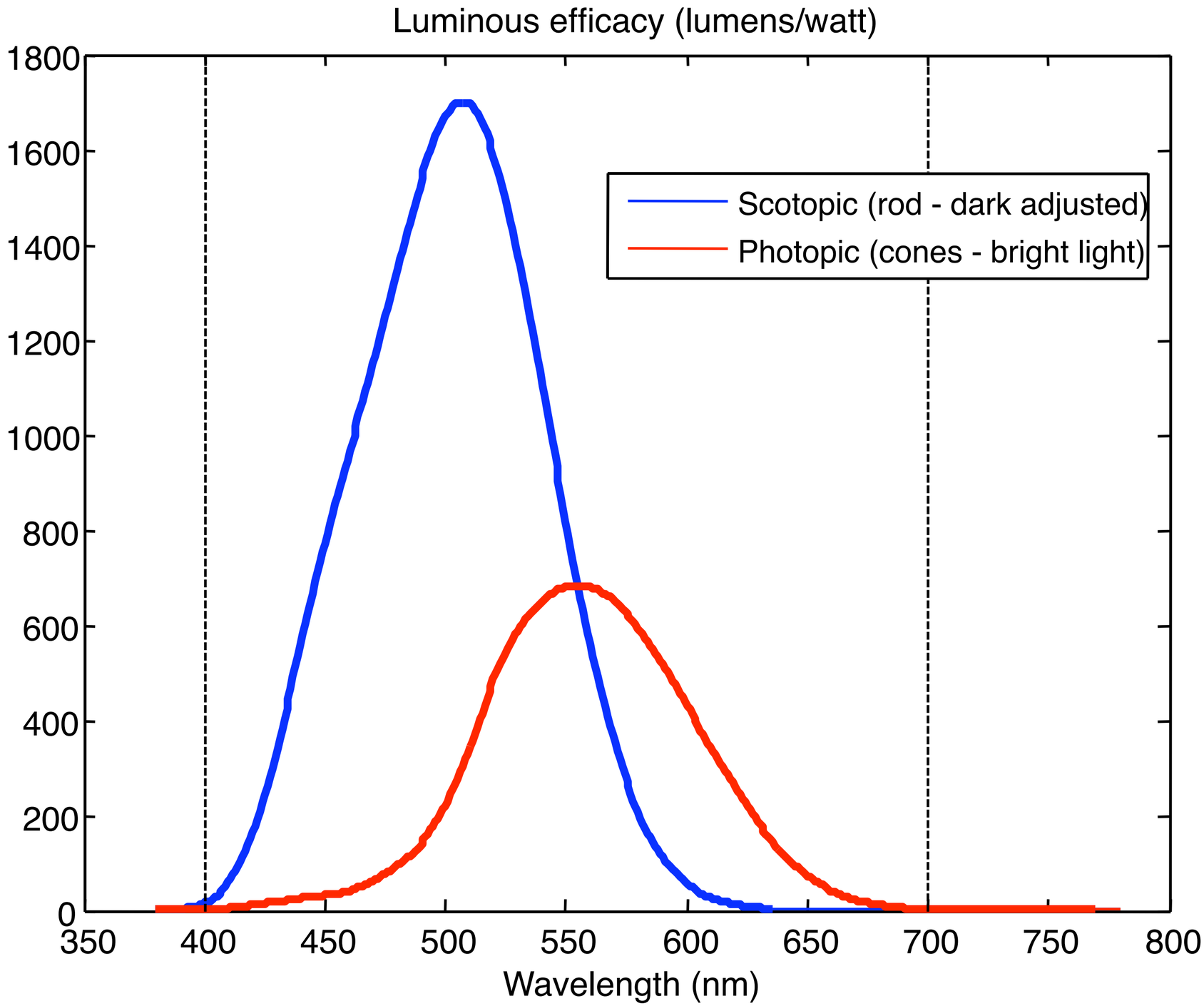
$$m = \int \Phi(\lambda) \times M(\lambda) \ \ \mathrm{d}\lambda$$

$$s = \int \Phi(\lambda) \times S(\lambda) \ \ \mathrm{d}\lambda$$

Marc Levoy

Rods vs Cones



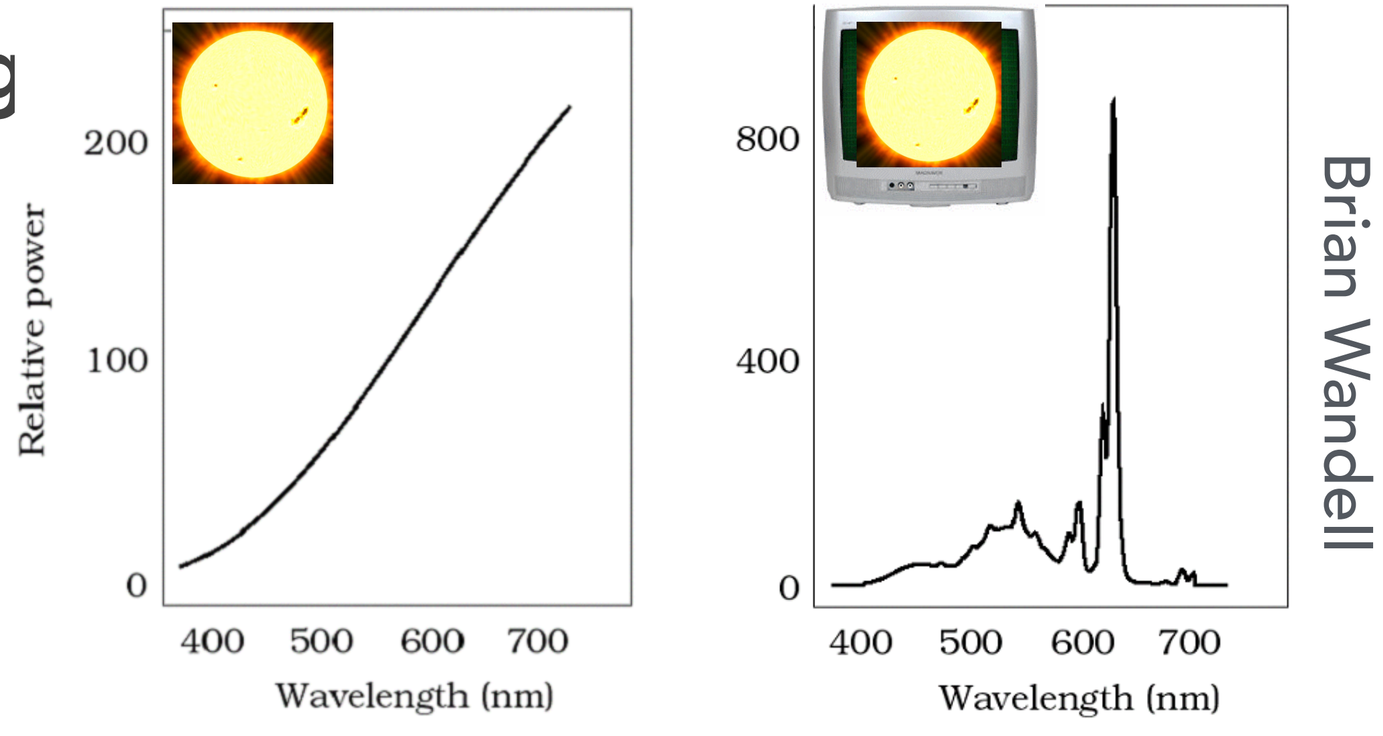
Cones and Metamers

Cone response is an integral

$$L = \int \Phi(\lambda) L(\lambda) \mathrm{d}\lambda \quad M = \int \Phi(\lambda) M(\lambda) \mathrm{d}\lambda \quad S = \int \Phi(\lambda) S(\lambda) \mathrm{d}\lambda$$

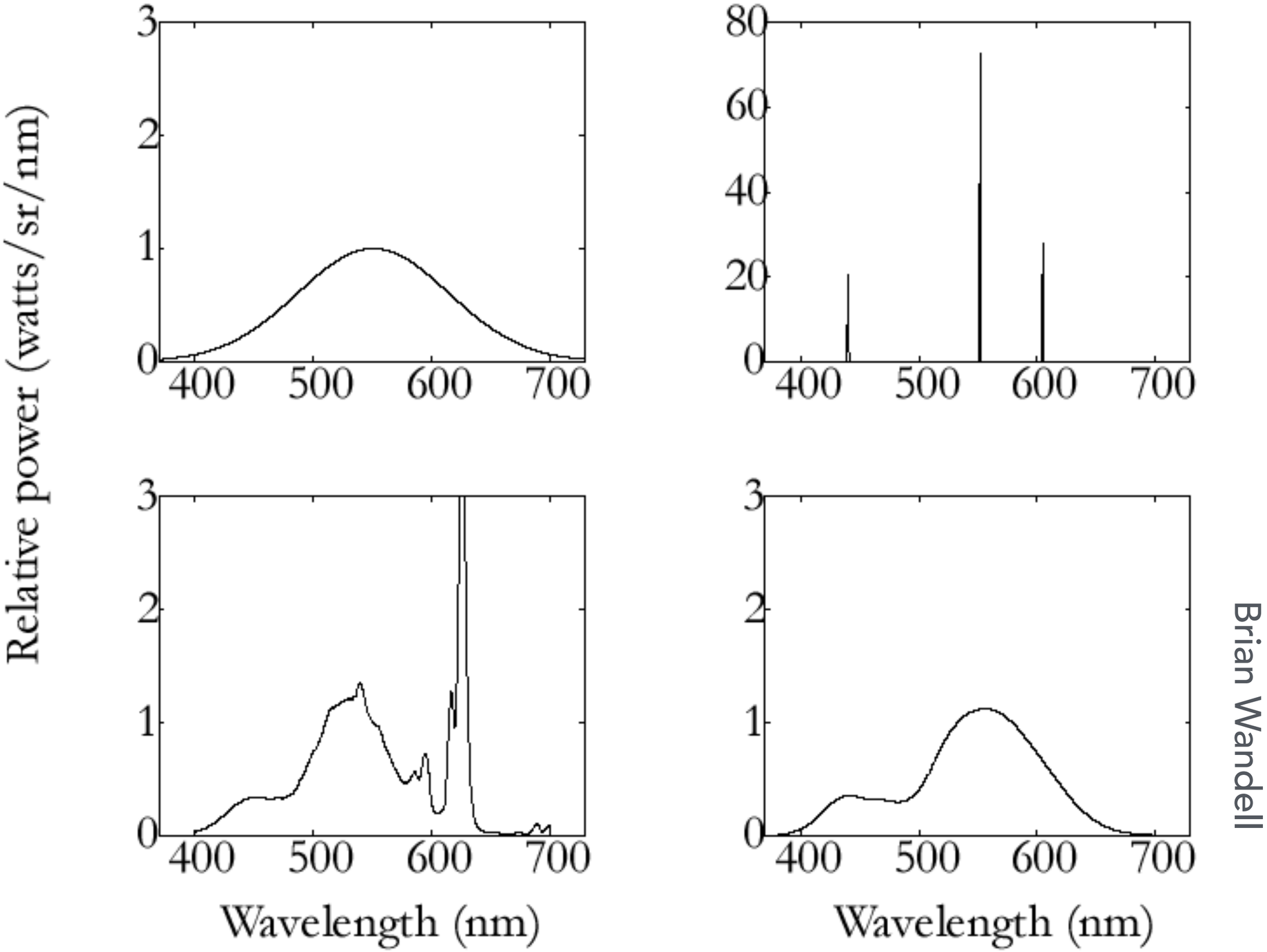
Metamers: Different light input $\Phi_1(\lambda), \Phi_2(\lambda)$ produce same L, M, S cone response

- **Different spectra look the same**
- **Useful for measuring**



Brian Wandell

Metamerism is a Big Effect



Brian Wandell

Trichromaticity

Eye records color by 3 measurements
We can "fool" it with combination of 3 signals

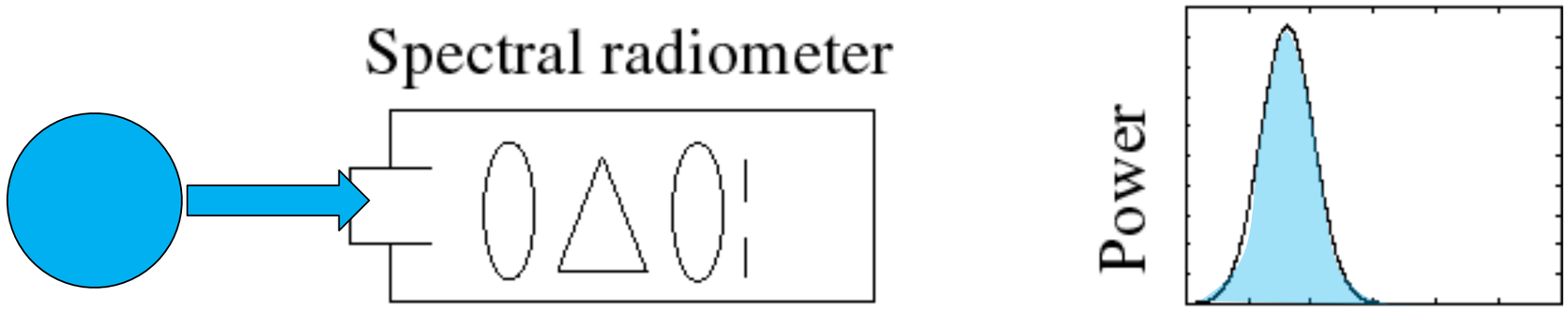
So display devices (monitors, printers, etc.) can generate perceivable colors as mix of 3 primaries

Cone Responses are Linear

- Response to stimulus Φ_1 is (L_1, M_1, S_1)
- Response to stimulus Φ_2 is (L_2, M_2, S_2)
- Then response to $\Phi_1 + \Phi_2$ is $(L_1 + L_2, M_1 + M2, S_1 + S_2$

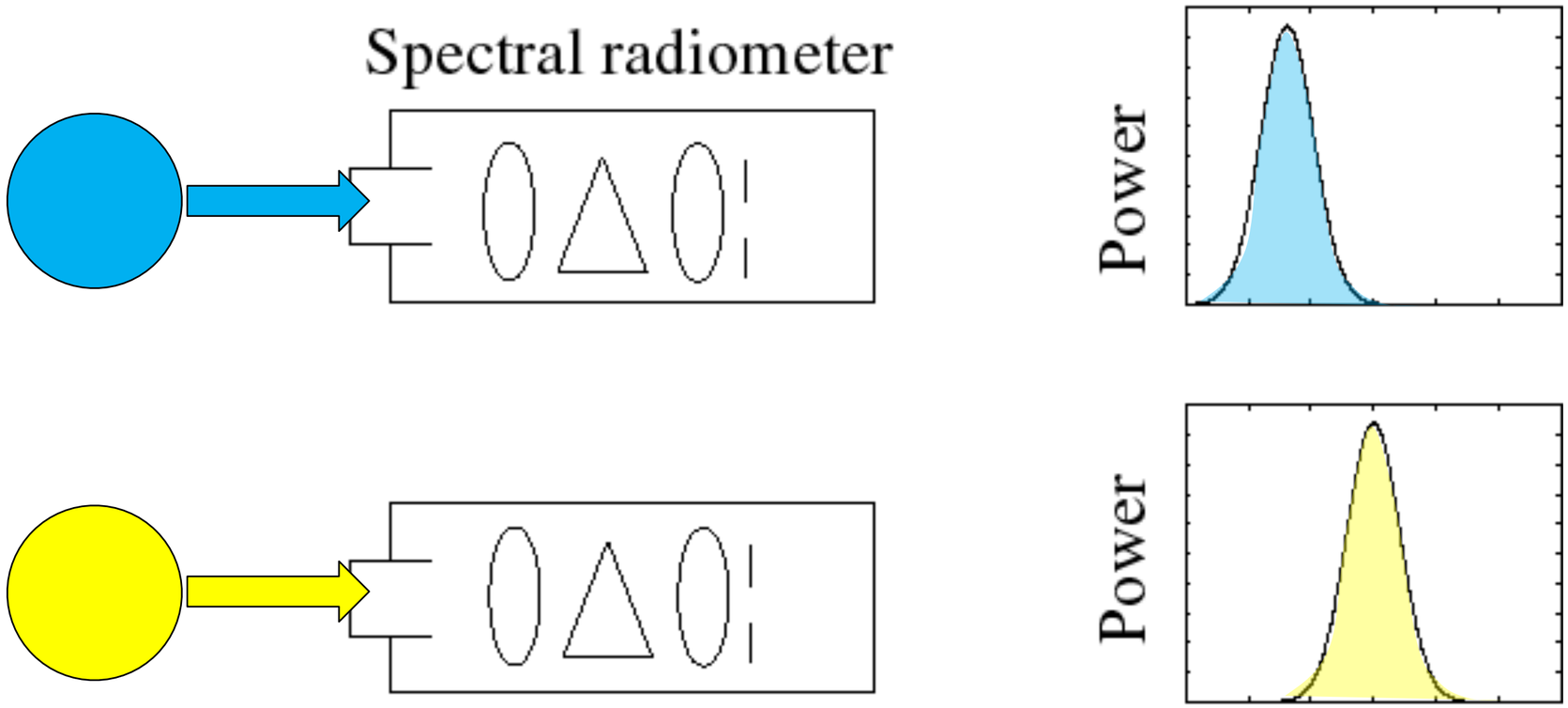
- Response to $n\Phi_1$ is (nL_1, nM_2, nS_1)

Superposition (Linearity) of Spectral Power Distributions



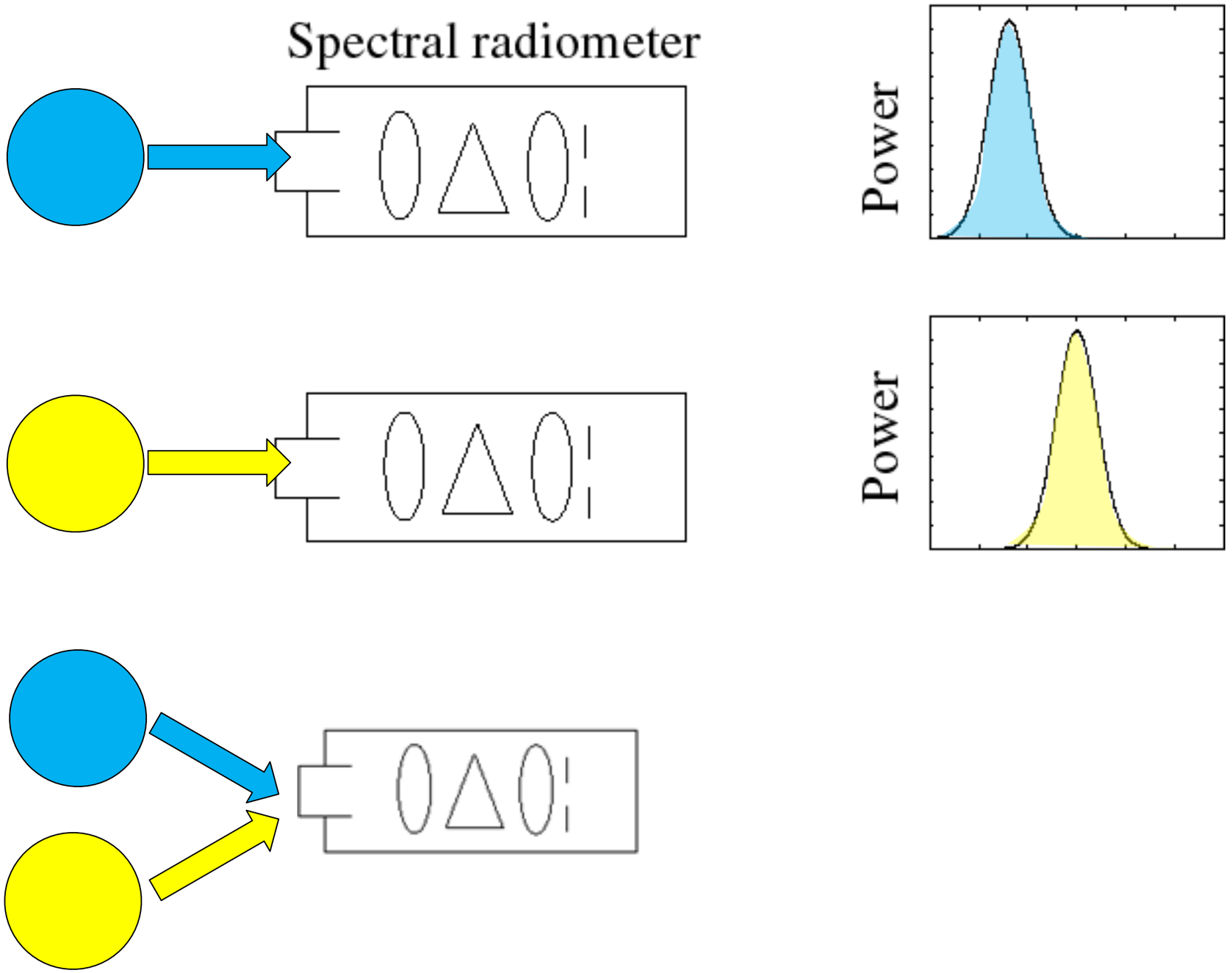
[Brian Wandell]

Superposition (Linearity) of Spectral Power Distributions



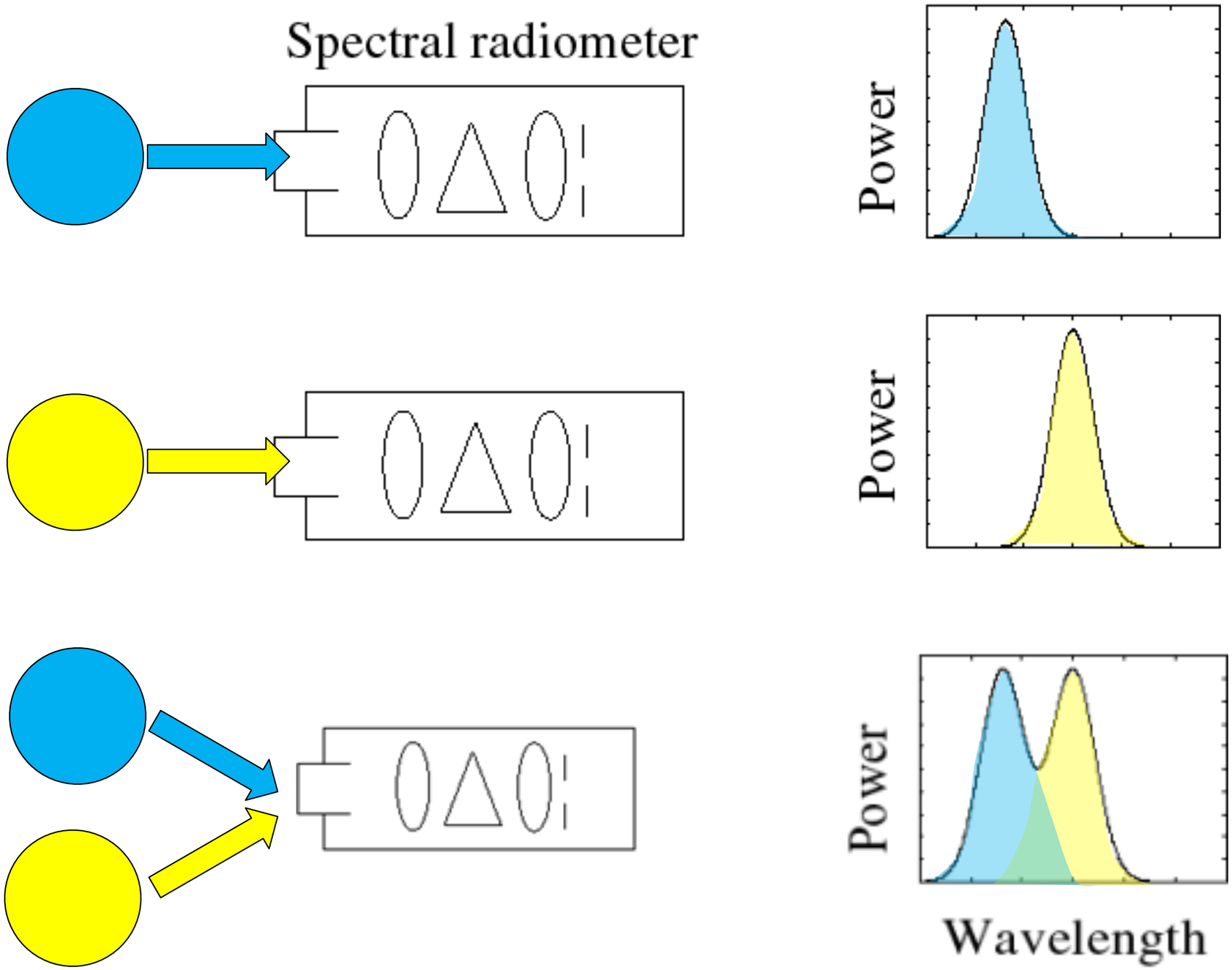
[Brian Wandell]

Superposition (Linearity) of Spectral Power Distributions



[Brian Wandell]

Superposition (Linearity) of Spectral Power Distributions

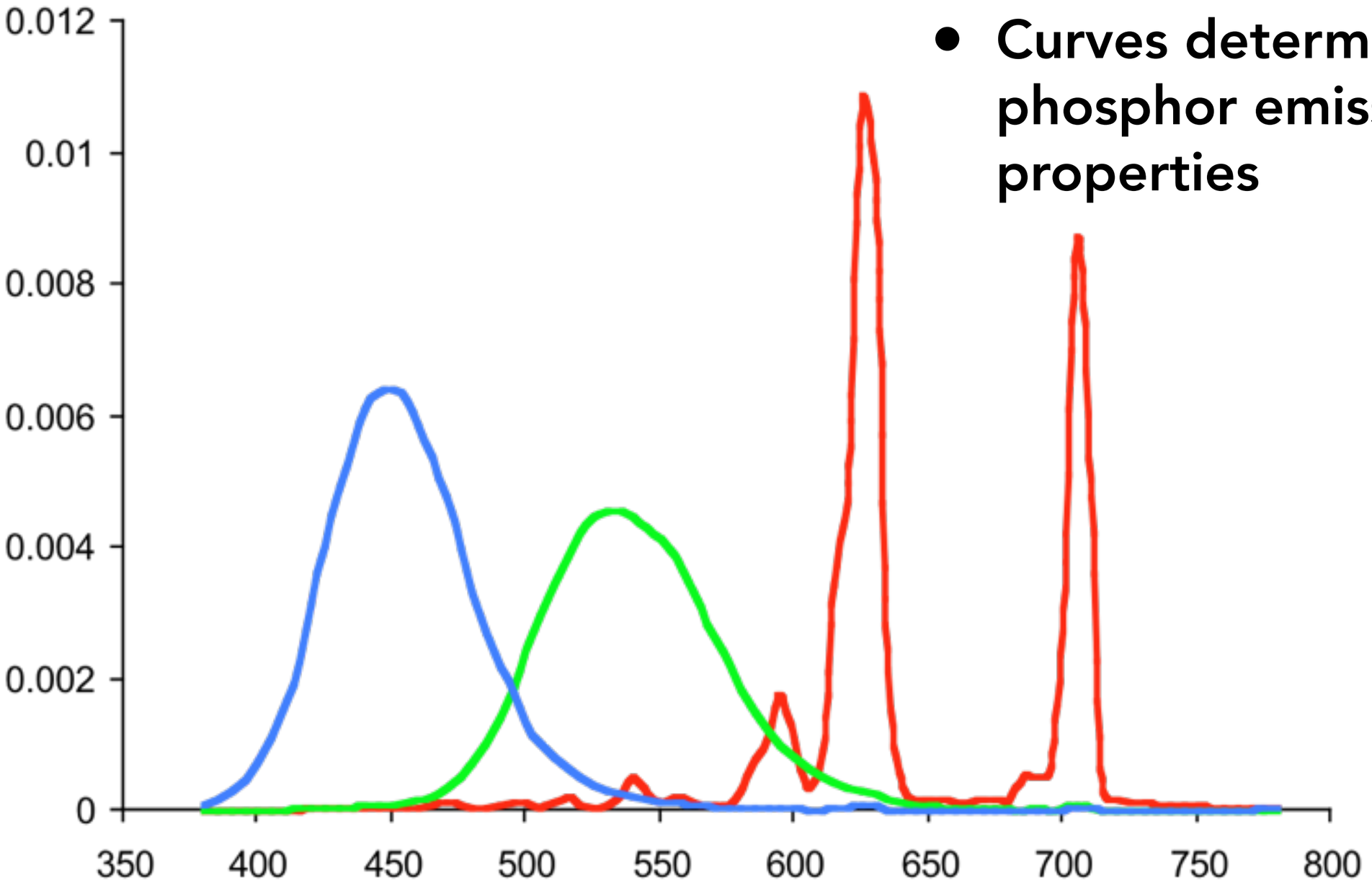


[Brian Wandell]

Additive Mixing

- Given three primaries we agree on
- Match generic input light with p_1, p_2, p_3
- Negative not realizable, but can add primary to test light $\Phi = \alpha p_1 + \beta p_2 + \gamma p_3$
- Color now described by $\alpha,\ \beta,\ \gamma$
- Example: computer monitor [RGB]

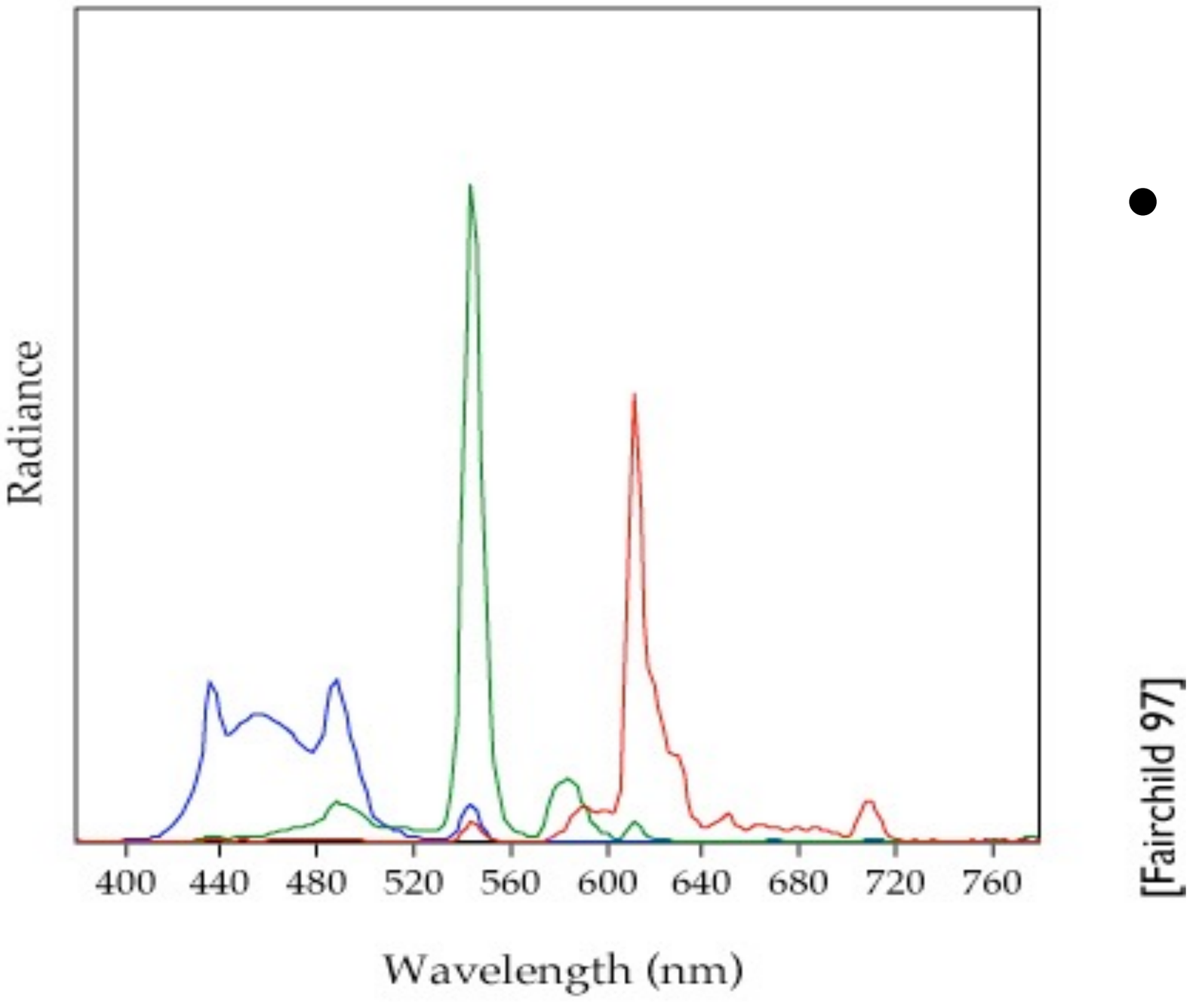
CRT Primaries



- Curves determined by phosphor emission properties

- Curves determined by phosphor emission properties

LCD Primaries

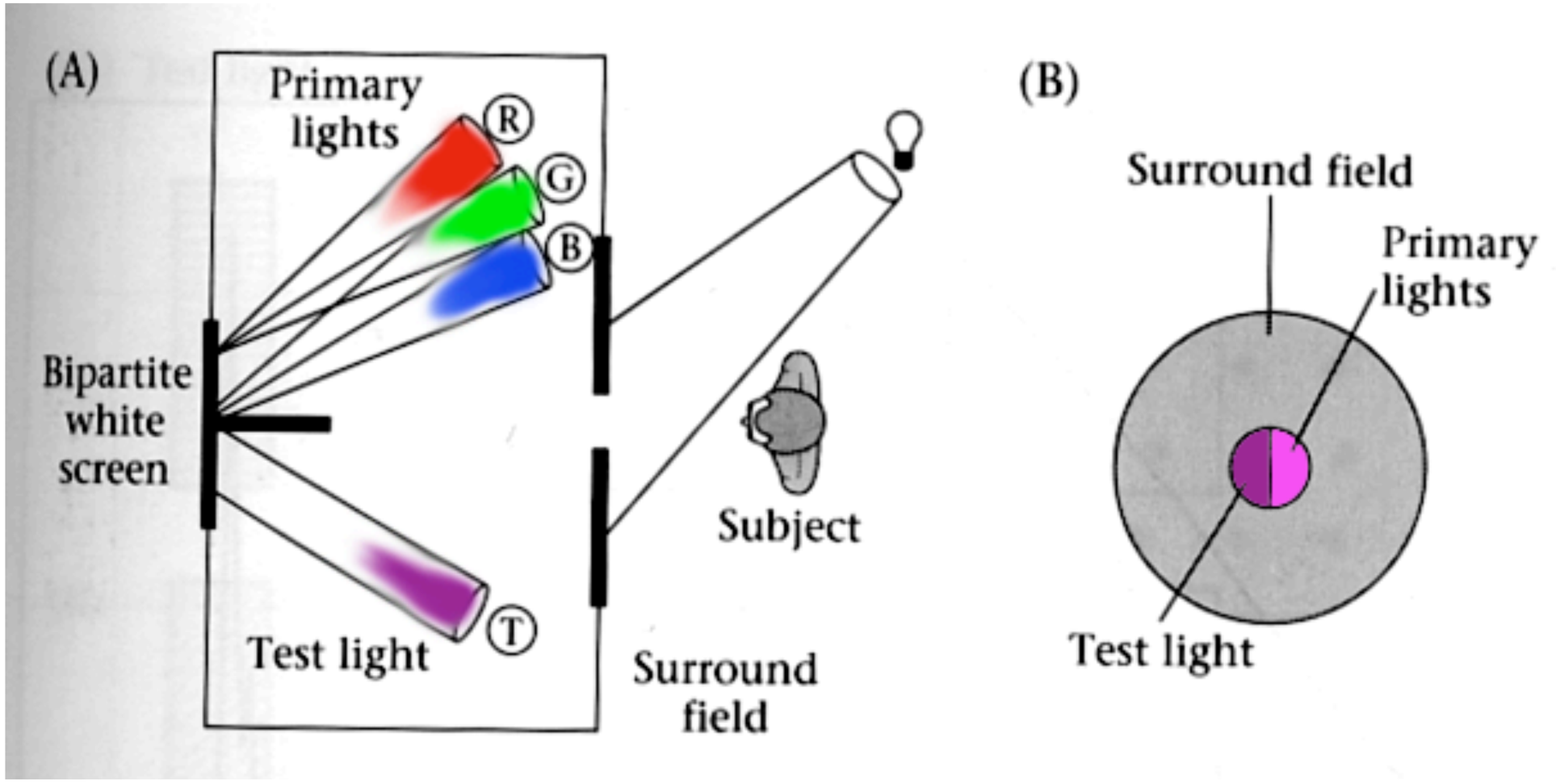


[Fairchild 97]

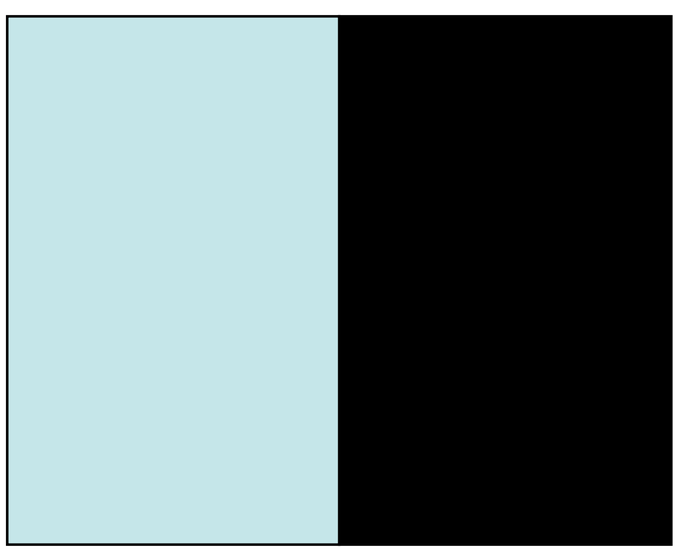
- Curves determined by (fluorescent) backlight and filters

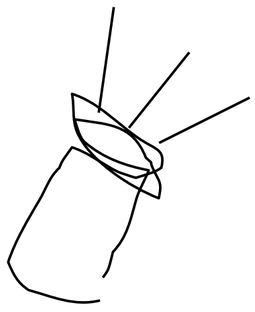
- **Curves determined by phosphor emission properties**

Additive Color Matching

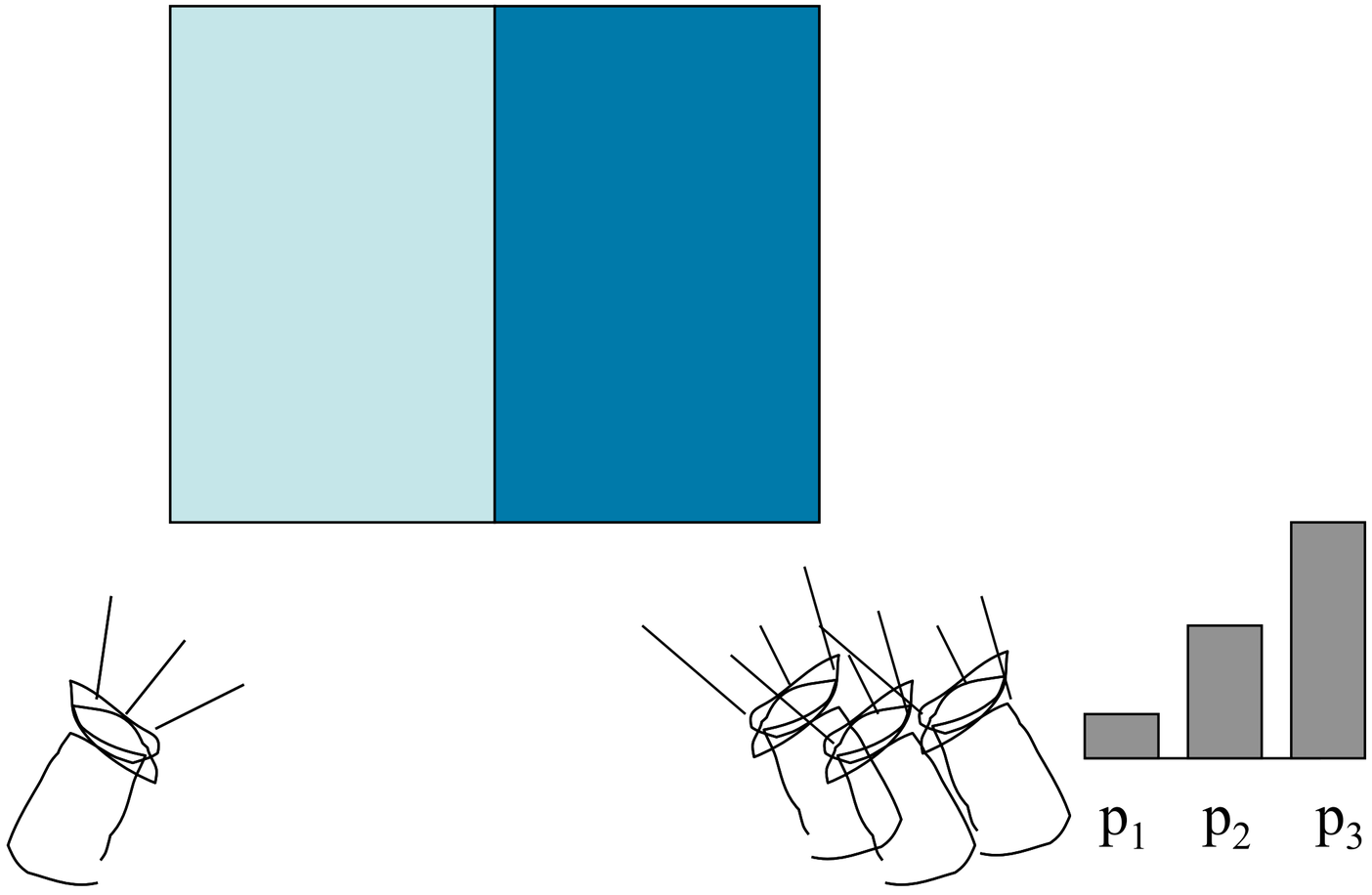


Experiment 1

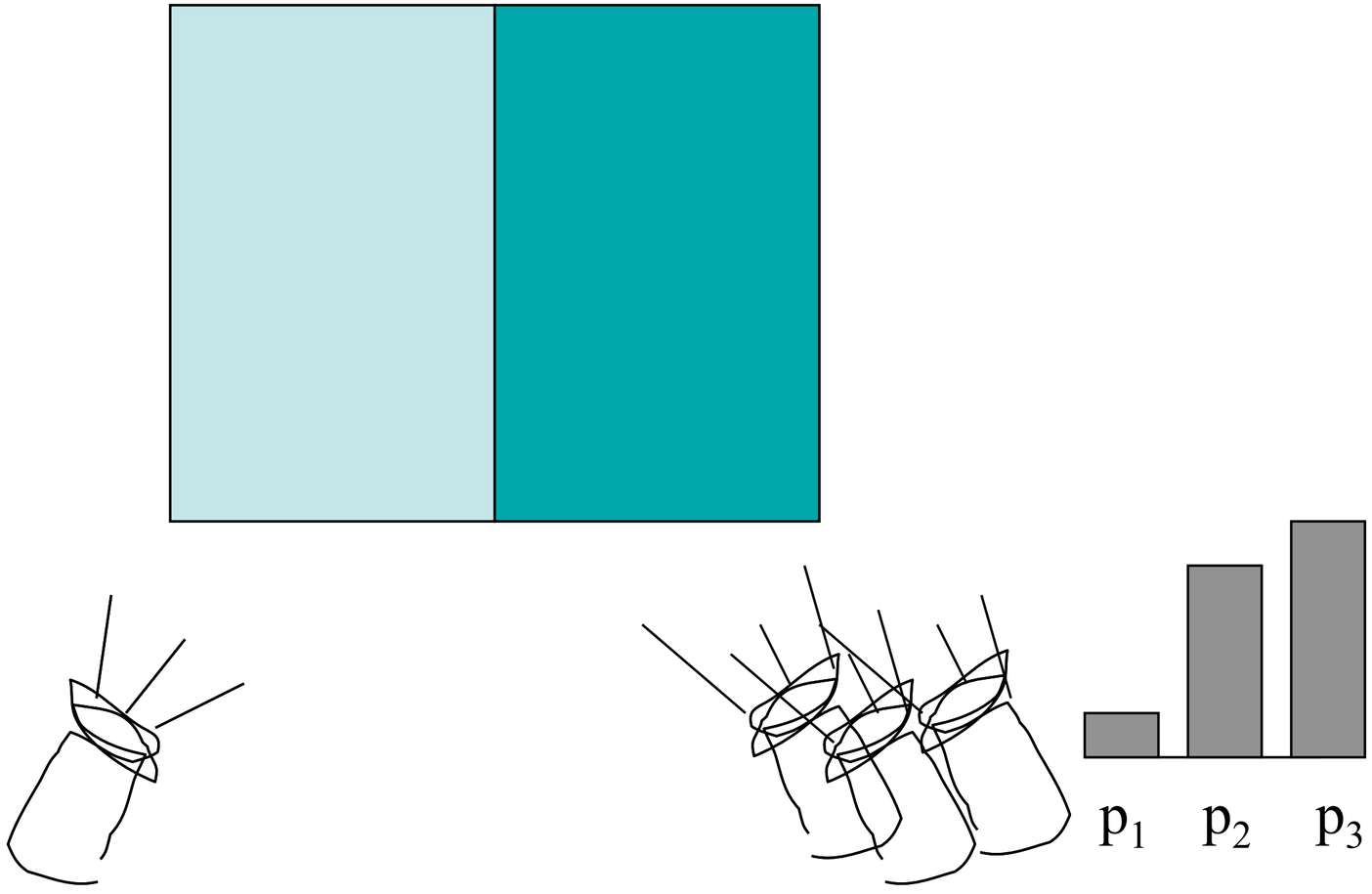




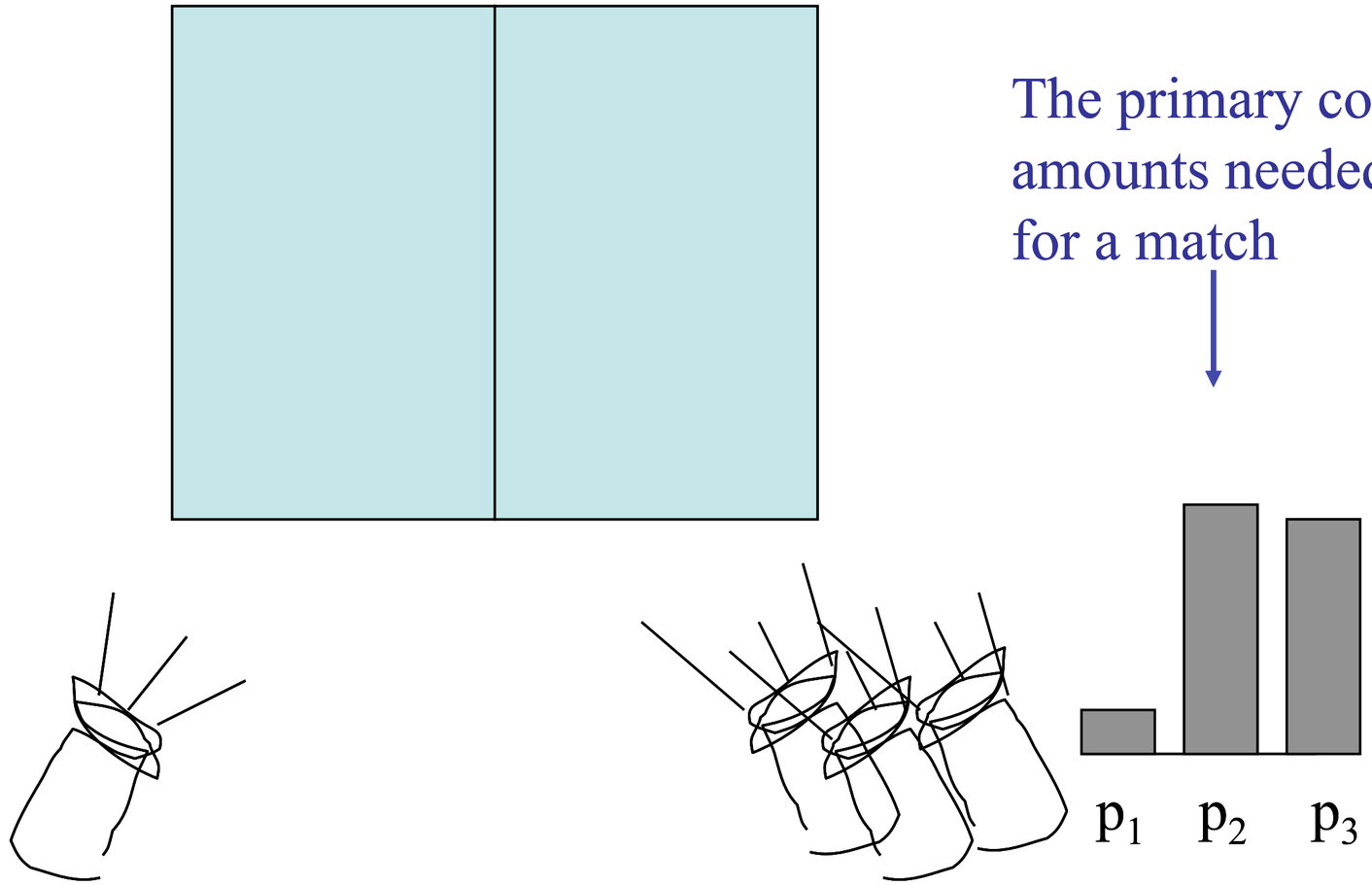
Experiment 1



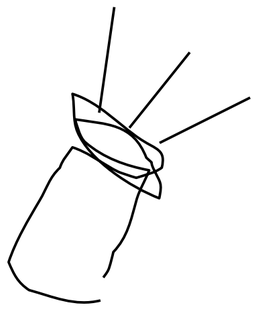
Experiment 1



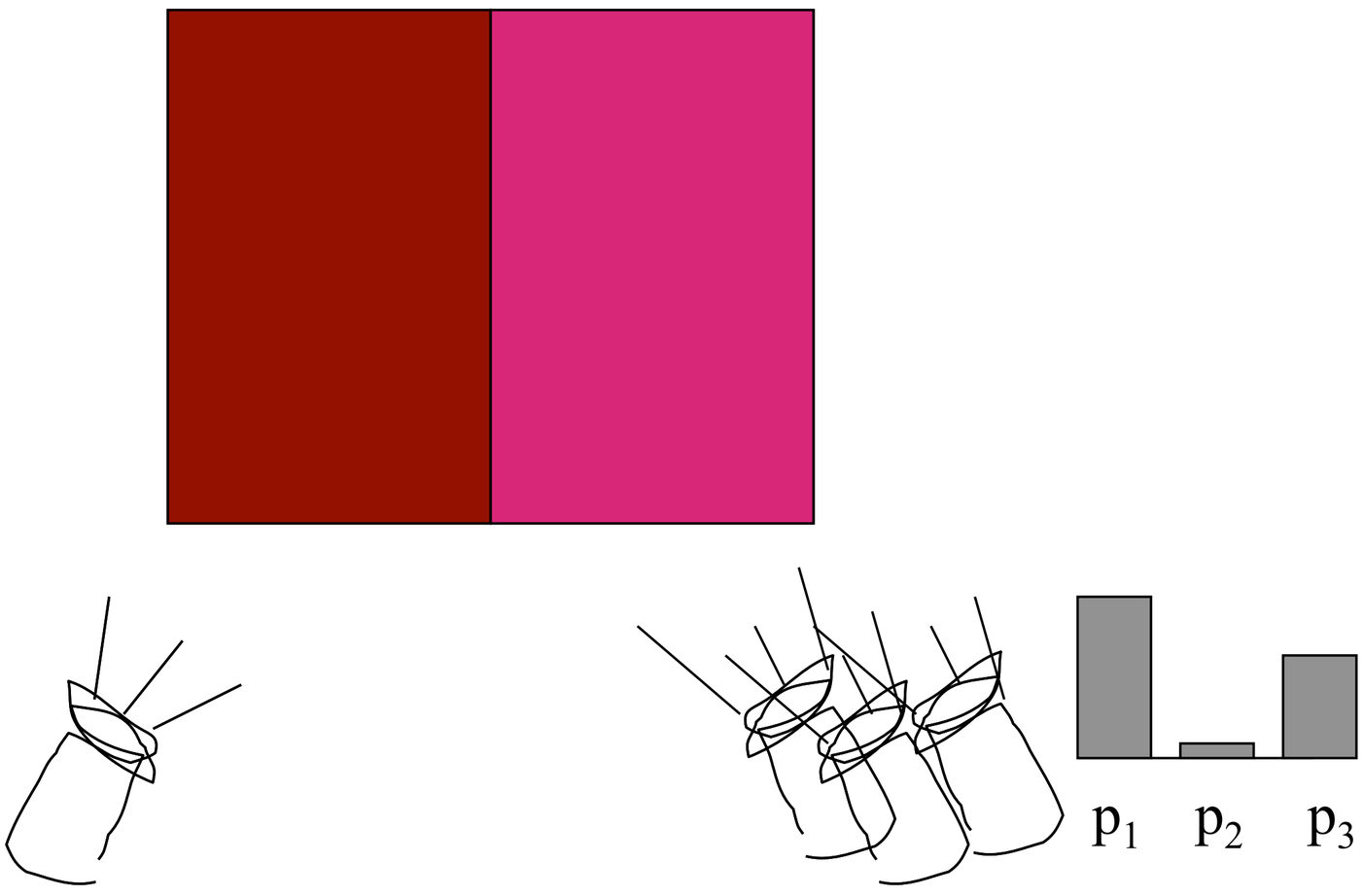
Experiment 1



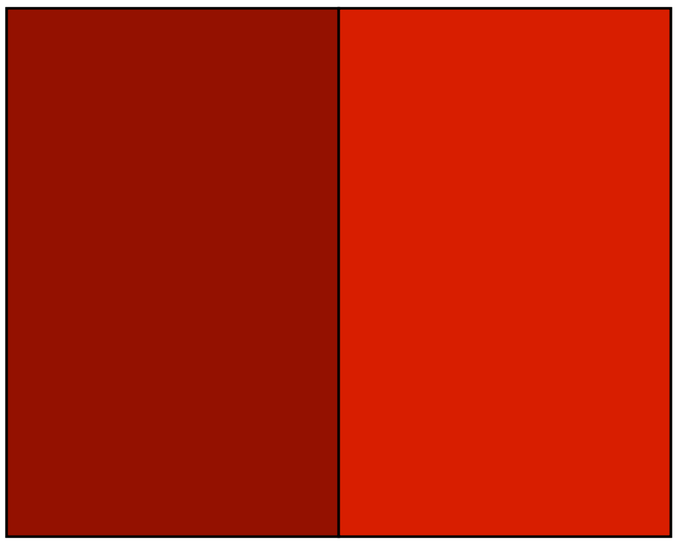
Experiment 2

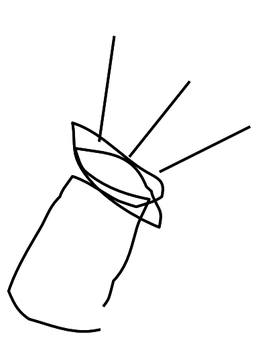


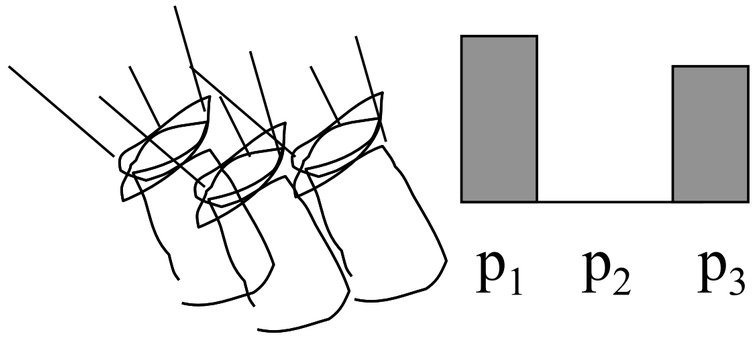
Experiment 2



Experiment 2

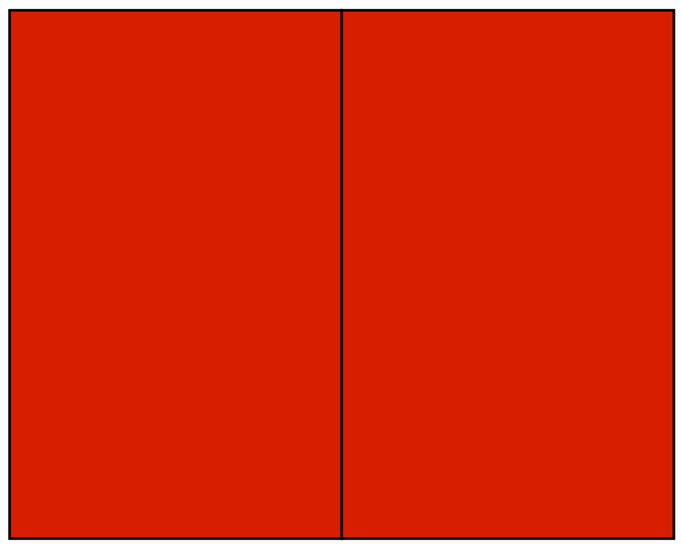


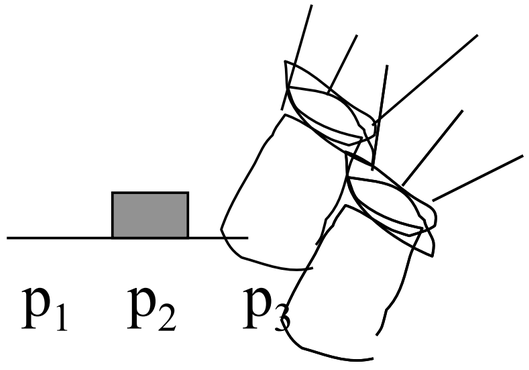




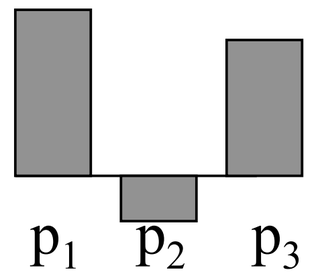
Experiment 2

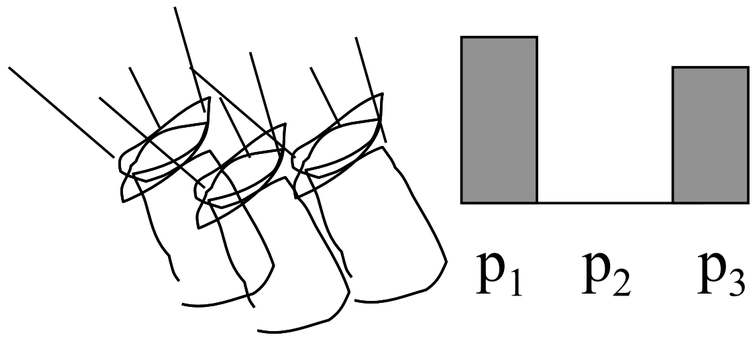
We say a "negative" amount of p_2 was needed to make the match, because we added it to the test color's side.



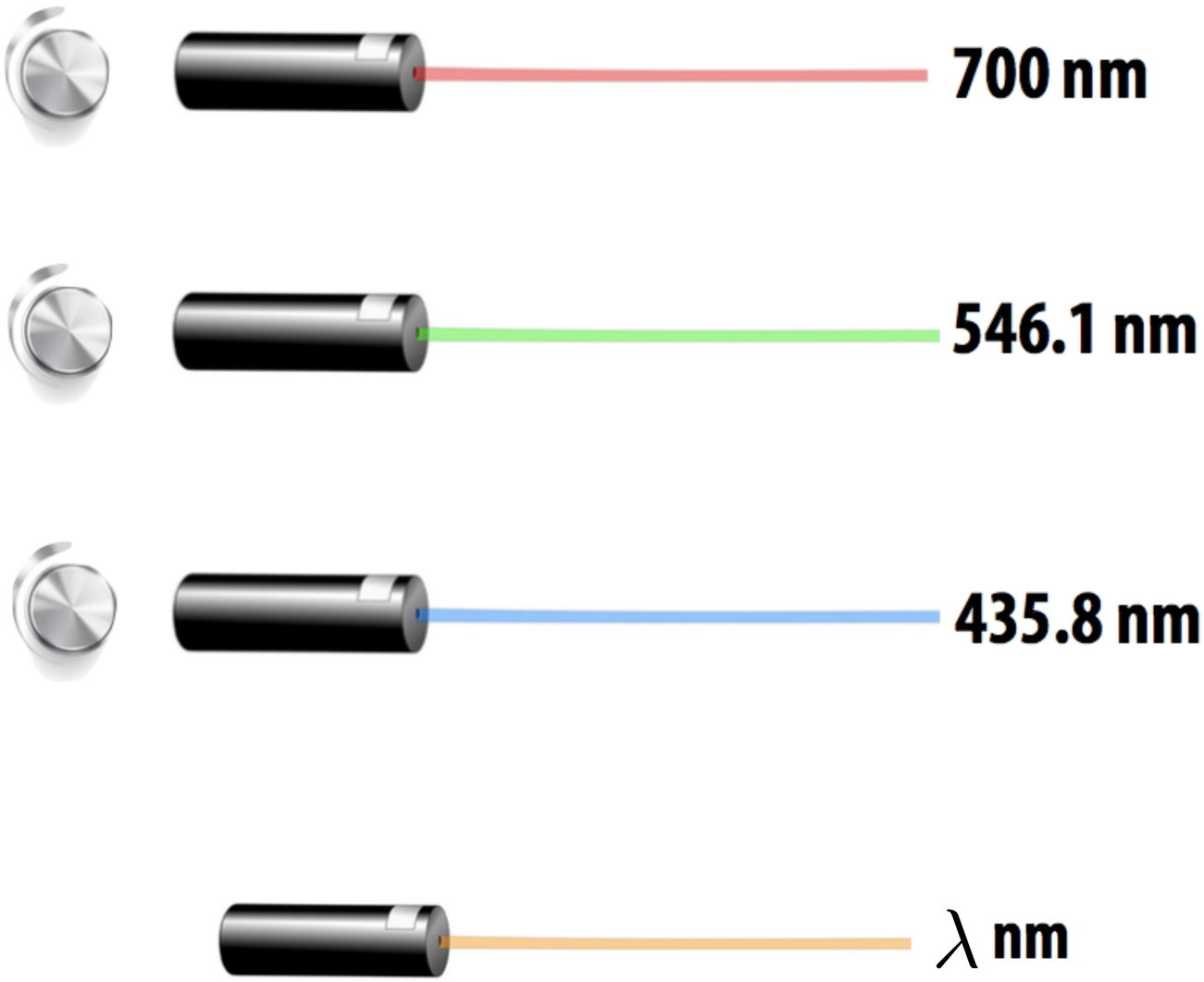


The primary color amounts needed for a match:





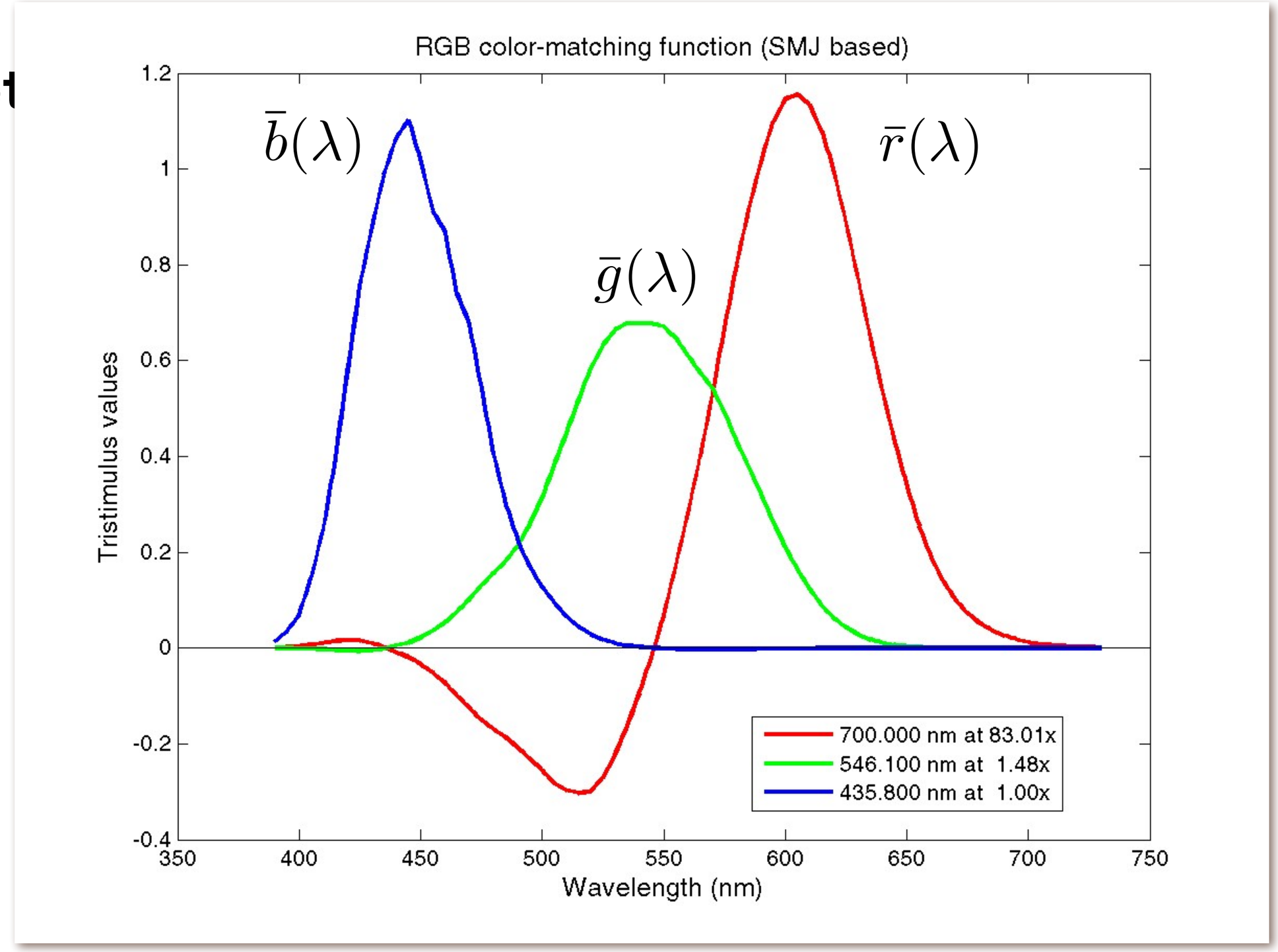
Color Matching Functions



Kayvon Fatahalian

Color Matching Functions

Not

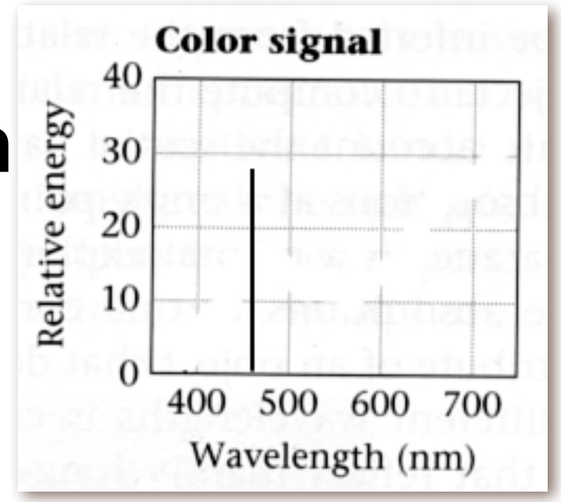


Input wavelengths are CIE 1931 monochromatic primaries

Using Color Matching Functions

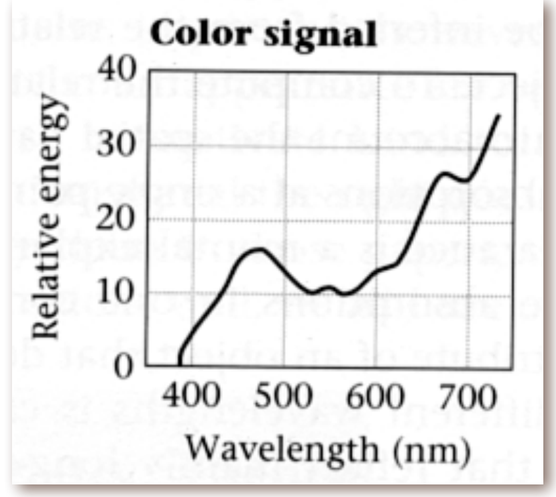
- For a monochromatic light of wavelength λ_i we know the amount of each primary necessary to match it:

$$\bar{r}(\lambda_i), \bar{g}(\lambda_i), \bar{b}(\lambda_i)$$



- Given a new light input signal

$$\Phi = \begin{pmatrix} \phi(\lambda_1) \\ \vdots \\ \phi(\lambda_N) \end{pmatrix}$$



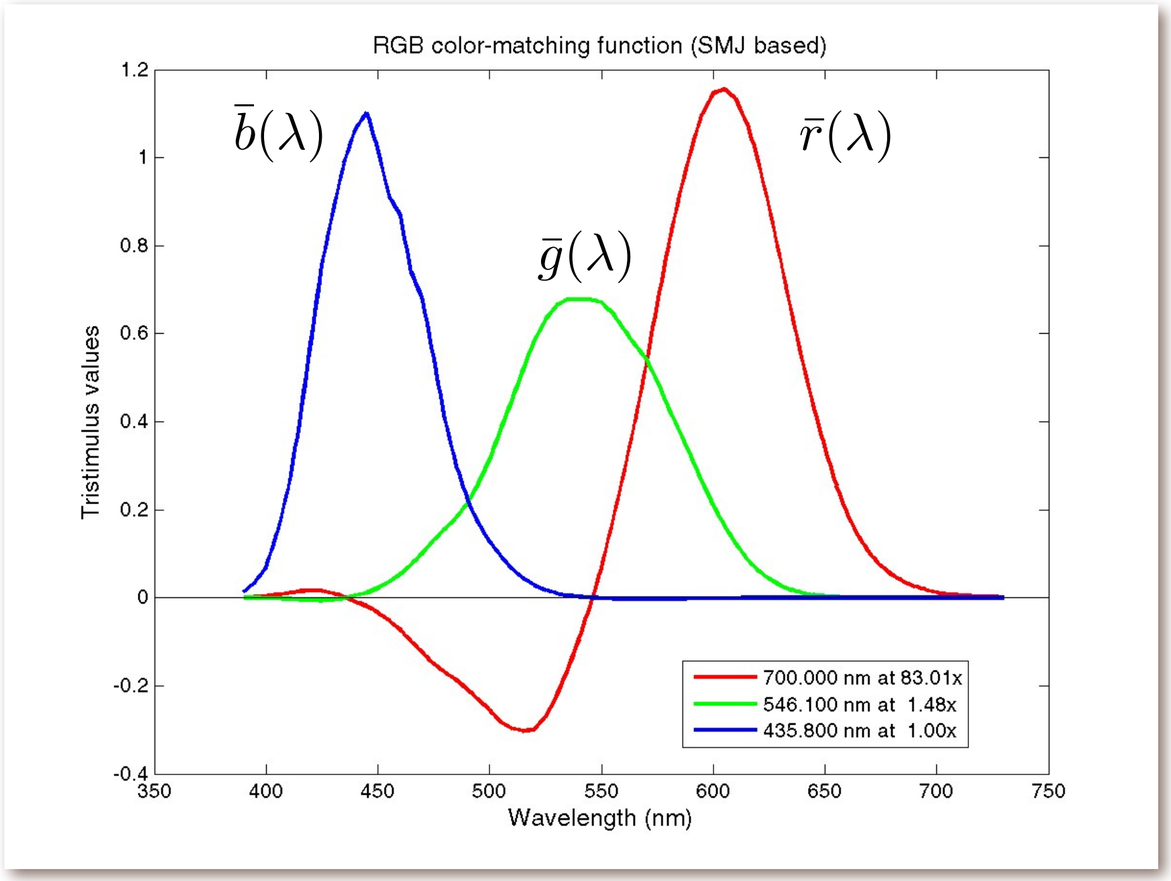
- Compute the primaries necessary to match it

Using Color Matching Functions

- Given color matching functions in matrix form and new light

$$C = \begin{pmatrix} \bar{r}(\lambda_1) & \dots & \bar{r}(\lambda_N) \\ \bar{g}(\lambda_1) & \dots & \bar{g}(\lambda_N) \\ \bar{b}(\lambda_1) & \dots & \bar{b}(\lambda_N) \end{pmatrix}$$

$$\Phi = \begin{pmatrix} \phi(\lambda_1) \\ \vdots \\ \phi(\lambda_N) \end{pmatrix}$$



- amount of each primary necessary to match is given by

$$\begin{pmatrix} r \\ g \\ b \end{pmatrix} = \mathbf{C} \cdot \Phi$$

Using Color Matching Functions

- Given color matching functions in matrix form and new light

$$\begin{pmatrix} r \\ g \\ b \end{pmatrix}_{\text{CRT}} = \mathbf{C}_{\text{CRT}} \cdot \Phi \qquad \begin{pmatrix} r \\ g \\ b \end{pmatrix}_{\text{LCD}} = \mathbf{C}_{\text{LCD}} \cdot \Phi$$

$$\begin{pmatrix} r \\ g \\ b \end{pmatrix}_{\text{LCD}} = \mathbf{C}_{\text{LCD}} \cdot \mathbf{C}_{\text{CRT}}^{-\text{P}} \cdot \begin{pmatrix} r \\ g \\ b \end{pmatrix}_{\text{CRT}}$$

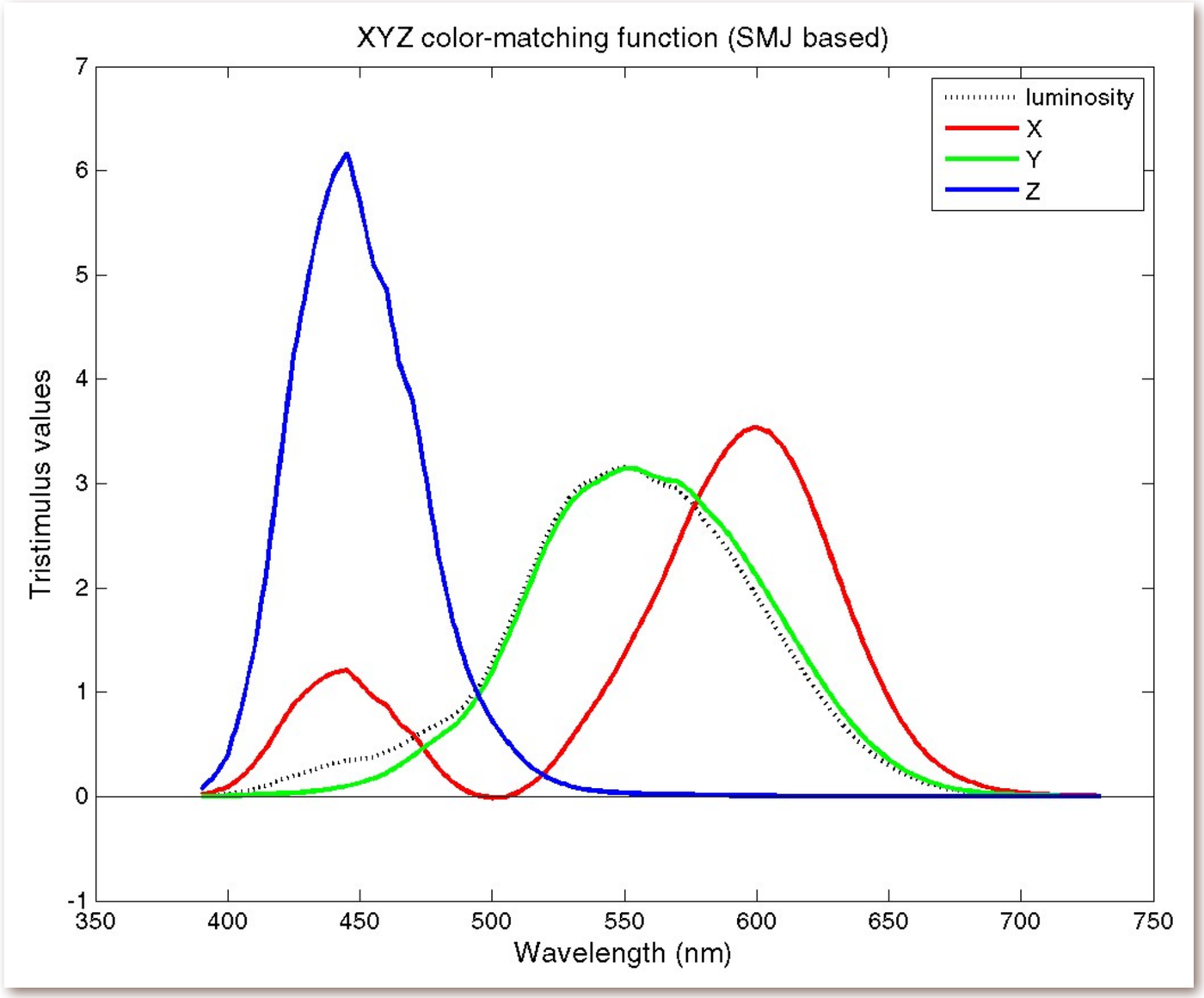
$$\begin{pmatrix} r \\ g \\ b \end{pmatrix}_{\text{LCD}} = \mathbf{C}_{\text{CRT}\to\text{LCD}} \cdot \begin{pmatrix} r \\ g \\ b \end{pmatrix}_{\text{CRT}}$$

- amount of each primary necessary to match is given by

$$\begin{pmatrix} r \\ g \\ b \end{pmatrix}_{\text{display}} = \mathbf{C}_{\text{display}} \cdot \begin{pmatrix} r \\ g \\ b \end{pmatrix}_{\text{sRGB}}$$

CIE XYZ

Imaginary set of color primaries with positive values, X, Y, Z



Rescaled XYZ to xyz

Rescale X, Y, and Z to remove luminance, leaving chromaticity:

$$x = X / (X+Y+Z)$$

$$y = Y / (X+Y+Z)$$

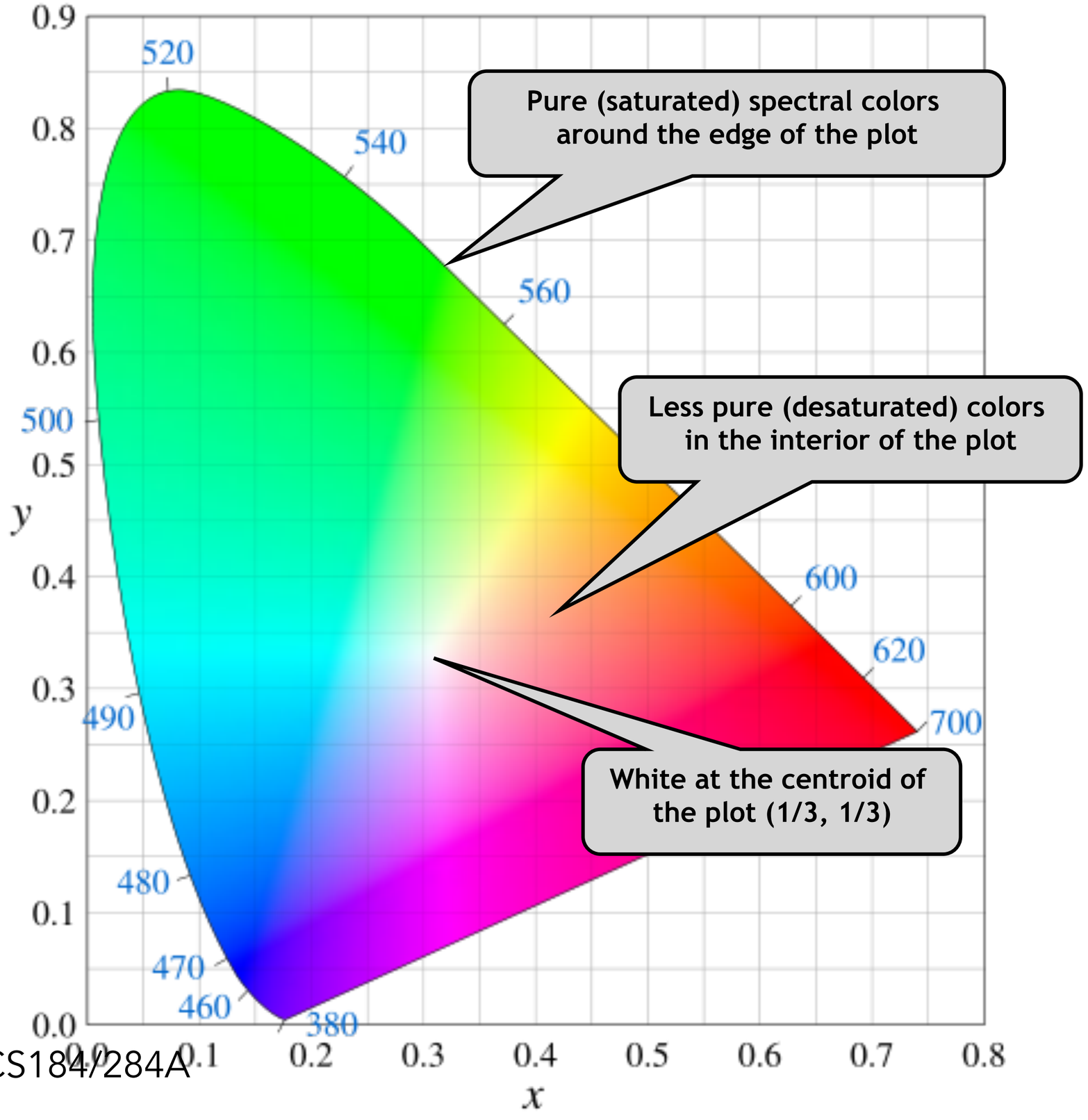
$$z = Z / (X+Y+Z)$$

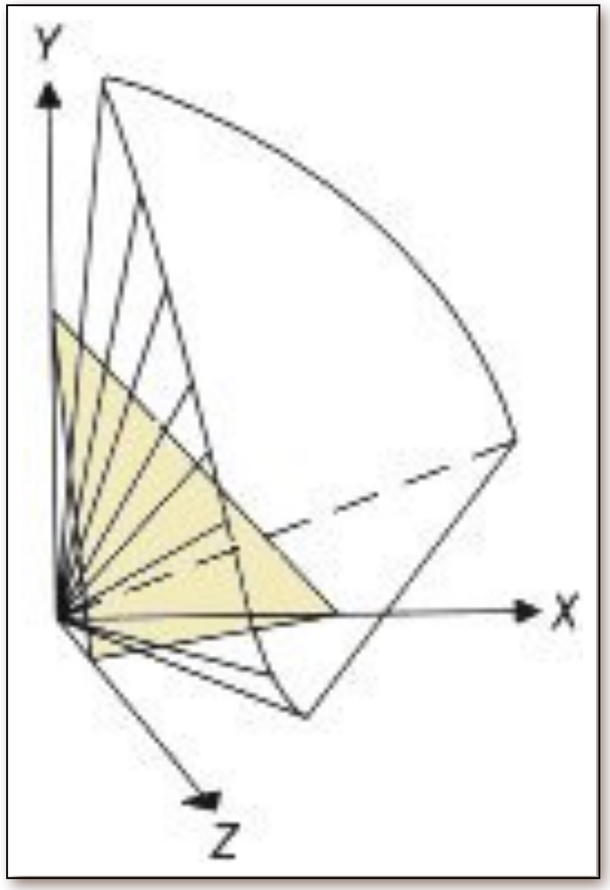
Because the sum of the chromaticity values x, y, and z is always 1.0, a plot of any two of them loses no information

$$x+y+z = 1$$

— Chromaticity Diagram

CIE Chromaticity Diagram

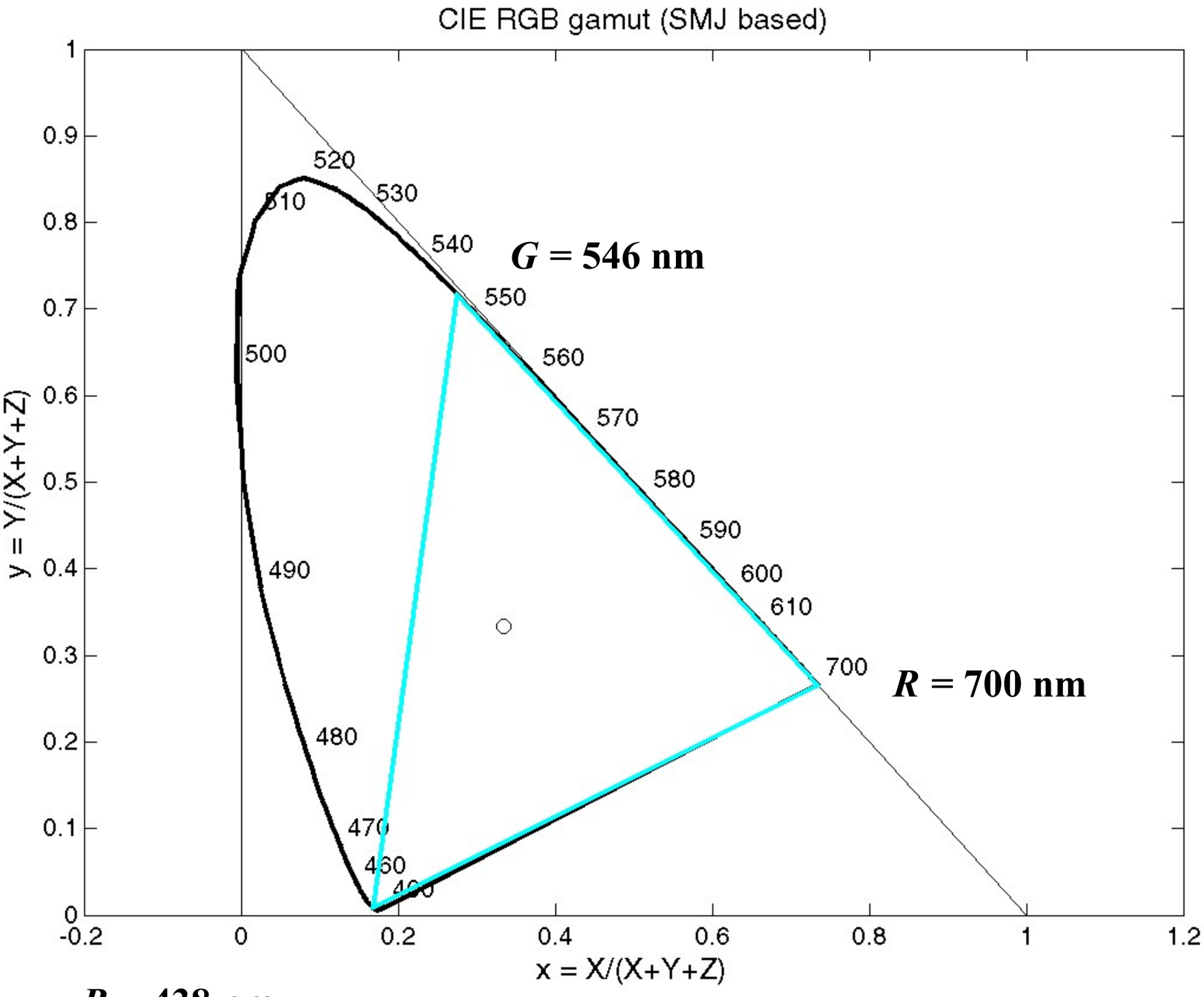




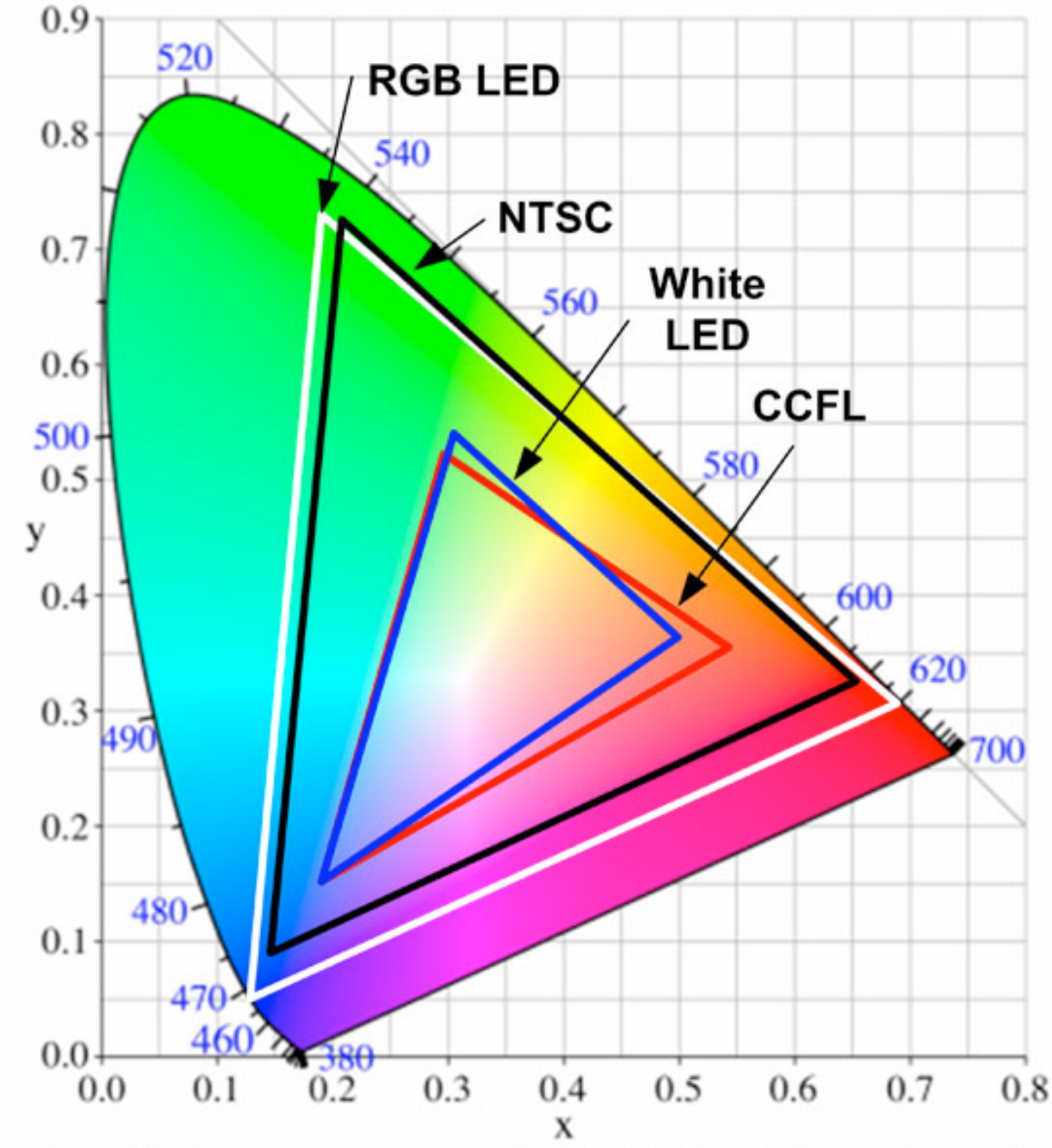
Gamut

- Gamut is the chromaticities generated by a set of primaries
- Because everything we've done is linear, interpolation between chromaticities on a chromaticity plot is also linear
- Thus the gamut is the convex hull of the primary chromaticities
- What is the gamut of the CIE 1931 primaries?

CIE 1931 RGB Gamut



Other Gamuts (LCDs and NTSC)



Subtractive Mixing

- Given three primaries we agree on p_1, p_2, p_3
- Make generic color with $\Phi = W - (\alpha p_1 + \beta p_2 + \gamma p_3)$
- Max limited by W
- Color now described by $\alpha,\ \beta,\ \gamma$
- Example: ink [CMYK]

Subtractive Mixing

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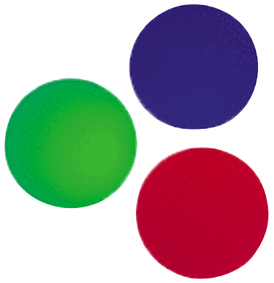
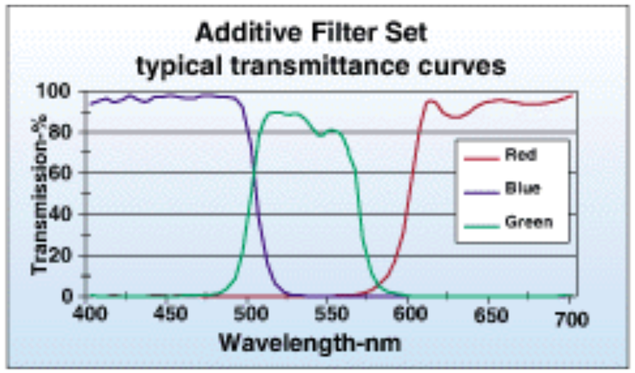
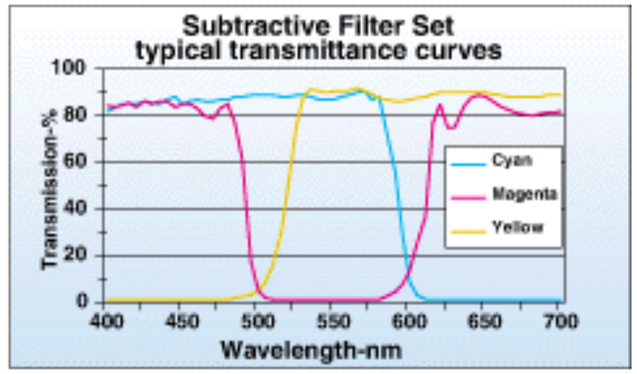
- Max limited by W
- Color now described by $\alpha,\ \beta,\ \gamma$
- Example: ink [CMYK]

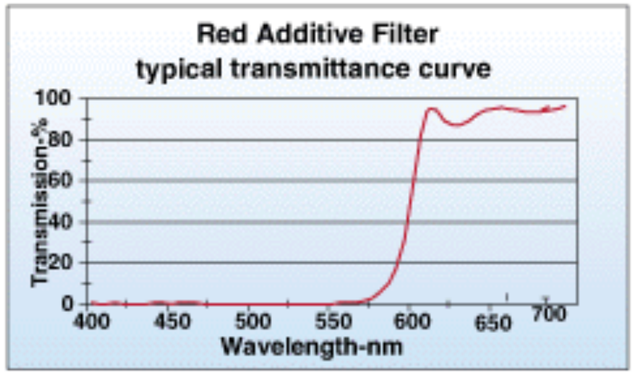
Why 4th ink for black?

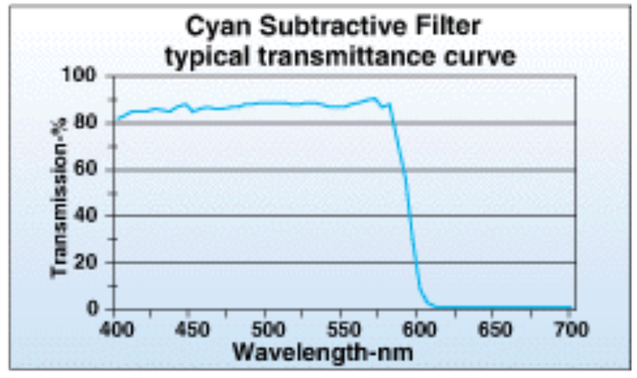
Additive & Subtractive Primaries

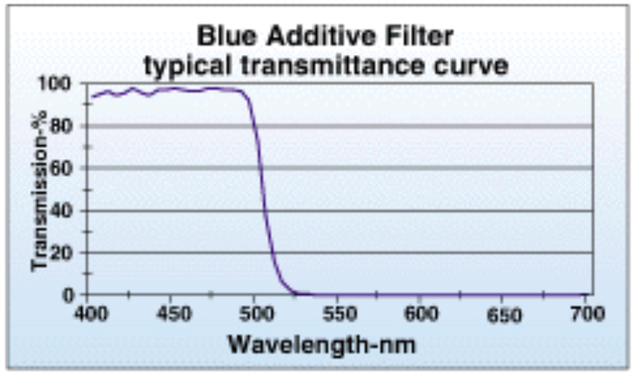


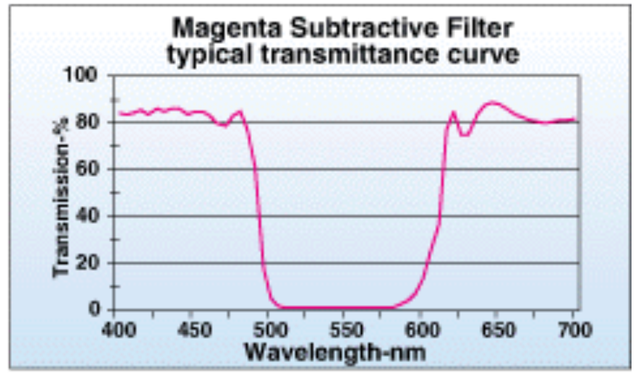


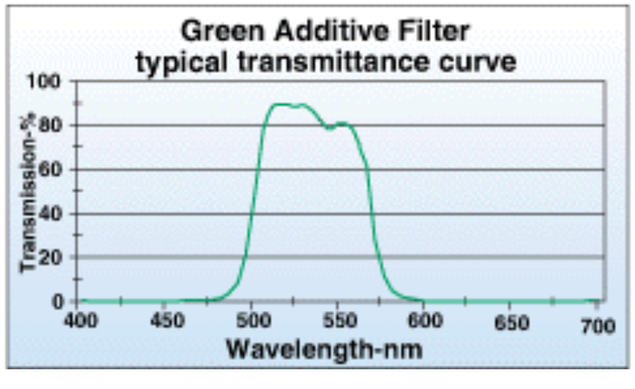


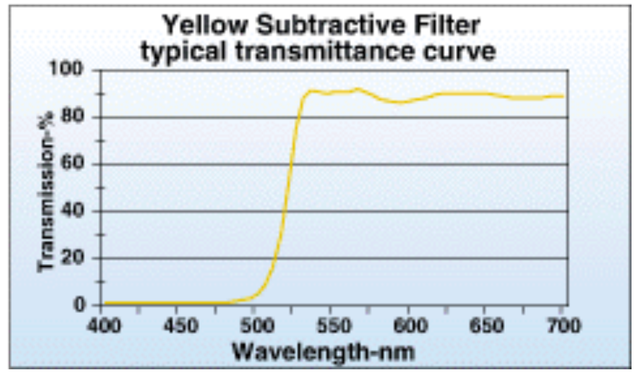












CS184/284A

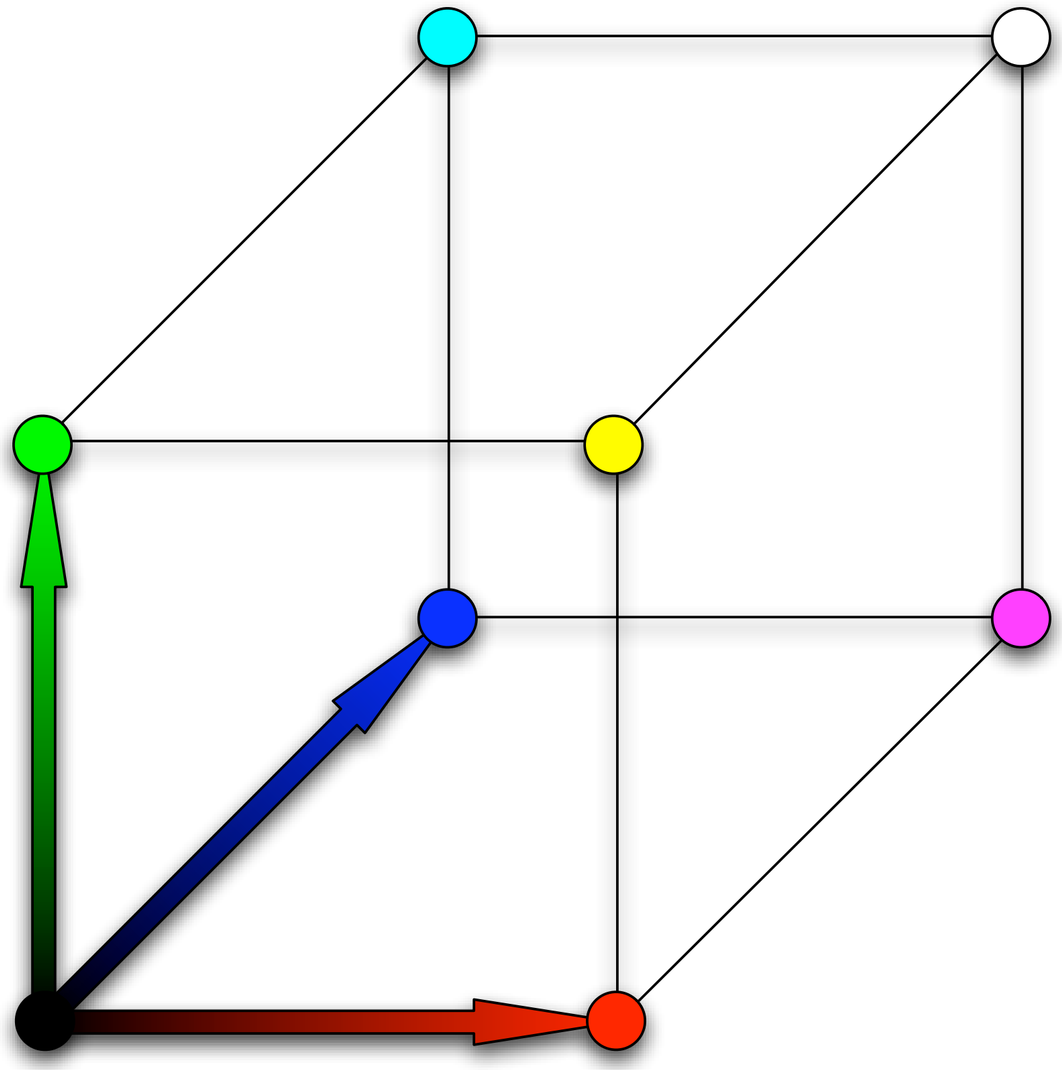
Additive & Subtractive Primaries

- Incorrect to say "the additive primaries are red, green, and blue"
  - Any set of three non-collinear primaries yields a gamut
  - Primaries that appear red, green, and blue are a good choice, but not the only choice
  - Are additional (non-collinear) primaries always better?
- Similarly saying "the subtractive primaries are magenta, cyan, and yellow" is also incorrect, for the same reasons
  - Subtractive primaries must collectively block the entire visible spectrum, but many sets of blockers that do so are acceptable "primaries"
  - The use of black ink (the K in CMYK) is a good example
  - Modern ink-jet printers often have 6 or more ink colors

Color Spaces

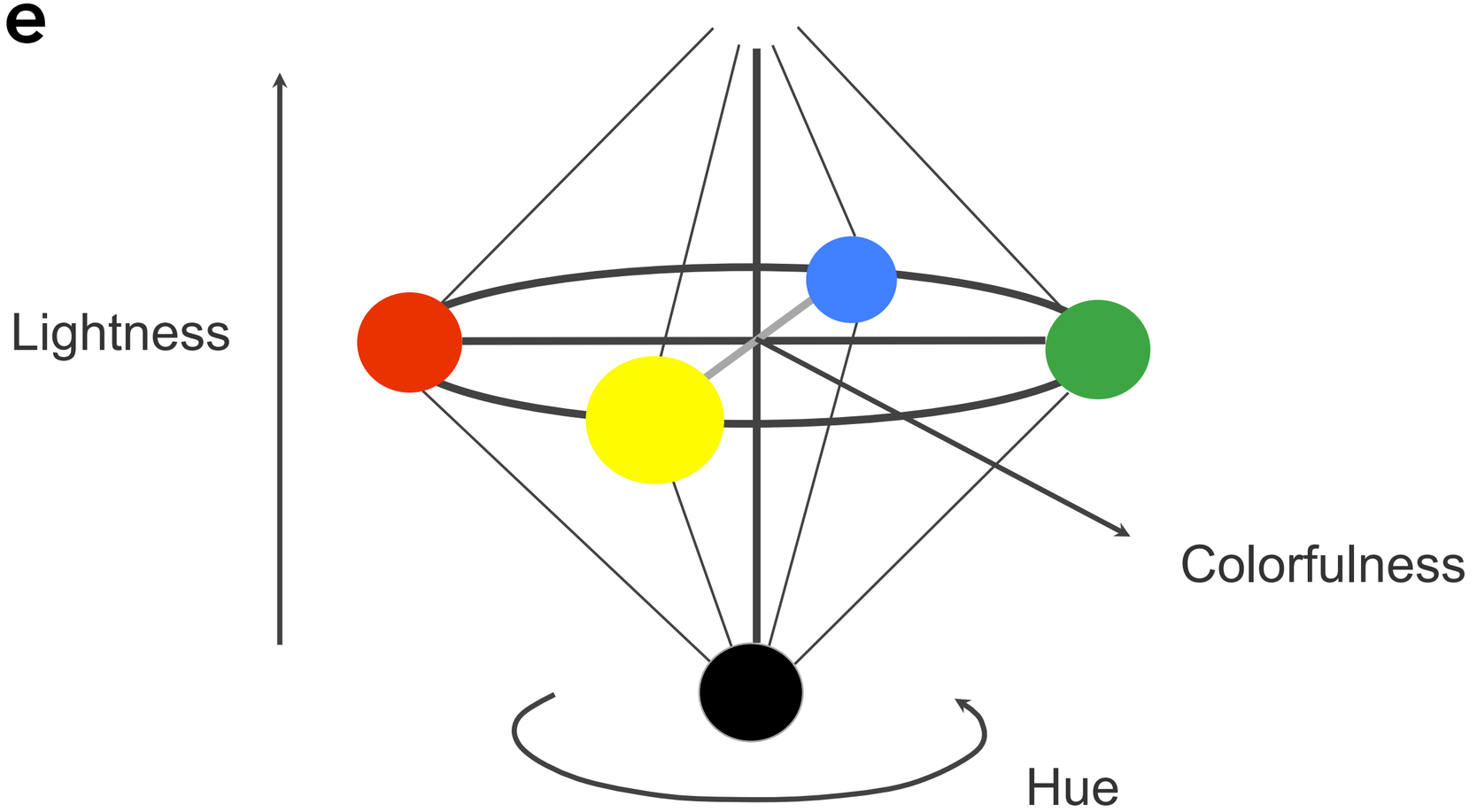
RGB color cube

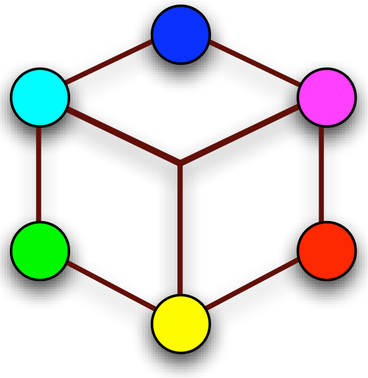
- Does not correspond very well to perception (e.g. distance between two points has little meaning)

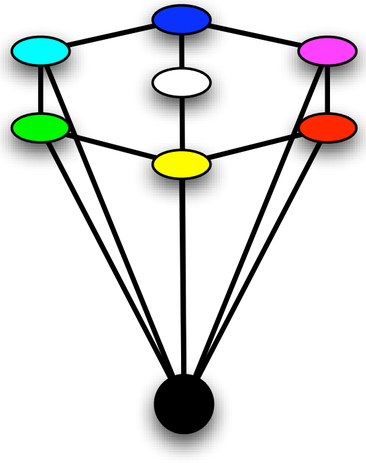


Color Spaces

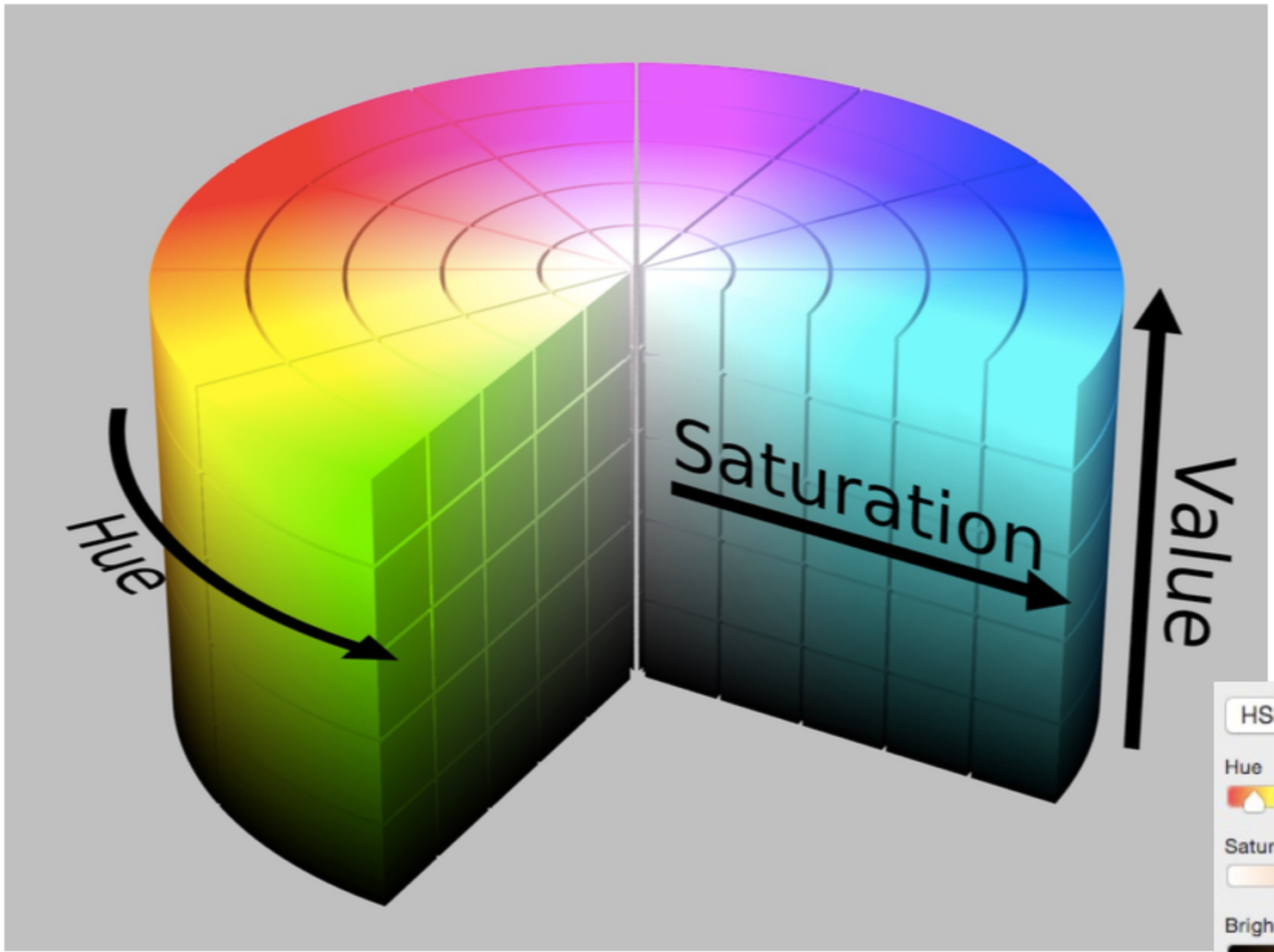
HSV color cone

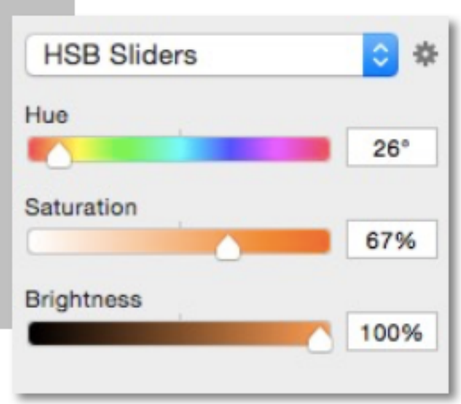




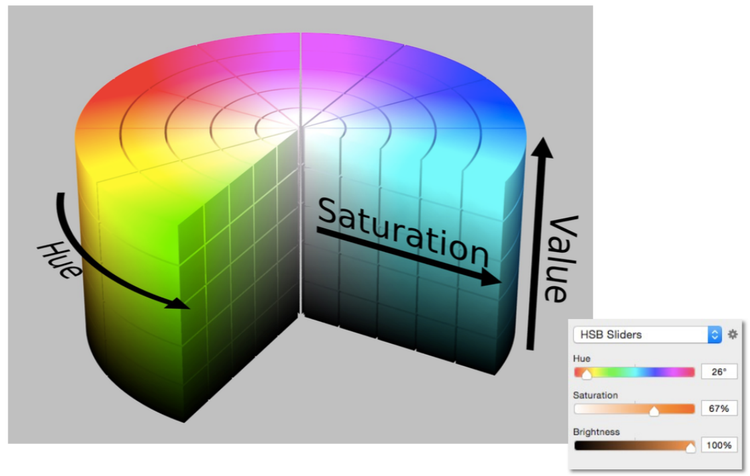


HSV Color Space (Hue-Saturation-Value)





HSV Color Space (Hue-Saturation-Value)



Perceptual dimensions of color

Hue

- the “kind” of color, regardless of attributes
- colorimetric correlate: dominant wavelength
- artist’s correlate: the chosen pigment color

Saturation

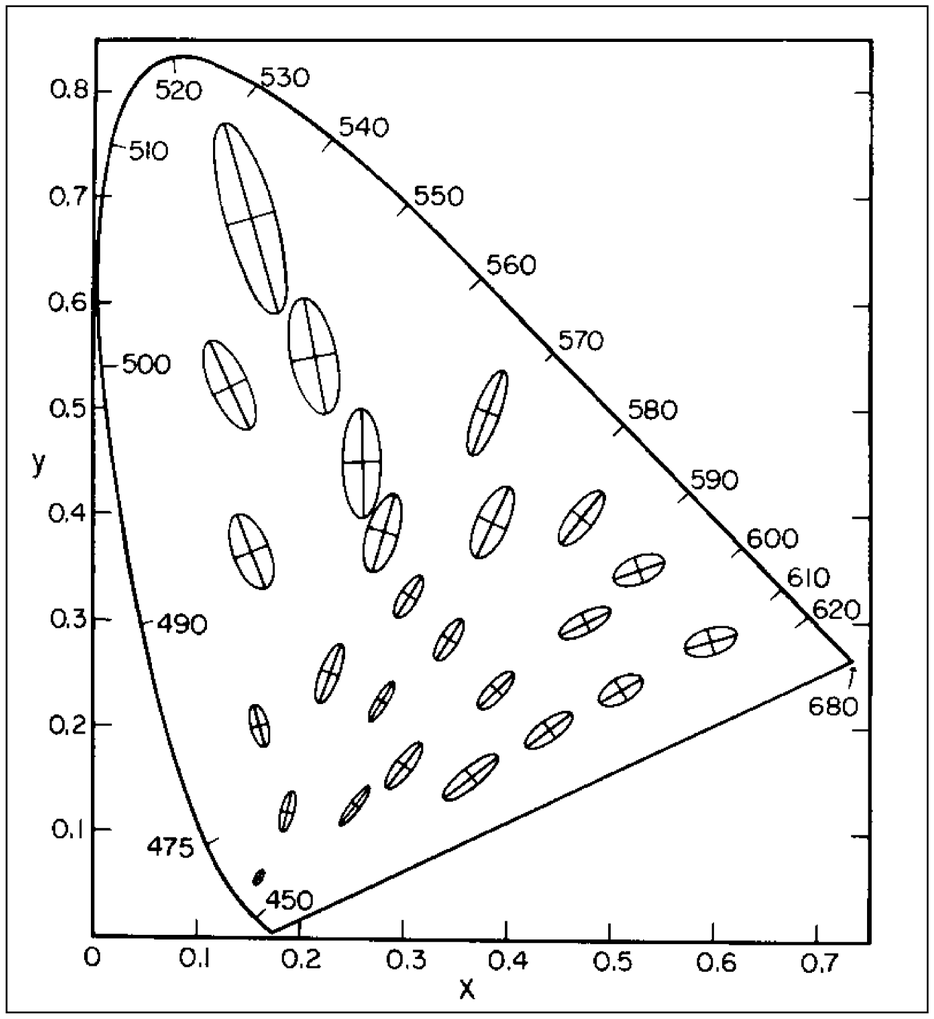
- the “colorfulness”
- colorimetric correlate: purity
- artist’s correlate: fraction of paint from the colored tube

Lightness (or value)

- the overall amount of light
- colorimetric correlate: luminance
- artist’s correlate: tints are lighter, shades are darker

Color Spaces

RGB color cube
HSV color cone
CIE (x,y)

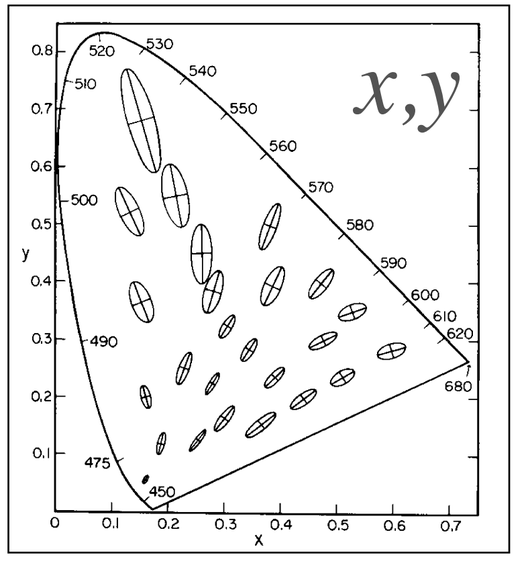


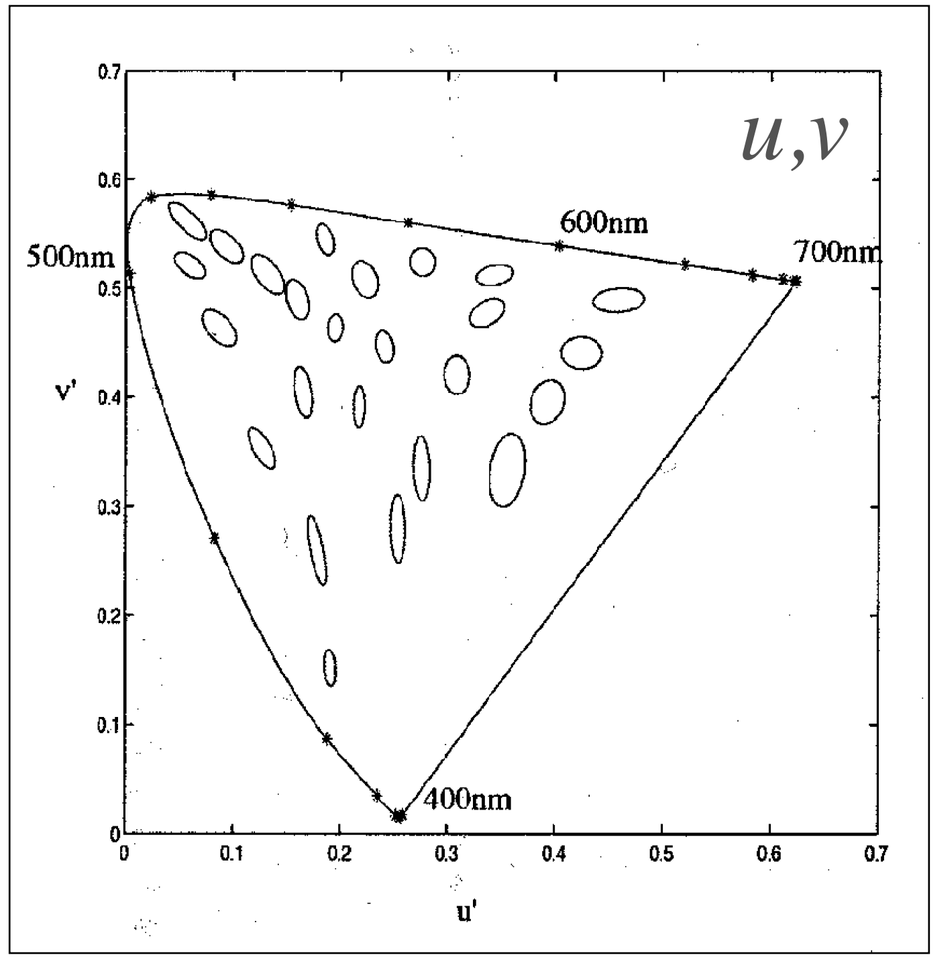
MacAdam Ellipses (10x)

Colors in ellipses indistinguishable from center.

Color Spaces

RGB color cube
HSV color cone
CIE (x,y)
CIE (u,v)





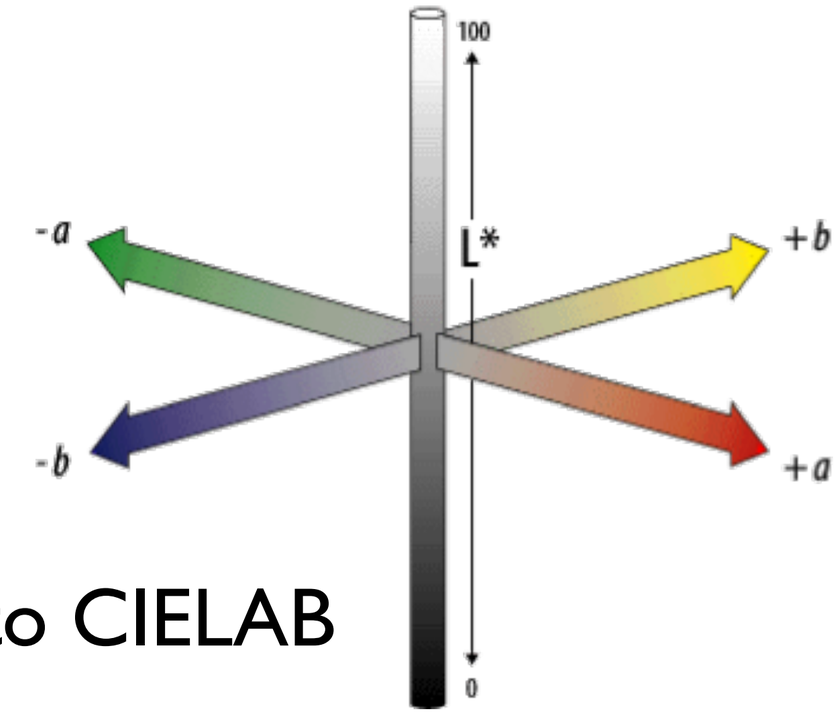
Scaled to be closer to circles.

$$\begin{bmatrix} u' \\ v' \end{bmatrix} = \frac{1}{X + 15Y + 3Z} \begin{bmatrix} 4X \\ 9Y \end{bmatrix}$$

CIELAB Space (AKA L*a*b*)

A commonly used color space that strives for perceptual uniformity

- L* is lightness
- a* and b* are color-opponent pairs
  - a* is red-green
  - b* is blue-yellow
- Nonlinear transformation from CIEXYZ to CIELAB
- Visual system appears to encode color this way (opponent color theory)
  - Afterimages
  - No “redish green” or “blueish yellow”



Monitor Intensity and Gamma

- Monitors convert **pixel value into intensity level**
  - 0.0 maps to zero intensity = black (well not quite)
  - 1.0 maps to full intensity = white

- Monitors are not linear
  - 0.5 does not map to “halfway” gray, (e.g. 0.5 might map to 0.217)
  - Nonlinearity characterized by exponential function

$$I = a^{\gamma}$$

    where I = displayed intensity and a = pixel value (between 0 and 1) γ
  - For many monitors γ is near 2 (often between 1.8 and 2.2)

Determining Gamma

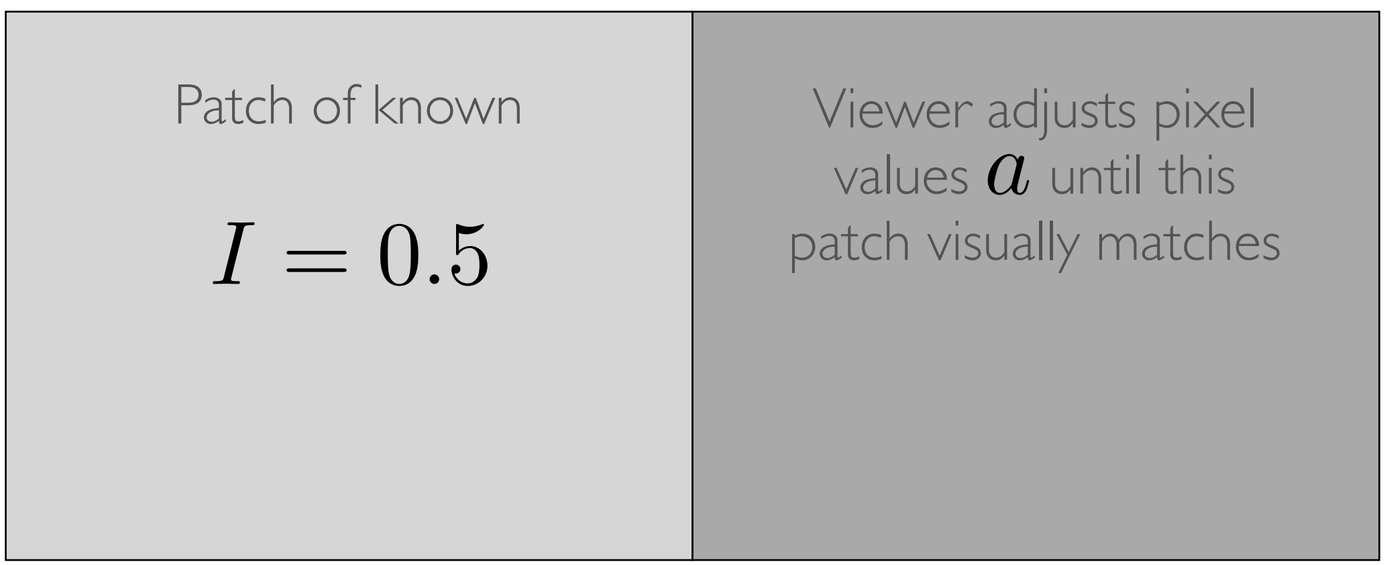
$$I = a^\gamma$$

- Suppose I know displayed intensity of a patch $I = 0.5$

$$0.5 = a^\gamma$$

- Let viewer adjust pixel value a of nearby patch until match

$$\gamma = \frac{\ln 0.5}{\ln a}$$



- How do we make a patch of known intensity?

Determining Gamma

http://www.cs.cornell.edu/Courses/cs4620/2008fa/homeworks/gamma.htm

Dynamic Range

Max/min values also limited on devices

- “blackest black”
- “brightest white”

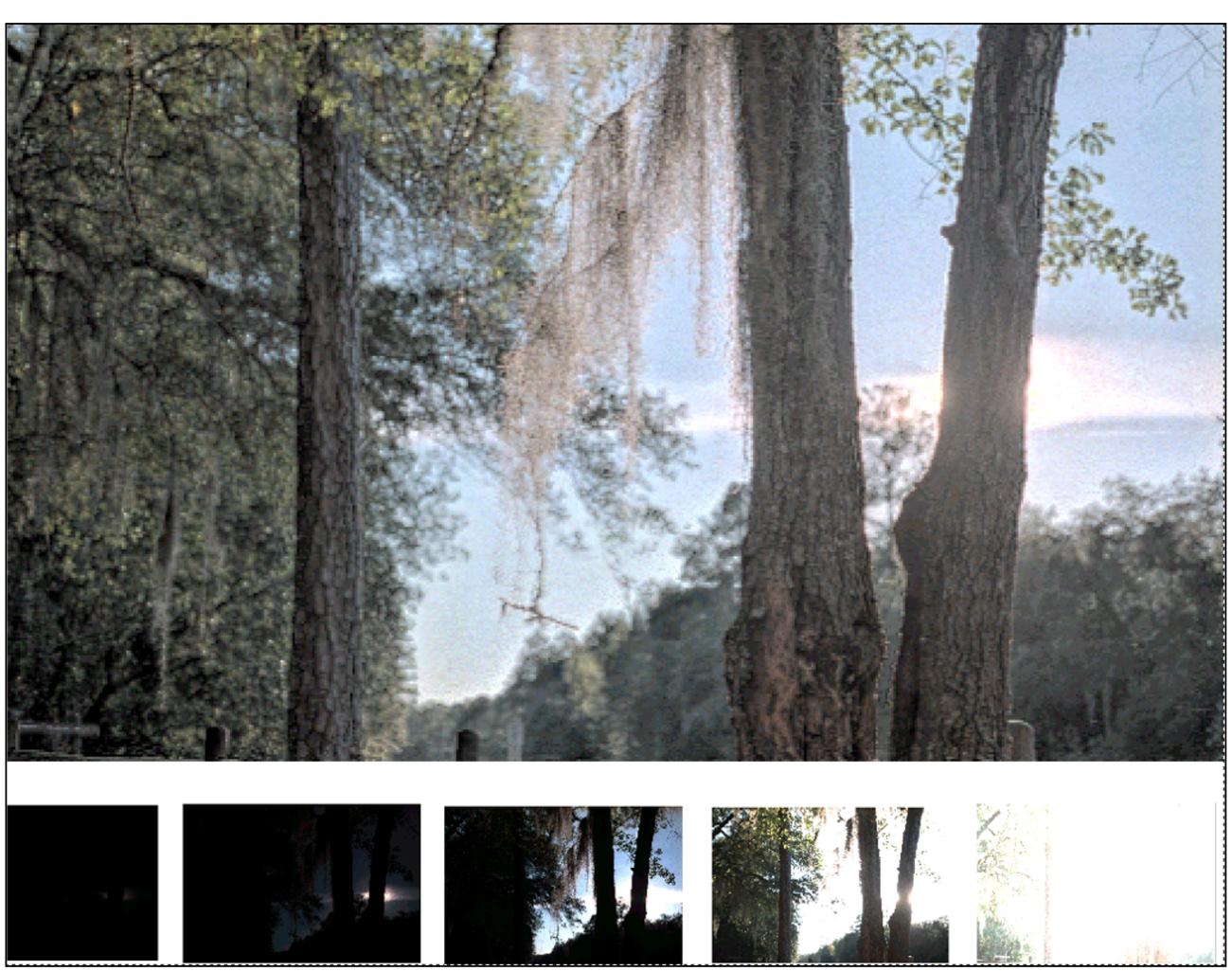
Surface of the sun	**2,000,000,000 nits**
Sunlight clouds	**30,000**
Clear sky	**3,000**
Cellphone display	**500**
Overcast sky	**300**
Scene at sunrise	**30**
Scene lit by moon	**0.001**
Threshold of vision	**0.000001**

1nit = 1 candela / m^2

Dynamic Range

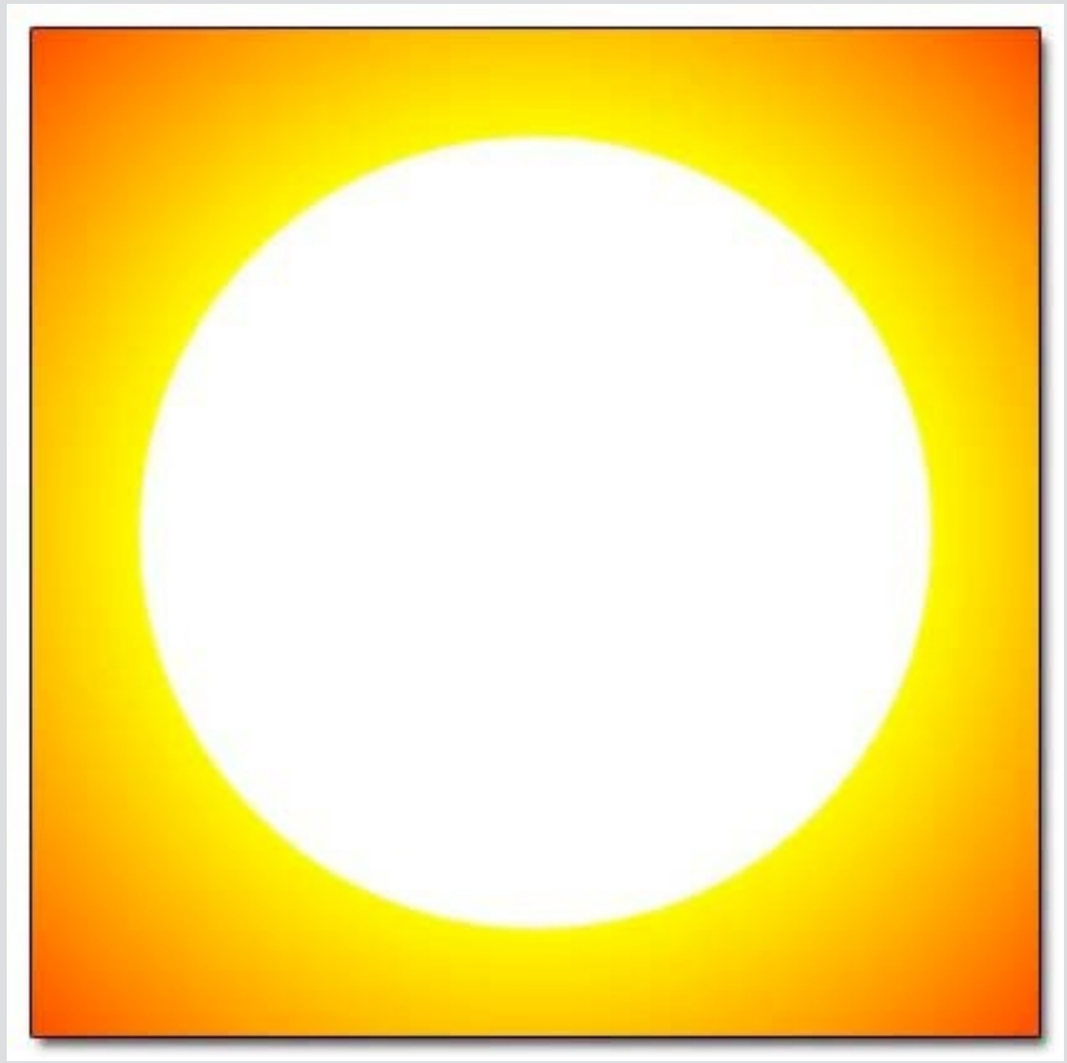
Max/min values also limited on devices

- “blackest black”
- “brightest white”



Jack Tumblin

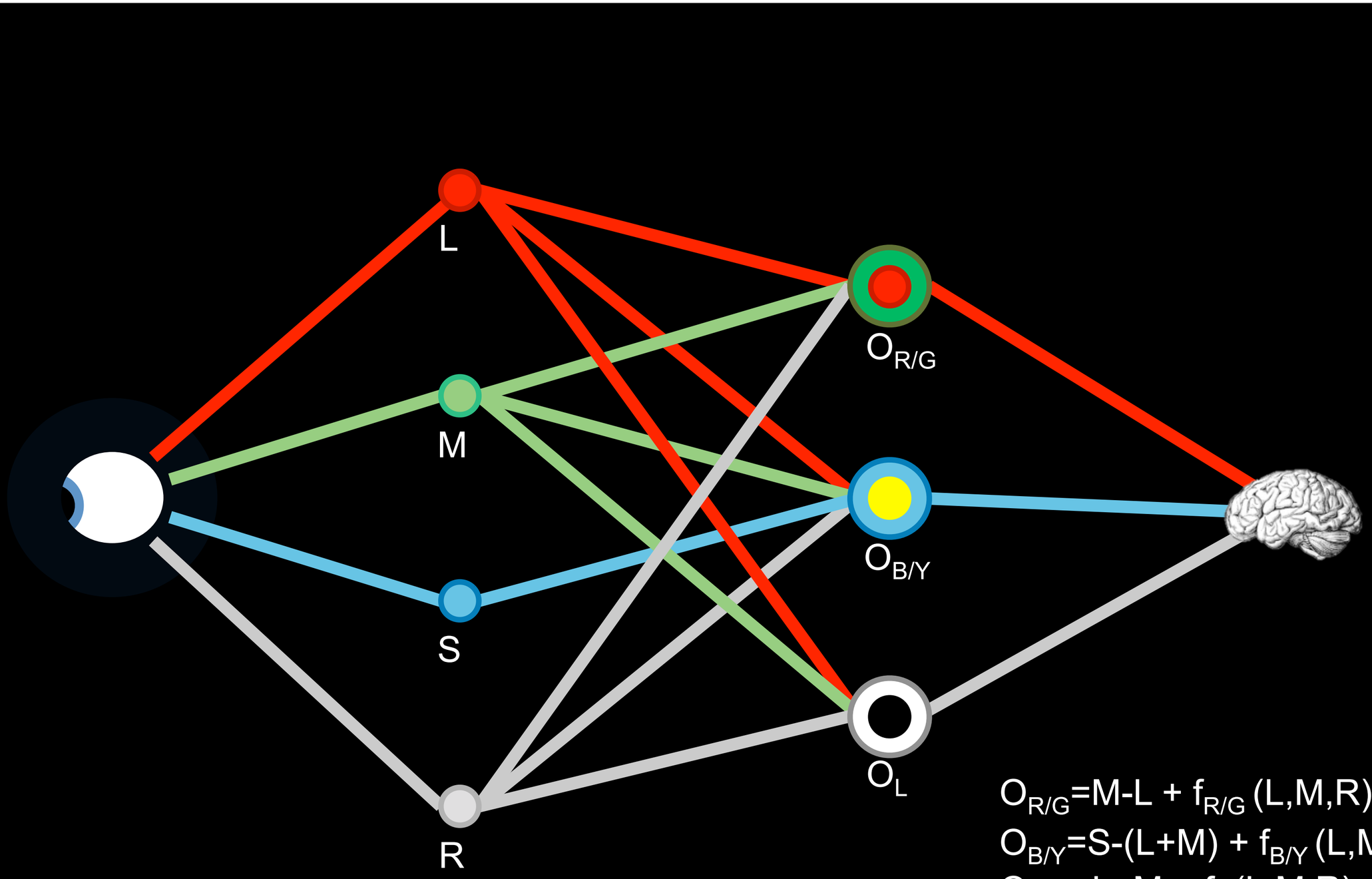
Fake High Dynamic Range



Tone Mapping

Kirk and O'Brien 2011

Rods Contribute to Color



$O_{R/G}$=M-L + $f_{R/G}$ (L,M,R)

$O_{B/Y}$=S-(L+M) + $f_{B/Y}$ (L,M,S,R)

O_L =L+M + f_L (L,M,R)

Color Phenomena

Light sources seldom shine directly in eye

Light follows some transport path, *i.e.*:

- **Source**
- **Air**
- **Object surface**
- **Air**
- **Eye**

Color effected by interactions

Reflection

Light strikes object
Some frequencies reflect
Some adsorbed
Reflected spectrum is light times
Recall metamers...

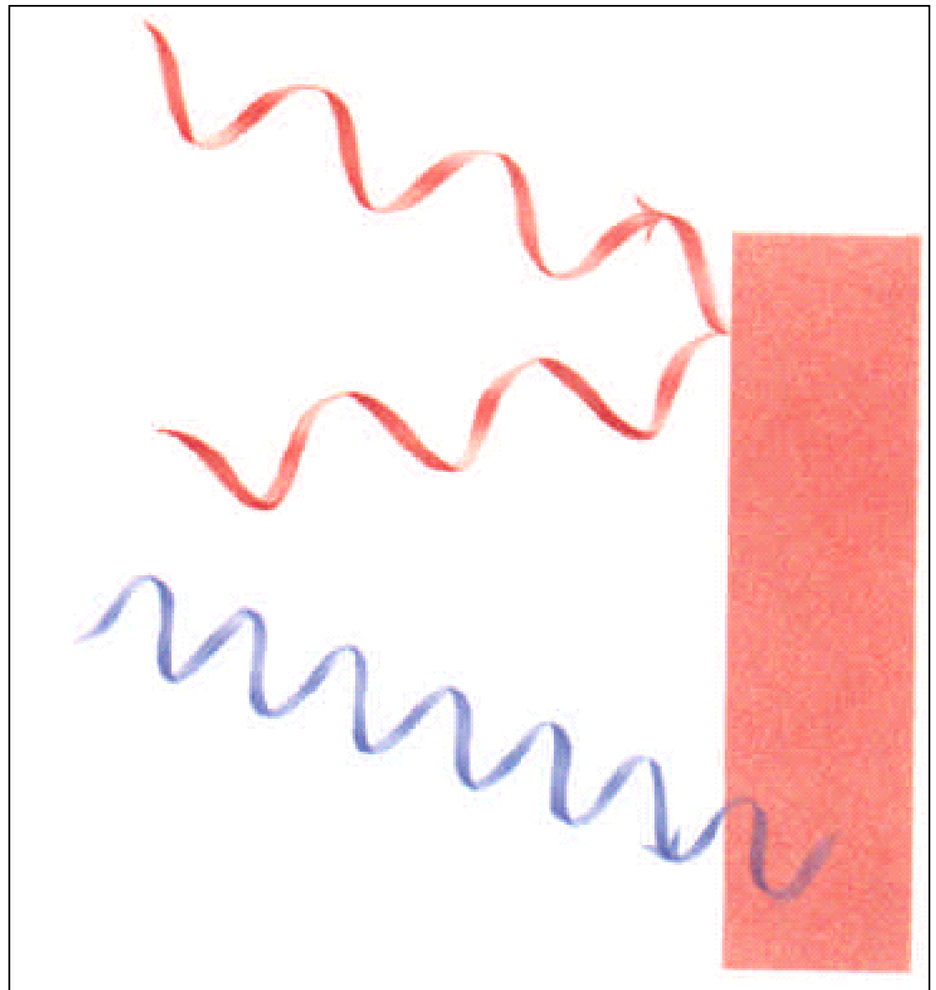


Fig. 1.18 Reflection: red light bounces off an opaque red object, while light of other colours is absorbed.

Unknown?

Transmission

Light strikes object
Some frequencies pass
Some adsorbed (or reflected)

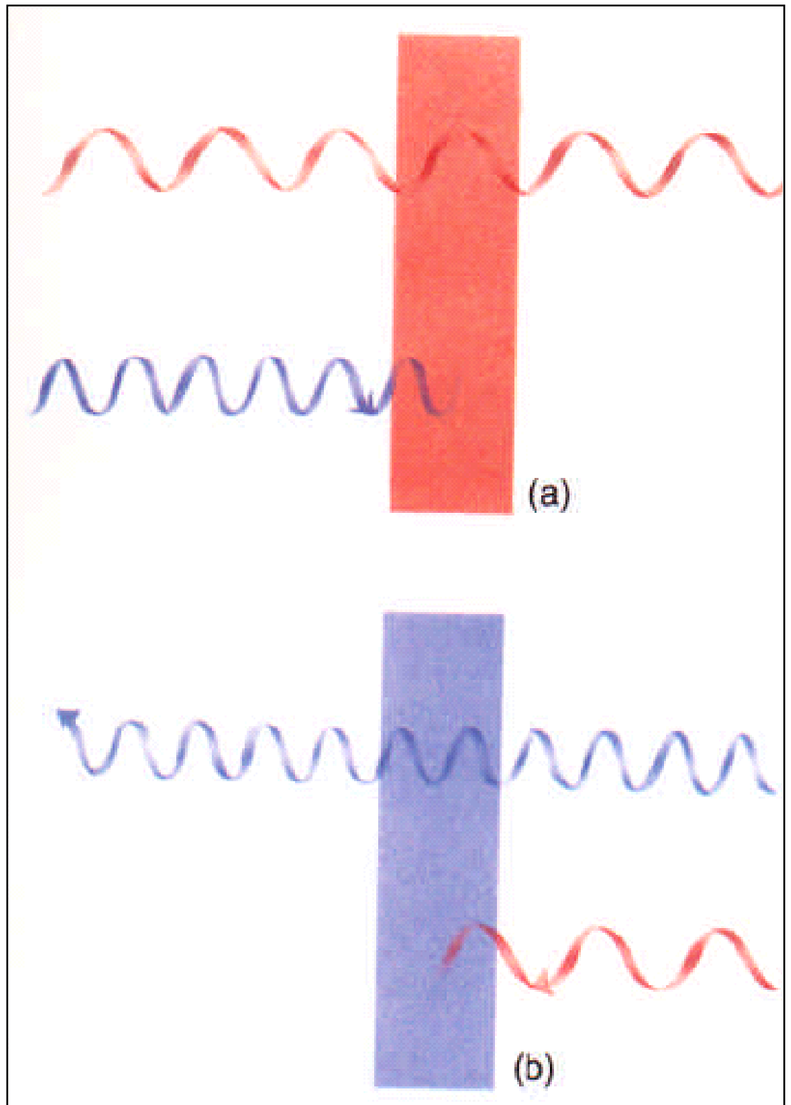


Fig. 1.17 Absorption: a red transparent medium absorbs all wavelengths of light except red (a); a blue transparent medium absorbs all wavelengths except blue (b)

Scattering

Interactions with small particles in medium
Long wavelengths ignore
Short ones scatter

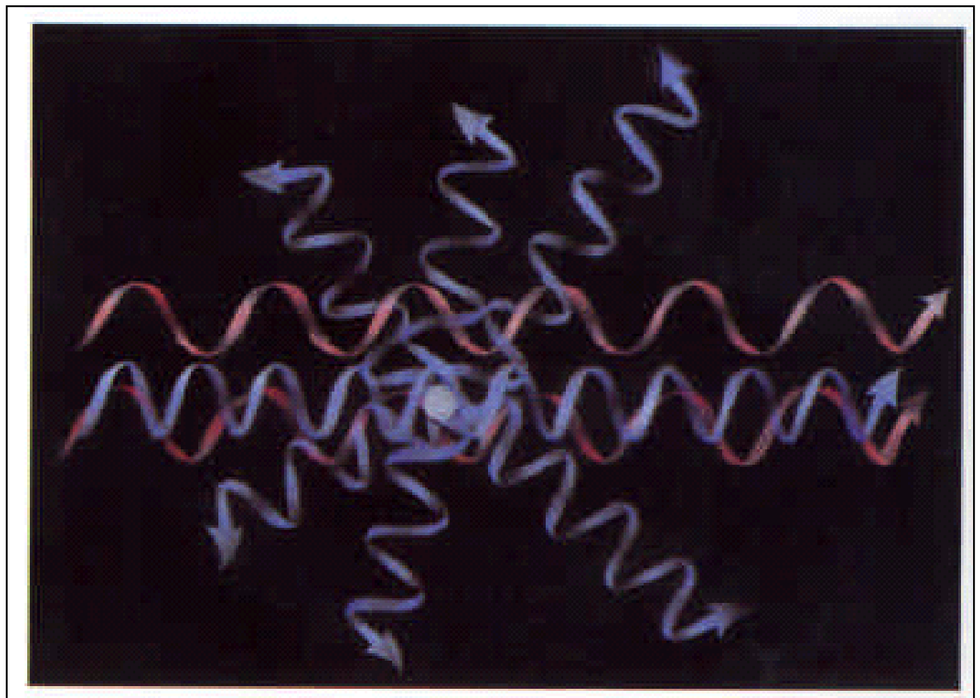


Fig. 1.25 Rayleigh scattering: when particles in air or water are small relative to light wavelength they scatter blue light preferentially.

Unknown?

Interference

Wave behavior of light

- Cancelation
- Reinforcement

Wavelength dependent

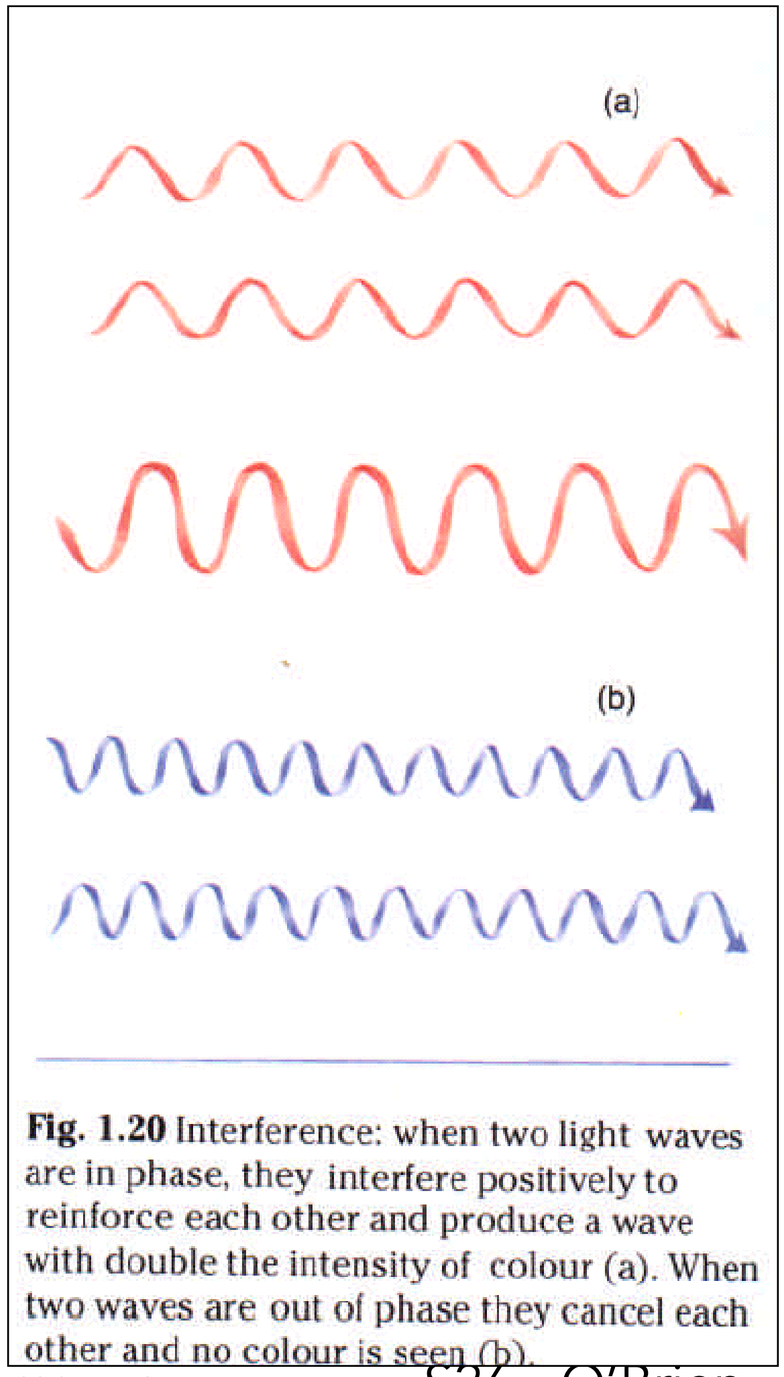


Fig. 1.20 Interference: when two light waves are in phase, they interfere positively to reinforce each other and produce a wave with double the intensity of colour (a). When two waves are out of phase they cancel each other and no colour is seen (b).

Unknown?

Iridescence

Interaction of light with
- Small structures
- Thin transparent surfaces

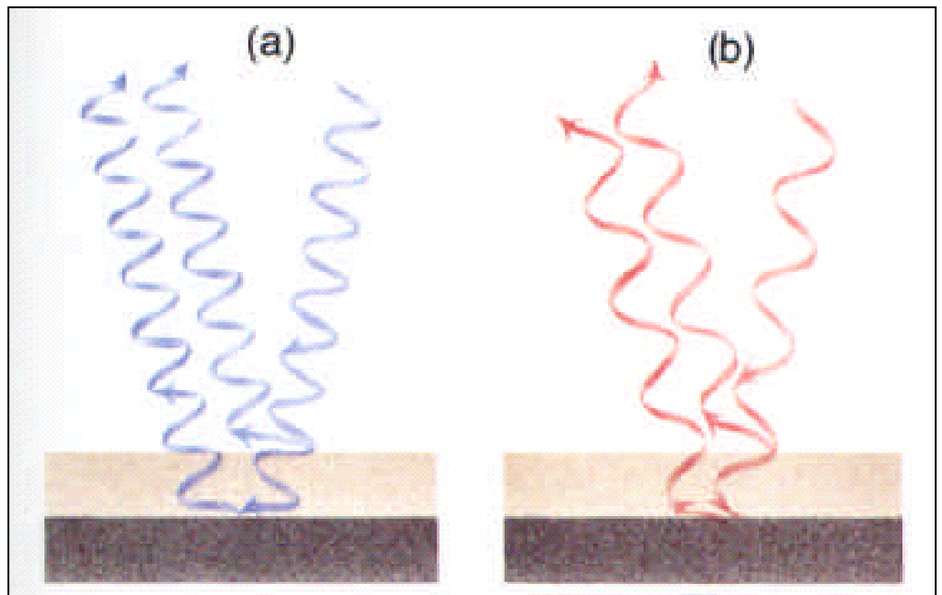


Fig. 1.22 Iridescence: when a light wave is partially reflected and partially transmitted at the surface of a thin layer of transparent material (e.g. a bubble), the two parts of the original wave may interfere with each other when the transmitted wave is reflected from a lower layer and re-emerges at the surface. In this case the blue waves are in phase and their colour is reinforced (a) but the red waves are out of phase and their colour is cancelled (b).

Unknown?

Iridescence

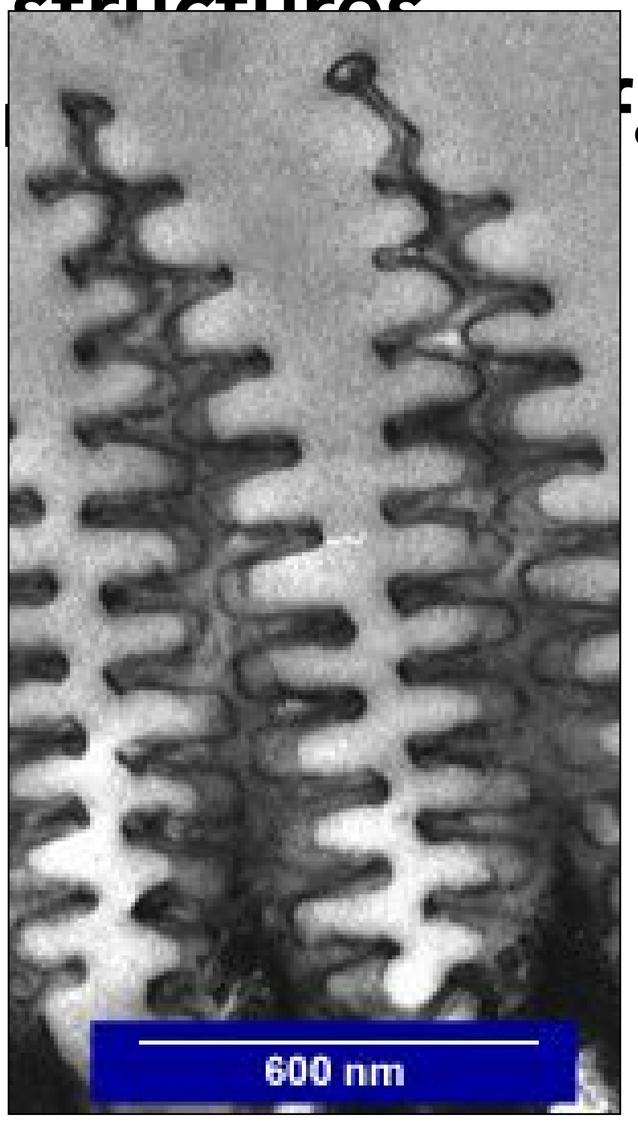
Inte

- Sn
- Th

Iridescence

Interaction of light with

- Small structures
- Thin t faces



Fluorescence / Phosphorescence

Photon come in, knocks up electron
Electron drops and emits photon at other frequency
May be some latency

Radio active decay can also emit visible photons

Fluorescence / Phosphorescence

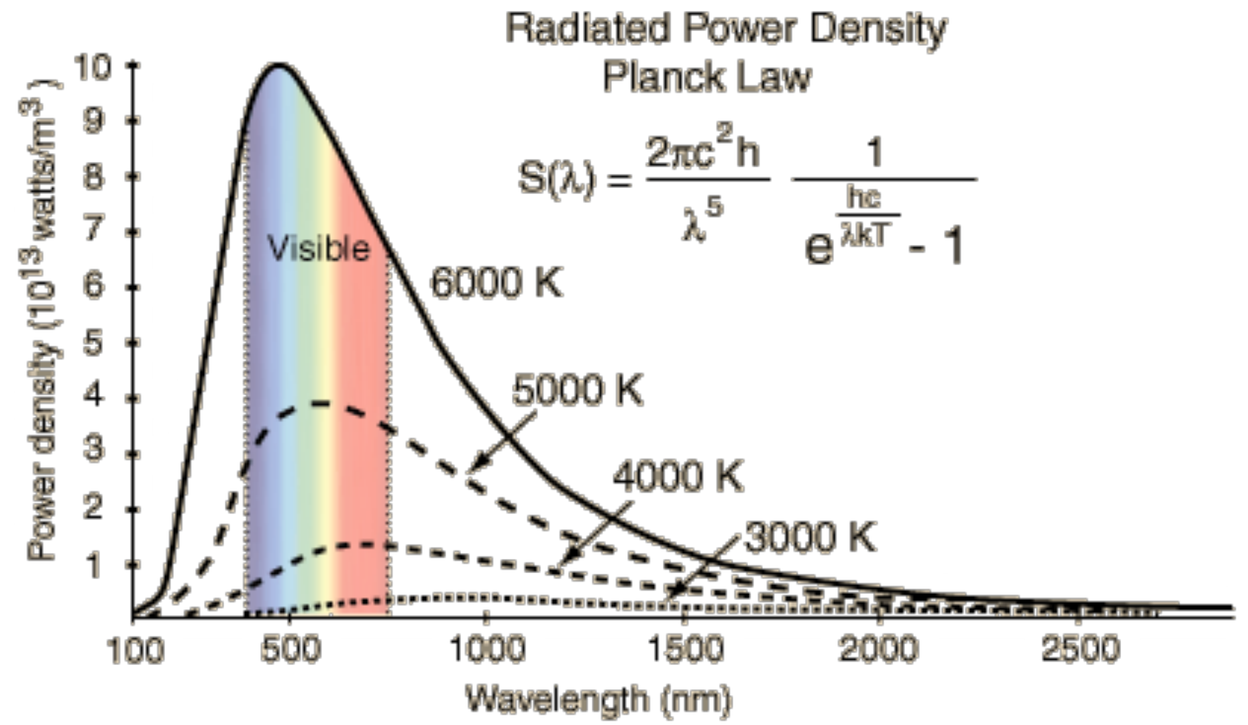
Black Body Radiation

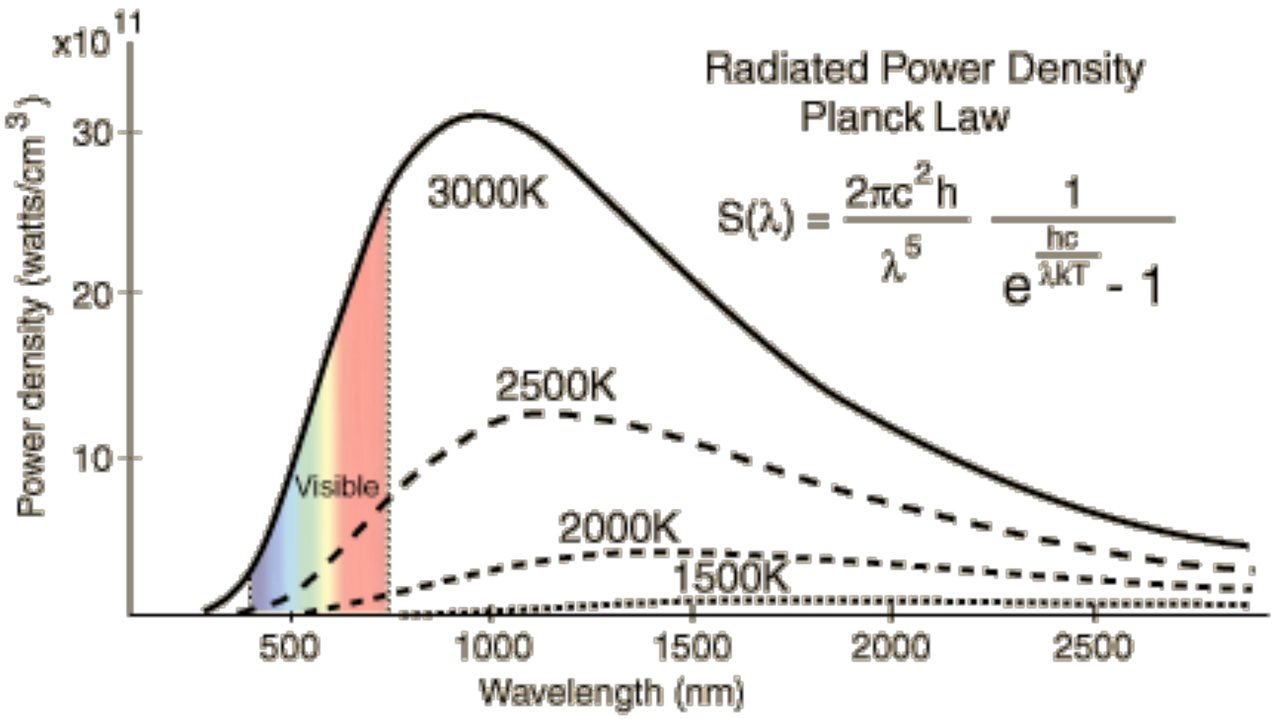
Hot objects radiate energy
Frequency is temperature dependent
Moderately hot objects get into visible range
Spectral distribution is given by

Leads to notion of “color temperature”

$$E(\lambda) \propto \left(\frac{1}{\lambda^5}\right)\left(\frac{1}{\exp(hc/k\lambda T)-1}\right)$$

Black Body Radiation





HyperPhysics

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